Positive scalar curvature and uniform codimension-two width

OpenAI

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Abstract

We prove the quantitative continuous form of Gromov's scalar-curvature conjecture in every dimension $n \geq 4$. Every complete connected smooth boundaryless n-manifold with scalar curvature at least one admits a continuous map to a simplicial complex of dimension at most $n-2$ whose entire fibers have diameter bounded only in terms of n. We also obtain the continuous macroscopic-dimension conclusion for universal covers of closed manifolds with positive scalar curvature in every dimension $n \geq 2$.

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1 Introduction

A lower bound for scalar curvature is a local condition. Gromov's macroscopic dimension conjecture predicts a global consequence: at a controlled scale, two dimensions can be collapsed. We prove the quantitative form of this assertion for arbitrary complete manifolds in dimensions at least four.

For a metric space X and an integer $d \geq 0$, define its *Urysohn d-width* by

$$\mathrm{UW}_d(X) = \inf_f \sup_{y \in K} \mathrm{diam}_X\, f^{-1}(y),$$

where the infimum runs over continuous maps $f : X \to K$ to simplicial complexes of dimension at most d. The complexes have their usual polyhedral topology. The diameter is that of the entire fiber, including all its components; empty fibers may be omitted.

Theorem 1.1. *For every integer $n \geq 4$, there is a finite constant C_n such that every connected complete smooth Riemannian n-manifold (M, g), without boundary and with $\mathrm{Scal}_g \geq 1$, admits a continuous map $f : M \to K$ to a simplicial complex of dimension at most $n-2$ satisfying*

$$\mathrm{diam}_g\, f^{-1}(y) \leq C_n \qquad (y \in K).$$

Thus $\mathrm{UW}_{n-2}(M, g) \leq C_n$. The constant is independent of M, its topology, and its geometry. The theorem applies to compact and noncompact manifolds. If $\mathrm{Scal}_g \geq \sigma^2 > 0$, applying the theorem to $\sigma^2 g$ gives $\mathrm{UW}_{n-2}(M, g) \leq C_n/\sigma$.

For a closed connected Riemannian manifold, let $\mathrm{FillRad}(M, g)$ be Gromov's filling radius [17], defined by the Kuratowski embedding and the fundamental class, with coefficients in $\mathbb{Z}$ if M is orientable and in $\mathbb{Z}_2$ otherwise [32, Definition 3.1]. The next corollary gives the closed-manifold case in dimensions at least four of Gromov's filling-radius Conjecture A^- [20, §3].

Corollary 1.2 (Closed-manifold filling-radius bound)**.** *Let (M, g) be a closed connected smooth Riemannian n-manifold with $n \geq 4$. If $\mathrm{Scal}_g \geq \sigma^2$, where $\sigma > 0$, then*

$$\mathrm{FillRad}(M, g) \leq \frac{C_n}{2\sigma},$$

where C_n is the constant in Theorem 1.1.

For the closed-cover consequence we also use the *continuous macroscopic dimension*: $\dim_{\mathrm{mc}} X \leq d$ means that there is a continuous map to a simplicial complex of dimension at most d, with its uniform simplex metric and a finite uniform bound on the diameters of its entire fibers. Each simplex has the same standard Euclidean shape and edge length.

Corollary 1.3 (Continuous macroscopic dimension of universal covers)**.** *Let (M, h) be a closed connected smooth Riemannian manifold of dimension $n \geq 2$ with positive scalar curvature. Let $\widetilde{h}$ be the lift of h to its universal cover $\widetilde{M}$. Then*

$$\dim_{\mathrm{mc}}(\widetilde{M}, \widetilde{h}) \leq n - 2.$$

Corollary 1.4 (Aspherical obstruction)**.** *A closed smooth aspherical manifold of dimension $n \geq 4$ admits no Riemannian metric of strictly positive scalar curvature.*

An independent proof in every dimension follows from the stronger rational-inessentiality theorem of OpenAI [34, Theorem 1.1 and Corollary 1.2].

1.1 History and significance

Gromov introduced macroscopic dimension in his study of positive scalar curvature [18, §2]. His quantitative conjecture already asks whether a complete manifold with a positive scalar-curvature lower bound can be mapped continuously to a polyhedron of codimension two with all point inverses bounded at the curvature scale [18, §$2\frac{1}{2}$]; see also [22, p. 3180, Comments after Conjecture 3.2]. The diameter condition concerns each entire fiber, including all its components. In the universal cover of a closed manifold the corresponding macroscopic-dimension assertion asks only for some finite bound. The continuous invariant and the version requiring a Lipschitz map to a uniformly metrized complex are distinct [13, Definitions 1.1–1.2, Theorem 6.3 and the following Example].

The pointwise curvature hypothesis cannot be replaced by its fixed-scale macroscopic counterpart [24, Definition 1.3]: at a prescribed radius $r > 0$, the macroscopic scalar curvature at a point is the scalar curvature of the simply connected space form of the same dimension whose r-balls have the volume of the r-ball about a lift of the point in the universal cover of the concentric $2r$-ball. Kumar and Sen construct closed manifolds in every dimension at least four with a prescribed positive fixed-scale macroscopic scalar-curvature lower bound and arbitrarily large Urysohn codimension-two width [24, Theorem A].

The low-dimensional cases have a substantial earlier history. Gromov and Lawson obtained geometric width bounds for closed orientable three-manifolds [19, Corollary 10.11], and Bolotov proved that the universal cover of a closed three-manifold cannot have macroscopic dimension two [8]. For closed three-manifolds with positive scalar curvature, Liokumovich and Maximo constructed Morse functions with curvature-scale bounds for the areas and ambient diameters of connected level components [25, Theorem 1.1]. Liokumovich and Wang constructed proper Morse functions on complete noncompact three-manifolds with uniformly positive scalar curvature, with uniform diameter bounds for connected level components [26, Theorem 1.1]. We use that result and an explicit passage to the graph obtained by collapsing each connected level component to a point for the three-dimensional closed-cover consequence; the surface case follows from the Bonnet–Myers argument.

The aspherical obstruction in dimensions four and five was established by Chodosh and Li [9, Theorem 2], with an independent proof in dimension five by Gromov [23, §7, Main Corollary].

For the macroscopic-dimension conjecture, Bolotov proved the closed spin case under the Strong Novikov Conjecture and a cohomological-dimension condition [5]. Bolotov and Dranishnikov then obtained the codimension-two conclusion for closed spin manifolds under the Strong Novikov Conjecture and injectivity of the connective-to-periodic real K-homology map [6, Theorem 4.3]. For closed orientable totally non-spin manifolds of dimension at least five, they also showed that a macroscopic dimension drop by one implies a drop by two [7, Theorem 3.4]. Daher and Dranishnikov established the analogous implication for closed orientable totally non-spin four-manifolds with residually finite fundamental group [10, Theorem 3.3]. These topological advances concern universal covers and finite bounds; the present quantitative statement is formulated for arbitrary complete manifolds with a bound depending only on dimension after scalar normalization.

The geometric method uses the minimal-hypersurface approach of Schoen and Yau, in which stability and the Gauss equation pass scalar-curvature information to hypersurfaces [35, Section 1], and the torical symmetrization mechanism described by Gromov [21, Section 2]. Gromov's μ-bubble method uses prescribed mean curvature to localize the hypersurfaces entering scalar-curvature descent [18, §$5\frac{5}{6}$]. The weighted variational formulation and its relation to warped-circle geometry and stability are developed by Chodosh and Li [9, Section 3 and Remark 11].

Removing singular regions while retaining scalar-curvature information is also central to Lohkamp's splitting theory [27, §1.2 and Theorem 3.15]. In the construction below, function

updates exclude singular points from the regions where the next scalar bound is required; the support of the minimizer's perimeter measure and both retained sides are then regularized together.

Recent results also address quantitative width and singular descent. Antonelli proves width bounds under a positive scalar-curvature lower bound, an almost-nonnegative Ricci lower bound, and uniform noncollapse, with constants depending on the volume lower bound [3, Theorem 1.4]. Bi and Zhu develop singular scalar-curvature descent for enlargeability obstructions and cubical-width inequalities [4, Theorem B and Section 4]. Rather than continuing the descent on a single hypersurface, the present construction keeps both sides of every cut as well as its boundary support, through countably many cuts.

1.2 The proof and its main constructions

The proof of Theorem 1.1 constructs separating sets with a bound on their intersections. Fix a length D, depending only on n, and a separated net (x_i) whose closed D-balls cover M and whose closed $2D$-balls are locally finite. Give each center a label. Since the cuts insert cylinders, we record original positions by a continuous collapse $q : M \to M$ with $d_g(q(x), x) < 1$. The required separating set for label i separates the q-preimages of the inner ball and the exterior of the outer ball; no $n-1$ distinct separating sets may meet. These are the partitions of Proposition 2.1. To see why they suffice, thicken the separating sets simultaneously without increasing their multiplicity (Lemma 2.2), and choose continuous sign coordinates $t_i : M \to [0, 1]$ with transitions inside these neighborhoods, equal to one on the inner preimage and zero on the outer exterior preimage. At most $n-2$ coordinates are fractional at any point, so the coordinate map lands in an $(n-2)$-dimensional cubical complex. Some coordinate is always one; equality of coordinate vectors therefore confines the collapse images of an entire fiber to the same outer ball. Section 2 triangulates this map and obtains

$$C_n = 4D + 2.$$

Each label carries a function μ, which supplies the prescribed weighted mean curvature of its cut and obeys $2|\operatorname{grad}\mu| \le a\mu^2 + \Pi$, with a coefficient $a \in [.60, .63]$ fixed at each stage and a positive *potential* Π; the scalar lower bound required for the cuts contains a multiple of the sum of these potentials (Definition 6.2). A function with constant potential p needs length at least $2\pi/\sqrt{ap}$ to pass from $+\infty$ to $-\infty$. Since the sum of the potentials must be finite, for infinitely many labels the potentials cannot have a common positive lower bound, so the corresponding required separations can be arbitrarily large. We obtain them by stretching the metric while keeping D fixed. For a metric h and a smooth function t, the weighted scalar curvature is

$$S_1^h(t) = \operatorname{Scal}_h + 2\Delta_h t - |\mathrm{d}t|_h^2. \tag{1.1}$$

By Lemma 5.3, $S_1^h(t) - |\mathrm{d}t|_h^2/k$ is the scalar curvature of $h + e^{-2t/k}(\mathrm{d}\theta_1^2 + \cdots + \mathrm{d}\theta_k^2)$ on $M \times \mathbb{T}^k$; thus $S_1^h(t)$ is the limit of these scalar curvatures as $k \to \infty$. The local graph deformation of Theorem 4.1 increases h and t, retaining a positive quadratic term t^2 in a lower bound for $S_1^h(t)$. Its loss is charged to the decrease of $\operatorname{tr}_h h_0$, for a background metric h_0 fixed throughout a round, plus an arbitrarily small quadratic loss. The deformation is driven by the forcing function $F(x, t)$ of Theorem 3.1, whose height derivative F_x is either at most $b_0 t^2$ or compensated by its t-dependence, $-FF_t \ge P_0 F_x$; the constants of F are fixed before the local geometry and the error tolerance. The trace starts at n and remains positive, so its decrements telescope independently of the number of overlapping balls. Rescaling between rounds restores the positive quadratic term while keeping a fixed positive reserve in original units. Each label stops after finitely many rounds, and the changes are supported in its original outer ball, so the construction stabilizes on compact sets (Theorem 5.1). Splitting the

increments of t among finitely many circle warping functions on each compact region converts the resulting bound into ordinary scalar-curvature control.

The descent uses weighted μ-bubbles and torical symmetrization. To obtain partitions, we retain both sides of each cut as well as its boundary support, attaching cylinders to the sides to replace their boundaries by complete ends. The connected components in Definition 6.1 have product identifications that make later cuts compatible at intersections (Lemma A.1). Besides the usual boundary stability inequality, this requires a bound on each retained side. Deleting subsets from a minimizing Caccioppoli set controls the mean-curvature term at its cylinder interface and gives the required scalar form bound (Lemma B.9 and Proposition B.10). The first cut uses an extra radial label to give compact boundary and side components before cylinder attachment (see below); after it, a common constant perturbation of the prescribed weighted mean curvature ensures uniqueness of minimizing Caccioppoli sets on components and their Euclidean products. Uniqueness then forces the cuts to converge to the corresponding end base products at infinity (Proposition B.5 and Corollary B.7), allowing the subsequent regularization to make their ends exactly compatible.

The curvature requirement is adapted to the intersections we must exclude. Each remaining label has a function μ and two prescribed open sets whose closures lie in the interiors of $\{\mu = +\infty\}$ and $\{\mu = -\infty\}$, respectively, where its future sign is fixed; call the label undecided outside these prescribed open sets. On a component of base dimension m, we require a positive lower bound for the stabilized scalar curvature, the scalar curvature of the component metric warped with all circle factors accumulated so far (Lemma 5.3), only where at least $m - 1$ labels remain undecided. A side keeps its dimension and loses the processed label, whereas a boundary loses both one dimension and that label. Thus every new required region projects into the old one (Definition 6.2 and Lemma 6.3). This restriction also permits singular cuts: their singular sets have dimension at most $m-8$. Smooth approximations to the remaining functions admit generic thresholds with at most $m - 8$ equalities at any singular point. The strict inequalities select signs to force nearby, leaving fewer than $m - 2$ labels undecided. Enlarging enough of the corresponding prescribed open sets therefore removes the singular set from the region with at least $m - 2$ remaining undecided labels, which contains the projections of all new required regions (Lemma C.1 and Proposition D.2).

Forcing these additional signs costs potentials concentrated in thin annuli around singular pieces. On the boundary and on both joined sides, with matching interface traces, estimates for inverse-square potentials bound each such potential by a fixed fraction of the reserved gradient form plus a local L^2 error (Proposition C.5); the reserved form is a fixed part of the gradient terms left over in the boundary and side stability forms (Definition B.11). To combine the additions without spending this reserve once per piece, Lemma D.4 uses compactness: on a compact region the bounded old potential acts on the limiting test function, while the shrinking potential acts on the concentrating remainder. The total form coefficient is therefore bounded by the larger of the previous and local coefficients, up to an arbitrarily small error. The extensions in Proposition D.2 preserve this bound throughout the component diagram. The regularization in Proposition E.1 can then approximate the cut closely near the required regions, smooth the interfaces, and use compatible positive supersolutions to restore the pointwise scalar bound by adding circle factors. These constructions prove the finite cutting theorem, Theorem 6.5.

For the countable iteration, the analysis on each component needs one finite warping function list on the entire complete component: it supplies the positive injectivity radius and uniform bounds on curvature and all its covariant derivatives on the whole component and polynomial weighted volume growth used for the barriers, the uniqueness of minimizing Caccioppoli sets, and the bounded solution of the elliptic equation (Lemmas 7.4 and A.2). The first, radial cut gives compact boundary and side components before cylinder attachment; on a noncompact manifold, alternating signs along a proper radial function enforce this confinement (Lemma 7.3). Their collapse images stay in those

compact regions, so only finitely many original outer balls and supports of the logarithms of warping functions meet each image. Lemma 7.4 and Proposition 7.5 preserve finite lists through the incident end base diagrams: all other labels remain forced, and their potentials form a summable list of spatial constants which the approximations reproduce exactly. The finite cutting argument can therefore be iterated with summable stage losses. An intersection of $n-1$ distinct separating sets would give a complete one-dimensional component. Its required count is zero, so the positive scalar bound holds everywhere. The finite warped-product formula then yields a differential inequality with a negative quadratic upper bound on its arclength universal cover which is impossible on the whole line (Lemma 7.6). This gives the intersection bound in Proposition 7.2.

Section 8 puts these partitions on one fixed copy of M. Truncate the analytic cylinders and compress each inserted finite collar into an arbitrarily thin old collar, preserving all earlier separating sets and signs. The displacement errors are summable in the original metric, and the change for a center label is confined to a fixed-radius original ball about that center. Local finiteness makes both the collapse maps and the partitions eventually unchanged near each point. This gives the continuous collapse and locally finite separating sets used above; locally finite coordinate supports ensure continuity of the width map in the ordinary polyhedral topology.

The three corollaries are proved in Section 9. For Corollary 1.3 in dimensions at least four, apply the theorem to the normalized lifted metric and realize the coordinate complex with the uniform simplex metric; local finiteness preserves continuity and the conversion preserves fibers (Lemma 9.1). Dimension two uses Bonnet–Myers. In the noncompact three-dimensional case, Lemma 9.2 makes the bounded level components supplied by Liokumovich and Wang into entire fibers of a map to the graph obtained by collapsing each connected level component to a point. The filling-radius corollary follows from Gromov's comparison $\mathrm{FillRad} \le \frac{1}{2}\,\mathrm{UW}_{n-1}$. For the aspherical obstruction, the width bound on the universal cover contradicts Gromov's observation that a complete uniformly contractible n-manifold, such as the universal cover of a closed aspherical n-manifold, admits no continuous map with uniformly bounded fibers to a polyhedron of dimension less than n.

Organization. Section 2 deduces Theorem 1.1 from the partition statement, Proposition 2.1. Sections 3, 4 and 5 stretch the metric: the function with quadratic derivative bounds, the local stretching step, and the rounds of Theorem 5.1. Section 6 introduces components, the analytic invariant and the finite cutting theorem, Theorem 6.5. Its proof occupies the appendices: component geometry and finite reconstructions in Appendix A, weighted minimizing cuts in Appendix B, exclusion of singular points from the required regions in Appendices C and D, and regularization in Appendix E. Section 7 adapts this construction to countably many labels and proves the intersection bound, Proposition 7.2. Section 8 proves Proposition 2.1, and Section 9 proves the three corollaries.

Convention 1.5. The Laplacian is $\Delta = \operatorname{div}\operatorname{grad}$, and round spheres have positive scalar curvature. For a hypersurface with unit normal ν, its second fundamental form is $B(v,w) = \langle \nabla_v \nu, w\rangle$, and its mean curvature is $\operatorname{tr} B$. Integration and divergence use Riemannian densities. Distances in the final fiber bound are always measured in the original metric g.

2 From partitions to the width map

The geometric construction will produce signed partitions whose separating sets have no $(n-1)$-fold intersection. We first state this output and deduce the width bound from it. Its proof is in Section 8, after the stretching and cutting constructions.

2.1 The original scale and the partition statement

Let $D = WR_0$ be the length fixed in (5.2), at the start of Section 5; it depends only on n, and it is the only constant of the stretching construction used in this section.

Now fix $n \geq 4$ and (M, g) satisfying Theorem 1.1. Distances are measured in this original metric: write $d_0 = d_g$ and $B_0(x, R) = B_g(x, R)$. Let the centers x_i be those supplied by Theorem 5.1; its proof chooses them as a maximal D-separated set. Each center carries a label i, with the ball pair

$$A_i = \overline{B}_0(x_i, D), \qquad B_i = M \setminus B_0(x_i, 2D).$$

The A_i cover M. Every family of balls with these centers and a fixed finite radius is locally finite: a compact set has a compact neighborhood of that radius by completeness, and a compact metric space contains only finitely many D-separated points. The family is countable, since M is second countable. If some B_i is empty, $\operatorname{diam}_g M \leq 4D$ and a constant map gives the required bound. We therefore assume that all B_i are nonempty.

Proposition 2.1 (Reconstructed partitions)**.** *For the manifold, scale, centers, and nonempty ball pairs fixed above, there are a continuous proper map $q : M \to M$ and closed sets P_i, N_i, W_i for each label i such that*

$$d_0(q(x), x) < 1, \tag{2.1}$$

$$P_i \cup N_i = M, \qquad P_i \cap N_i = W_i, \qquad W_i \subset q^{-1}\big(B_0(x_i, 2D)\big), \tag{2.2}$$

$$q^{-1}(A_i) \subset P_i \setminus W_i, \qquad q^{-1}(B_i) \subset N_i \setminus W_i. \tag{2.3}$$

The separating set family (W_i) is locally finite, and no $n-1$ distinct separating sets have a common point.

The proof of Proposition 2.1 uses Theorem 5.1 to obtain the separated ball pairs, Proposition 7.2 to obtain the finite-stage separating sets, and collar compressions to put their locally stationary limit on M. For the width map, the properties in the statement are sufficient.

2.2 Thickening the separating sets without increasing their multiplicity

Put $k = n - 2$. We need open neighborhoods of the whole separating sets, rather than independently chosen neighborhoods of their individual compact pieces.

Lemma 2.2 (Simultaneous separating set neighborhoods)**.** *For the map and separating sets in Proposition 2.1, there are open sets $T_i \supset W_i$ forming a locally finite family and satisfying*

$$T_i \subset q^{-1}\big(B_0(x_i, 2D)\big) \setminus q^{-1}(A_i \cup B_i), \qquad \#\{i : x \in T_i\} \leq k \quad (x \in M). \tag{2.4}$$

Proof. Set $O_i = q^{-1}(B_0(x_i, 2D))$ and $F_i = q^{-1}(A_i \cup B_i)$. By Proposition 2.1, W_i is disjoint from the closed set F_i and lies in the open set O_i. At a point x, local finiteness and closedness of the separating sets give an open neighborhood meeting only separating sets that contain x. Shrink it to a neighborhood U_x contained in $O_i \setminus F_i$ for every such separating set. Therefore U_x meets at most k separating sets, and

$$U_x \cap W_i \neq \varnothing \quad \Longrightarrow \quad U_x \subset O_i \setminus F_i. \tag{2.5}$$

Choose an open cover $\mathcal{V}$ such that the union of its members meeting any fixed member is contained in some U_x. For completeness, one is given by the balls $V_x = B_0(x, \rho(x)/8)$, where

$$\rho(x) = \sup_U \min\{1, d_0(x, M \setminus U)\}, \qquad U \in \{U_x : x \in M\},$$

with distance to the empty set truncated to 1. This function is positive and 1-Lipschitz. If V_y meets V_x, the Lipschitz inequality gives $d_0(x,y) < 2\rho(x)/7$, and hence $V_y \subset B_0(x, 25\rho(x)/56)$. Thus the union of all balls meeting V_x is contained in $B_0(x, \rho(x)/2)$, which lies in a member of the original cover. Take $\mathcal{V} = \{V_x : x \in M\}$.

Define

$$T_i = \bigcup \{V \in \mathcal{V} : V \cap W_i \neq \varnothing\}.$$

It contains W_i, and (2.5) implies the containment in (2.4). If z belongs to several T_i, choose $V_0 \in \mathcal{V}$ containing z. For each such i there is a $V_i \in \mathcal{V}$ containing z and meeting W_i. All the V_i meet V_0, so their union lies in a single U_x. That U_x meets each of these separating sets, and therefore there are at most k of them. Finally, (T_i) is locally finite because $T_i \subset O_i$ and the family (O_i) is locally finite. □

2.3 Proof of the width bound

Proof of Theorem 1.1. For each i the closed sets

$$C_i^+ = P_i \setminus T_i, \qquad C_i^- = N_i \setminus T_i$$

are disjoint and cover $M \setminus T_i$. For a closed set C write $\delta_C(x) = \min\{1, d_0(x, C)\}$, with $\delta_\varnothing = 1$, and put

$$t_i(x) = \frac{\delta_{C_i^-}(x)}{\delta_{C_i^-}(x) + \delta_{C_i^+}(x)}. \tag{2.6}$$

The denominator is positive because the two closed sets are disjoint. These continuous coordinates have the following properties:

(i) $t_i = 1$ on $q^{-1}(A_i)$ and $t_i = 0$ on $q^{-1}(B_i)$;

(ii) $0 < t_i(x) < 1$ implies $x \in T_i$;

(iii) $\operatorname{supp} t_i \subset q^{-1}(\overline{B}_0(x_i, 2D))$, so the coordinate supports are locally finite;

(iv) at every point at least one coordinate equals 1, and at most k coordinates lie strictly between 0 and 1.

Here the last assertion uses the inner-ball cover and Lemma 2.2.

Consider the cubical set of finitely supported coordinate vectors described by these properties. Its cubes are

$$Q(S, A) = \{t : t_i = 1 \ (i \in S),\ 0 \le t_i \le 1 \ (i \in A),\ t_i = 0 \ (i \notin S \cup A)\},$$

where S, A are disjoint finite sets of labels, $S \neq \varnothing$, and $|A| \le k$. Triangulate each cube as follows: for every ordering $a_1, \dots, a_j$ of A, take the simplex with vertices

$$\mathbf{1}_S, \quad \mathbf{1}_{S \cup \{a_1\}}, \quad \dots, \quad \mathbf{1}_{S \cup A}. \tag{2.7}$$

The triangulations agree on faces: setting a free coordinate to zero removes it, while setting it to one moves it to S. They therefore define a simplicial complex K of dimension at most k.

We use the usual polyhedral topology on $|K|$. To verify continuity, fix a point of M and a neighborhood meeting only finitely many coordinate supports. On that neighborhood all other coordinates are identically zero. The vector (t_i) thus takes values in the finite cubical complex using this finite set of labels, and is a continuous map to that complex. Its triangulation is a

finite geometric simplicial complex, whose inclusion in $|K|$ is continuous. The coordinate map consequently defines a continuous map

$$F : M \longrightarrow |K|, \qquad \dim K \le n-2. \tag{2.8}$$

The triangulation retains the entire coordinate vector: the affine map from each simplex in (2.7) to its coordinate simplex agrees on faces and is injective on their union. One can see injectivity by restricting any two finite supports to their finite union; there it is the usual triangulation of a cubical complex. Thus $F(x) = F(y)$ implies $t_i(x) = t_i(y)$ for every i.

Choose, for any two points of the same fiber, a coordinate that equals 1 at the first point. It also equals 1 at the second. Since $t_i = 0$ on $q^{-1}(B_i)$ by (i), neither point lies in $q^{-1}(B_i)$. Thus both collapse images lie in the same open outer ball $B_0(x_i, 2D) = M \setminus B_i$, and hence

$$d_0(x,y) \le d_0(x,q(x)) + d_0(q(x),x_i) + d_0(x_i,q(y)) + d_0(q(y),y) < 4D+2.$$

This applies to every pair of points in the fiber, regardless of its number of components. Consequently

$$\sup_{z\in|K|} \operatorname{diam}_g F^{-1}(z) \le 4D+2 = 4WR_0+2. \tag{2.9}$$

The length D depends only on n, so this proves Theorem 1.1. □

3 A function with quadratic derivative bounds

The local metric deformation will use a forcing function depending both on a height variable x and on the parameter t. Its derivative in x must be large, while a large positive derivative in x must be accompanied by a favorable derivative in t. The following construction supplies these properties with constants chosen before any local geometric data.

Theorem 3.1 (Quadratic derivative bounds)**.** *For every $A_0, P_0, \varepsilon, b_0 > 0$ there exist $0 < S_0 \le 1$, $T \ge 1$, and a smooth function*

$$F : (-S_0, S_0) \times [T, \infty) \longrightarrow \mathbb{R}$$

such that, for $|x| < S_0$ and $t \ge T$, F is odd in x, $F_x > 0$, and

$$\begin{gathered} F_x \ge A_0 t, \qquad |F_t| \le \varepsilon |F|, \qquad F F_t \le \varepsilon F_x, \\ F_x \le b_0 t^2 \quad \text{or} \quad -F F_t \ge P_0 F_x, \\ F_x \le b_0 t^2 + \varepsilon F^2. \end{gathered} \tag{3.1}$$

For each fixed $t \ge T$,

$$\lim_{x\to -S_0} F(x,t) = -\infty, \qquad \lim_{x\to S_0} F(x,t) = +\infty.$$

These limits are uniform when t ranges over a compact subinterval of $[T,\infty)$. The choices depend only on $A_0, P_0, \varepsilon, b_0$ and fixed numerical constants, independently of geometric data and later deformation parameters. Restriction to any larger lower bound T preserves all these conclusions.

In Section 4, the bound $F_x \ge A_0 t$ gives the gradient and decay estimates of Lemmas 4.2 and 4.3 and the gain at large slope in Lemma 4.4. In the scalar comparison of that lemma, the two bounds on F_t control the t-derivative terms, the alternative bounds the possible deficit at large slope, and the last inequality bounds the loss at small slope; the poles keep the graph height close to its cutoff.

Proof. We construct F by prescribing its inverse in the first variable. For the moment, let $k > 0$ and let g be a positive smooth function on $[0, \infty)$, both to be chosen. Write $f > 0$ for an independent forcing function value and, for $t > 0$, set

$$X(f, t) = \frac{1}{k} \arctan \frac{f^{1/3}}{g(y)}, \qquad q = f^{2/3}, \qquad y = \frac{t}{q}. \tag{3.2}$$

Put $A(y) = g(y) + 2yg'(y)$. Differentiation at fixed f or at fixed t, respectively, gives

$$fX_t = -\frac{g'(y)}{k(1 + g(y)^2/q)}, \qquad X_f = \frac{A(y)}{3kq^2(1 + g(y)^2/q)}. \tag{3.3}$$

Thus $A > 0$ will make X invertible in f. For the inverse, $F_x = 1/X_f$ and $-FF_t/F_x = fX_t$. We will arrange the favorable forcing function branch on a decreasing part of g, and obtain the quadratic branch from a lower bound for $t^2 X_f$. We choose g to decrease for small y and to equal y for large y, while keeping A positive through the transition.

We construct g as a smooth rounded maximum with a controlled negative derivative. Fix a nonnegative smooth probability density ρ supported in $[1, 2]$, positive in $(1, 2)$, and vanishing to infinite order at its endpoints. For example, normalize

$$\rho(r) = \begin{cases} c_\rho \exp(-1/((r-1)(2-r))), & 1 < r < 2, \\ 0, & r \notin (1, 2), \end{cases} \qquad \int_1^2 \rho(r)\, \mathrm{d}r = 1.$$

Write $\mu = \int_1^2 r\rho(r)\, \mathrm{d}r$, so $1 < \mu < 2$. For $D_g > 0$ set

$$h(y) = D_g(1 + y)^{-1/4}, \qquad g(y) = \int_1^2 \max\{y, rh(y)\}\rho(r)\, \mathrm{d}r, \qquad y \geq 0. \tag{3.4}$$

Here and below smoothness at $y = 0$ means smooth extendibility to a neighborhood of $[0, \infty)$. To verify smoothness, put

$$M(z) = \int_1^2 \max\{z, r\}\rho(r)\, \mathrm{d}r.$$

Then $M'' = \rho$, with ρ extended by zero, and $g(y) = h(y)M(y/h(y))$. Thus g is smooth; moreover $M(z) = \mu$ for $z \leq 1$ and $M(z) = z$ for $z \geq 2$. Consequently $g > 0$, $g \geq y$, and $g(y) = y$ for all sufficiently large y.

For a direct derivative estimate, let

$$L(z) = \int_{r \leq z} \rho(r)\, \mathrm{d}r, \qquad R(z) = \int_{r \geq z} r\rho(r)\, \mathrm{d}r.$$

At $z = y/h(y)$ the identities

$$g = yL(z) + hR(z), \qquad g' = L(z) - \frac{hR(z)}{4(1 + y)}$$

give

$$g \geq y, \qquad -\frac{g}{4(1 + y)} \leq g' \leq 1, \qquad A(y) = g(y) + 2yg'(y) \geq \frac{y + 2}{2(y + 1)} g(y) \geq \frac{1}{2} g(y). \tag{3.5}$$

In particular the lower bound for A/g is numerical and does not deteriorate as D_g increases.

We now specify the order of all choices. Choose

$$k \geq \max\{A_0, \varepsilon^{-1}, \pi/2\}, \qquad Y \geq 1 \quad \text{with} \quad Y^{-3} \leq \frac{b_0}{12k}. \tag{3.6}$$

Choose D_g only after k, Y, large enough that

$$D_g \geq \max\left\{Y(1+Y)^{1/4}, \frac{8kP_0(1+Y)^{5/4}}{\mu}\right\}. \tag{3.7}$$

Since $y(1+y)^{1/4}$ is increasing, $y/h(y) \leq 1$ on $[0, Y]$. Hence throughout this interval

$$g(y) = \mu D_g(1+y)^{-1/4}, \qquad g'(y) = -\frac{\mu D_g}{4(1+y)^{5/4}} \leq -2kP_0. \tag{3.8}$$

The equality cases at the endpoints are covered by the smooth matching already proved.

With this g fixed, define the finite constants

$$g_0 = \min_{0\leq y\leq Y} g(y) > 0, \qquad G_0 = \max_{0\leq y\leq Y} g(y), \qquad B = \sup_{y>Y} \frac{g(y)}{y} < \infty.$$

For example,

$$1 \leq B \leq \max\left\{1, \frac{2D_g}{Y(1+Y)^{1/4}}\right\},$$

because $g(y) \leq \max\{y, 2D_g(1+y)^{-1/4}\}$. Finally choose

$$T \geq \max\left\{1, \frac{6}{\varepsilon}, \frac{12kB^2}{b_0}, YG_0^2, \frac{12kY}{\varepsilon g_0}\right\}. \tag{3.9}$$

Thus the order is k, then Y, then D_g and g, and then T; none of the preceding choices depends on a later one.

We now invert the map X in Equation (3.2). By Equation (3.5), $X_f > 0$. As $f \downarrow 0$, the argument y tends to infinity, where $g(y) = y$, and $X(f,t) \to 0$. As $f \to \infty$, $y \to 0$, $g(y) \to g(0) > 0$, and $X(f,t) \to S_0 := \pi/(2k) \leq 1$. Thus for every $t > 0$, $X(\cdot, t)$ maps $(0,\infty)$ bijectively onto $(0, S_0)$ with positive derivative. Its inverse $f = F(x,t)$ is smooth by the implicit function theorem.

On a neighborhood of $f = 0$ the exact identity $g(y) = y$ gives

$$X(f,t) = \frac{1}{k}\arctan(f/t), \qquad F(x,t) = t\tan(kx).$$

Define $F(0,t) = 0$ and extend F oddly to $-S_0 < x < 0$. The displayed formula proves joint smoothness across $x = 0$, including at the endpoint $t = T$ by smooth extension to $t > 0$. The inverse construction also proves the asserted pole limits. For uniformity, if $I \subset [T,\infty)$ is compact and $H > 0$, then $\max_{t\in I} X(H,t) < S_0$. Whenever x exceeds this maximum, monotonicity gives $F(x,t) > H$ for every $t \in I$. The negative pole follows by oddness.

It remains to prove Equation (3.1). For $x > 0$ set $f = F(x,t)$ and use the variables in Equation (3.2). Implicit differentiation of $X(F(x,t),t) = x$ yields

$$F_x = \frac{1}{X_f}, \qquad F_t = -\frac{X_t}{X_f}, \qquad \frac{FF_t}{F_x} = \frac{g'}{k(1+g^2/q)}, \qquad \frac{F_t}{F} = \frac{3yg'}{t(g+2yg')}. \tag{3.10}$$

In particular the sign in the last identity is the sign of g'. Since $A \le g + 2y \le 3g$ and $y \le g$, we have

$$tX_f = \frac{yA}{3k(q+g^2)} \le \frac{1}{k}. \tag{3.11}$$

This proves $F_x \ge kt \ge A_0 t$ without a constant depending on g. Also $|yg'| \le g$: for $g' \ge 0$ this follows from $g' \le 1$ and $y \le g$, whereas for $g' < 0$ the lower derivative bound in Equation (3.5) even gives $|yg'| \le g/4$. Therefore

$$\left|\frac{F_t}{F}\right| \le \frac{6}{t} \le \varepsilon, \qquad \frac{FF_t}{F_x} \le \frac{1}{k} \le \varepsilon.$$

For the second inequality a negative numerator is already favorable; for a nonnegative numerator use $g' \le 1$.

Consider first $y > Y$. Using $A \ge g/2 \ge y/2$ and $g/y \le B$ gives

$$\begin{aligned} t^2 X_f &= \frac{y^2 A}{3k(1+yg^2/t)} \\ &\ge \frac{y^3}{6k(1+B^2y^3/T)} = \frac{1}{6k(y^{-3}+B^2/T)} \ge \frac{1}{6k(Y^{-3}+B^2/T)} \ge \frac{1}{b_0}. \end{aligned} \tag{3.12}$$

The last inequality uses Equations (3.6) and (3.9). This bound remains uniform as $y \to \infty$, and it proves $F_x \le b_0 t^2$ throughout this region.

If $0 < y \le Y$, then $q = t/y \ge T/Y$ and

$$\frac{g^2}{q} \le \frac{YG_0^2}{T} \le 1.$$

It follows from Equations (3.3) and (3.8) that

$$-\frac{FF_t}{F_x} = fX_t = \frac{-g'}{k(1+g^2/q)} \ge P_0. \tag{3.13}$$

The two regions therefore establish the alternative in Equation (3.1). In this second region we also have

$$F^2 X_f = \frac{qA}{3k(1+g^2/q)} \ge \frac{qg_0}{12k} \ge \frac{g_0 T}{12kY} \ge \frac{1}{\varepsilon}, \tag{3.14}$$

so $F_x \le \varepsilon F^2$. Together with the small-F_x bound for $y > Y$ this proves the final inequality of Equation (3.1).

Oddness preserves all five inequalities on $x < 0$, since F_x, F^2, and FF_t are even and $|F_t|, |F|$ are even. At $x = 0$ they follow by continuity; more explicitly, $F = F_t = 0$ and $F_x = kt$, while $T \ge 12kB^2/b_0 \ge k/b_0$ gives the small-F_x branch there. All claims follow. □

4 A compactly supported stretching step

Use $S_1^h(t)$ from (1.1), with the Laplacian convention in Convention 1.5. The next theorem enlarges the distance from a compact set while preserving a positive quadratic term in this quantity. Its error has two parts: a trace decrement, which telescopes under successive metric enlargements, and an arbitrarily small quadratic loss.

Theorem 4.1 (Local stretching)**.** *Fix $n \geq 2$ and $0 < \delta < 1/10$, and put $L_* = 1.004$. There are constants $W, T \geq 1$, depending only on n, δ, with the following property. Let h_0 be a complete smooth metric on a manifold M^n without boundary, let $h \geq h_0$ be smooth, and let $t \geq T$ be smooth. Let $A \subset M$ be nonempty and compact, and let $B \subset M$ be nonempty and closed, with*

$$\operatorname{dist}_{h_0}(A, B) \geq W.$$

For every $0 < d < \delta$ there are a smooth complete metric $h' \geq h$ and a smooth function $t' \geq t$, equal to h, t outside a compact subset of $\{\operatorname{dist}_{h_0}(A, \cdot) < W\}$, such that

$$S_1^{h'}(t') \geq S_1^h(t) + (t'^2 - t^2) - \delta(\operatorname{tr}_h h_0 - \operatorname{tr}_{h'} h_0)t'^2 - dt'^2, \tag{4.1}$$

and

$$\operatorname{dist}_{h'}(A, B) \geq L_* W. \tag{4.2}$$

Neither W nor T depends on d, the geometry of M, or the sets A, B.

The distance gain comes from a graph height close to a cutoff that is large on A and zero outside an intermediate neighborhood. Where the slope is large, the deformed metric is $h + \mathrm{d}u^2$, so the height drop adds length to every curve from A to B. Near zero slope we weaken the graph term by a parameter a, making the residual scalar loss arbitrarily small. The accompanying change in t cancels the tangential second-derivative terms in the scalar comparison. The modified graph equation chooses u using the forcing function from Theorem 3.1: its poles keep u close to the cutoff, while its derivative estimates provide the positive quadratic term in Lemma 4.4.

A positive floor r permits the existence argument on a closed auxiliary manifold (Lemma 4.2). Sending r to zero makes the height and all its derivatives tend to zero on a fixed outer annulus (Lemma 4.3). We can therefore cut off the metric and changes in t there at the remaining scalar cost. Section 4.6 performs this cutoff and verifies the distance gain by estimating the height variation of crossing curves. All geometric estimates take place on the compact auxiliary manifold; only the forcing function and scale constants must be independent of that geometry.

4.1 The slope switch and the order of constants

We first specify the deformation associated with a provisional smooth height u; the height itself will be chosen below. Write $s = |\mathrm{d}u|_h$. Fix $s_0 = 10^{-2}$ and a smooth nondecreasing function $\rho : [0, \infty) \to [0, 1]$ such that

$$\rho = 0 \text{ on } [0, s_0], \qquad \rho = 1 \text{ on } [5s_0, \infty), \qquad 0 \leq \rho' \leq 60.$$

For $0 < a \leq 1/2$ and $s \geq 0$ define

$$\mu = a^2 + (1 - a^2)\rho(s), \quad \theta = \frac{\mu s^2}{1 + s^2}, \quad l = (1 - \theta)^{1/2}, \quad \beta = \log l. \tag{4.3}$$

For $s > 0$ also put

$$\chi = \left(1 + \frac{a^2 s_0^2}{\theta}\right)^{1/2}, \qquad m = \frac{\mathrm{d}\log\chi}{\mathrm{d}\log\sqrt{\theta}} = -\frac{a^2 s_0^2}{\theta + a^2 s_0^2}. \tag{4.4}$$

Here θ is strictly increasing. The factor χ expresses the regularized forcing function multiplier as $J = \sqrt{\theta + a^2 s_0^2} = \sqrt{\theta}\,\chi$; after division by $\sqrt{\theta}$, it multiplies the forcing function in (4.15). m enters

the scalar comparison when this multiplier is differentiated. These two uses, and the tangential terms of that comparison, which involve $s\beta_s$, need the following bounds:

$$\begin{gathered} 0 \le -s\beta_s \le 2, \qquad -1 \le m \le 0, \qquad 1 \le \chi \le 2 \quad (s \ge s_0), \\ m \le -\tfrac{1}{2}, \quad \chi \ge s_0/s, \quad \theta\chi^2 \le 2a^2 s_0^2 \quad (0 < s \le s_0). \end{gathered} \tag{4.5}$$

Indeed, on $s \le 5s_0$,

$$-s\beta_s = \frac{s\theta_s}{2(1-\theta)} \le 30s^3 + \frac{s^2}{1+s^2} < 2,$$

and on $s \ge 5s_0$ it equals $s^2/(1+s^2)$. The remaining bounds follow directly from $\theta \ge a^2 s^2/(1+s^2)$ and the formula with $\mu = a^2$ on $s \le s_0$.

For $t \ge 1$ we also have

$$\frac{(t-\beta)^2 - t^2}{\theta} \le 4(1+s)t \qquad (s > 0). \tag{4.6}$$

To check this, write $b_* = -\beta$. If $\theta \le 1/2$, then $b_* \le \theta$, and the left side is at most $3t$. Otherwise use $b_* \le \frac{1}{2}\log(1+s^2) \le \log(1+s) \le \sqrt{s}$ to bound it by $4t\sqrt{s} + 2s \le 4(1+s)t$.

For the provisional height u, define the raw metric and function by

$$h_u = h + \frac{\theta}{(1-\theta)s^2}\,\mathrm{d}u^2 = h + \frac{\mu}{1+(1-\mu)s^2}\,\mathrm{d}u^2, \qquad t_u = t - \beta. \tag{4.7}$$

Where $s > 0$, choose level coordinates so that $h = v^2\mathrm{d}u^2 + \gamma_u$, with $v = 1/s$. Then $h_u = (v/l)^2\mathrm{d}u^2 + \gamma_u$: the deformation multiplies normal lengths by l^{-1} and leaves the level metric unchanged. This is the form used in the scalar comparison of Lemma 4.4. The expressions in (4.7) are smooth also at $s = 0$, and $h_u \ge h$, $t_u \ge t$. To see explicitly the smoothness needed at zero slope, note that near $s = 0$ the coefficient of $\mathrm{d}u^2$ is $a^2/[1+(1-a^2)s^2]$, while the increment of t is $-\frac{1}{2}\log([1+(1-a^2)s^2]/[1+s^2])$. Both are smooth functions of the first jet of u; therefore C^3 convergence of u to zero gives C^2 convergence of these increments of the metric and of t to zero. For $s \ge 5s_0$, $h_u = h + \mathrm{d}u^2$. At points of nonzero slope in the original manifold, let $e = \mathrm{grad}_h\, u/s$. With the given background metric h_0, the rank-one inverse formula gives

$$\mathrm{tr}_h\, h_0 - \mathrm{tr}_{h_u}\, h_0 = \theta b^2, \qquad b^2 = h_0(e,e) \le 1. \tag{4.8}$$

We now fix all constants that are required to be independent of the local geometry and of d. Let

$$A_0 = 1000, \qquad P_0 = 20, \qquad C_1 = s_0^{-1}, \qquad b_0 = \frac{\delta s_0^2}{1000}, \qquad 0 < \varepsilon \le \min\left\{\frac{s_0}{100}, \frac{1}{8n}, \frac{1}{8C_1}\right\}. \tag{4.9}$$

Apply Theorem 3.1 with these constants, obtaining F, S_0, T. In particular $S_0 \le 1$ and $T \ge 1$. Put $C_2 = 2C_1 b_0$. Only after a loss d has been specified do we choose a so that

$$0 < a \le \tfrac{1}{2}, \qquad (2C_2 + 3)a^2 \le d/2. \tag{4.10}$$

The width W will be a sufficiently large numerical constant; it may, and will, be at least 10^5. It is chosen to satisfy the explicit cutoff and distance inequalities below. Thus the choices of W, T precede d, a and all local geometric choices.

4.2 Cutoffs and a closed auxiliary manifold

Write $r_A(x) = \mathrm{dist}_{h_0}(A, x)$. There exist smooth functions $\psi \geq 0$ and $0 \leq \eta_1, \zeta \leq 1$ such that

$$\begin{gathered} \inf_A \psi > W/5, \qquad |\mathrm{d}\psi|_{h_0} \leq 1, \qquad \operatorname{supp}\psi \subset \{r_A < W/2\}, \\ \eta_1 = 1 \text{ on } \{r_A \leq .55W\}, \qquad \eta_1 = 0 \text{ on } \{r_A \geq .8W\}, \qquad |\mathrm{d}\eta_1|_{h_0} \leq s_0/8, \\ \zeta = 1 \text{ on } \{r_A \leq .82W\}, \qquad \zeta = 0 \text{ on } \{r_A \geq .98W\}. \end{gathered} \tag{4.11}$$

Choose also a fixed compact annulus $\mathcal{A}_\zeta$ in $\{.8W < r_A < W\}$ containing $\overline{\{0 < \zeta < 1\}}$. All cutoff error estimates below are imposed on this whole annulus, including any intermediate plateau of ζ.

For clarity, a smooth approximation to a Lipschitz function on the compact neighborhood in question can be made in finitely many coordinate charts. Shrink each chart until its metric distortion is as small as desired, convolve there, and combine with a smooth partition of unity. The gradient of the combination is the convex combination of the local gradients plus terms bounded by the chosen uniform approximation error times the derivatives of the partition. Choose the local errors small enough to make the latter sum arbitrarily small. One may preserve prescribed constant regions by using charts lying inside them and leaving a buffer between each such region and the transition. Apply this construction to $(W/3 - r_A)_+$, with value error less than $W/100$ and gradient bound 1.05, then divide by 1.05 and use the zero-region buffer to obtain ψ. Smooth cutoff functions of r_A/W give the other two functions; for η_1 the gradient can be bounded by $12/W < s_0/8$. This proves Equation (4.11) after increasing W once.

For $0 < r \leq 1/2$ define the positive-floor cutoff

$$\eta = r + (1 - r)\eta_1^2. \tag{4.12}$$

Then $\eta = 1$ near $\operatorname{supp}\psi$, $\eta = r$ on $\{r_A \geq .8W\}$, and, uniformly in r,

$$0 < \eta \leq 1, \qquad |\mathrm{d}\eta|_h \leq s_0/4, \qquad \frac{|\mathrm{d}\eta|_h^2}{\eta} \leq 1. \tag{4.13}$$

The same statements hold with h_0 in place of h; this follows from $h \geq h_0$ and $|\mathrm{d}\eta|^2/\eta \leq 4|\mathrm{d}\eta_1|^2$.

Completeness of h_0 and compactness of A make the closed neighborhoods used above compact. Enclose a slightly larger neighborhood, say $\{r_A \leq 1.1W\}$, in a smooth relatively compact domain. Outside this neighborhood, modify h and t near the boundary of the domain to be products, and double the domain. Convex interpolation for t preserves $t \geq T$, and interpolation of positive definite tensors gives the metric extension. We thereby obtain a closed auxiliary manifold N on which $h, t, \psi, \eta_1, \zeta$ are smooth, with the last three functions extended by zero where appropriate. The extensions of h, t are fixed independently of r. If the relevant component of M is already compact, it can simply be used as N. The construction can be made on finitely many components if A meets more than one. From now until the final cutoff, we work on N. Constants allowed to depend on this compact geometry and on a will be denoted by C; they will never enter W, T.

4.3 The modified graph equation

For a covector p with $s = |p|_h$, set

$$\mathcal{A}(p) = \frac{\sqrt{\mu}}{\sqrt{1 + s^2}} p^\sharp, \qquad J(s) = \sqrt{\theta + a^2 s_0^2}.$$

The symbols $\mu, \theta, l, \beta, \chi, m$ will always be evaluated at $s = |\mathrm{d}u|_h$ when a function u is under discussion. The maps $\mathcal{A}, J$ are smooth at $p = 0$: there $\mu = a^2$ is constant. The tangential and radial

eigenvalues of $\mathcal{A}_p$ are $\sqrt{\theta}/s$ and $(\sqrt{\theta})_s$, respectively; both are positive, and both have limit a at the origin.

For $\tau \in [0,1]$ consider

$$\operatorname{div}_h \mathcal{A}(\mathrm{d}u) - \mathrm{d}t(\mathcal{A}(\mathrm{d}u)) = J(|\mathrm{d}u|_h) F\left(\frac{u-\tau\psi}{\eta}, t\right). \tag{4.14}$$

The choice of operator is dictated by the scalar comparison. Where $s = |\mathrm{d}u|_h > 0$, put $e = \operatorname{grad}_h u/s$ and $H = \operatorname{div}_h e$, the mean curvature of the level sets with normal e. Since $\mathcal{A}(\mathrm{d}u) = \sqrt{\theta}\, e$ and $J(|\mathrm{d}u|_h) = \sqrt{\theta}\,\chi$ by (4.4), dividing (4.14) by $\sqrt{\theta}$ gives

$$H - et + e \log\sqrt{\theta} = \chi F\left(\frac{u-\tau\psi}{\eta}, t\right). \tag{4.15}$$

The weighted mean curvature $S = H - et$ is the combination entering the normal terms of $S_1^h(t)$. In the form $S + e\log\sqrt{\theta} = \chi F$, this identity lets the derivative bounds control their change under the graph deformation in the scalar comparison (4.24)–(4.25) in the proof of Lemma 4.4. The positive regularization in J keeps the original equation smooth with $J > 0$ also at $\mathrm{d}u = 0$; the divided identity is used only where $s > 0$.

The forcing function is evaluated only where its first argument lies in $(-S_0, S_0)$. The parameter τ connects the solution $u = 0$ at $\tau = 0$ to the required equation at $\tau = 1$. We first prove existence with bounds independent of r.

Lemma 4.2 (Height and gradient bounds)**.** *Fix the auxiliary manifold N, the extended data, and a as in* (4.10)*. For every floor $0 < r \le 1/2$, equation* (4.14) *at $\tau = 1$ admits a smooth solution on N. There are $S_1 < S_0$ and $G < \infty$, independent of r and of $\tau \in [0,1]$, such that every smooth solution u of* (4.14) *for $0 \le \tau \le 1$ satisfies*

$$|u - \tau\psi| \le S_1\eta, \qquad |\mathrm{d}u|_h \le G. \tag{4.16}$$

The constants S_1, G may depend on a and on the compact data.

Proof. For $S < S_0$ the functions $u_\pm = \tau\psi \pm S\eta$ have uniformly bounded first and second derivatives, independently of r, τ, S. Consequently the left side of Equation (4.14) at these functions is bounded. Since $J \ge as_0 > 0$ and $t(N)$ is compact, the uniform pole limits of F permit one choice $S_1 < S_0$ for which u_+ has strictly negative residual (left side minus right side) and u_- has strictly positive residual. Comparison at a maximum or minimum, using strict monotonicity in u, gives the height estimate.

One can make this argument before a solution has been constructed by extending F smoothly to all first arguments with $F_x > 0$, retaining its original values on a slightly larger interval than $[-S_1, S_1]$. For example, extend the positive even function F_x by a smooth cutoff to a positive constant and integrate from zero. Solve this globally defined equation. The same barriers force every solution into $[-S_1, S_1]$, so that the extension is never used in the resulting solution.

We give the gradient estimate, including the dependence of the forcing function on the graph angle. If $\max_N |\mathrm{d}u|_h \le 1$ there is nothing to prove. Otherwise, near a point of maximum slope, $\mu = 1$ and the graph in $(N,h)\times\mathbb{R}$ is an ordinary graph. Write

$$w = (1 + |\mathrm{d}u|_h^2)^{-1/2}, \qquad \nu = (N_u, -w), \qquad N_u = w \operatorname{grad}_h u,$$

for its downward unit normal. Denote its second fundamental form and mean curvature by II and H, with $H = \operatorname{div}\nu$. Extend t constantly in the vertical direction and put $X = -\operatorname{grad} t$ and $K = -\partial_z$.

The equation is

$$H + \langle X, \nu \rangle = p(x, z, w), \qquad p(x, z, \omega) = \sqrt{1 - \omega^2 + a^2 s_0^2}\, F\left(\frac{z - \tau\psi(x)}{\eta(x)}, t(x) \right).$$

Let $K^\top = K - w\nu$. Under a normal variation with speed v and parameter σ, with dots denoting $\mathrm{d}/\mathrm{d}\sigma$ at $\sigma = 0$,

$$\begin{aligned} \dot{H} &= -\Delta\, v - (|\mathrm{II}|^2 + \mathrm{Ric}(\nu, \nu))v, \\ \frac{\mathrm{d}}{\mathrm{d}\sigma}\langle X, \nu \rangle &= -\operatorname{Hess} t(\nu, \nu)v - \langle X^\top, \operatorname{grad} v \rangle, \\ \dot{w} &= -\langle K^\top, \operatorname{grad} v \rangle. \end{aligned}$$

Thus the linearization of the equation, with all terms moved to the left, is $-\mathcal{L}v$, where

$$\mathcal{L} = \Delta + \langle X^\top - p_\omega K^\top, \operatorname{grad} \cdot \rangle + |\mathrm{II}|^2 + \mathrm{Ric}(\nu, \nu) + \operatorname{Hess} t(\nu, \nu) + \partial_\nu^{\text{fixed } \omega} p. \tag{4.17}$$

The last derivative differentiates the base and height variables while holding the angle fixed. Translation in direction K leaves the angle, H, and $\langle X, \nu \rangle$ unchanged and differentiates the residual by p_z. Its normal speed is w; its tangential component differentiates an identically zero residual. We obtain the exact identity

$$\mathcal{L}w = -p_z. \tag{4.18}$$

Put $\xi = (u - \tau\psi)/\eta$. On the graph,

$$p_z = \frac{JF_x}{\eta}, \qquad \partial_\nu^{\text{fixed } \omega} p = J\left\{ \frac{F_x}{\eta}\left[-w - \tau \mathrm{d}\psi(N_u) - \xi \mathrm{d}\eta(N_u) \right] + F_t \mathrm{d}t(N_u) \right\}.$$

The height bound puts F, F_x, F_t in a fixed compact range. Moreover $|N_u| \le 1$, $|\mathrm{d}\psi|_h \le 1$, and $|\mathrm{d}\eta|_h \le s_0/4$. The zero-order coefficient in Equation (4.17), after omitting the nonnegative term $|\mathrm{II}|^2$, is therefore at least $-C/\eta$, with C independent of r, τ. At the minimum of w under consideration, the slope is at least one, so $J \ge 1/\sqrt{2}$ and $p_z \ge A_0 T/(\sqrt{2}\eta)$. Since $\operatorname{grad} w = 0$ and $\Delta\, w \ge 0$, Equation (4.18) implies

$$\frac{A_0 T}{\sqrt{2}\eta} \le \frac{C}{\eta} w.$$

This gives a positive lower bound for w, and hence the asserted G.

We finish by continuation at a fixed floor $r > 0$. Every classical $C^{2,\alpha}$ solution is smooth. Indeed, write the equation locally as $a^{ij}(x)\partial_{ij}u = f(x)$, with $a^{ij}, f \in C^{1,\alpha}$. Its coordinate difference quotients $u_h = (u(\cdot + h) - u)/|h|$, and likewise f_h and $(a^{ij})_h$, satisfy

$$a^{ij}(x)\partial_{ij}u_h = f_h - (a^{ij})_h \partial_{ij}u(x + h).$$

The right side has a uniform $C^{0,\alpha}$ bound as $h \to 0$, and $\sup |u_h| \le \sup |Du|$ on a smaller ball. The interior Schauder estimate [37, Lecture 12, Theorem 2] therefore bounds u_h in $C^{2,\alpha}$ on a further smaller ball. Passing to the derivative limit, including its Hölder bound, gives $u \in C^{3,\alpha}$; iteration gives smoothness. These qualitative estimates may depend on the existing $C^{2,\alpha}$ norm; the following a priori estimates do not.

For the a priori bounds, the height and gradient estimates place $(x, u, \mathrm{d}u)$ in a compact range on which the principal coefficients are uniformly elliptic. In coordinates write

$$\partial_i \mathfrak{A}^i(x, Du) = \mathfrak{B}(x, u, Du).$$

Differentiating with $v = \partial_k u$ gives

$$\partial_i(\mathfrak{A}^i_{p_j}\partial_j v) - \mathfrak{B}_{p_j}\partial_j v = \mathfrak{B}_u v + \mathfrak{B}_{x_k} - \partial_i \mathfrak{A}^i_{x_k}. \tag{4.19}$$

All explicit coefficient derivatives, the terms $\mathfrak{B}_u v + \mathfrak{B}_{x_k}$, and the vector field $\mathfrak{A}^i_{x_k}$ inside the last divergence are bounded on this compact range, without a second-derivative bound for u. The divergence-form Hölder estimate, with these bounded lower-order coefficients and divergence data, gives a uniform $C^{1,\alpha}$ bound for u on a finite cover. The nondivergence-form equation then has Hölder coefficients and right side, so Schauder estimates and differentiation give bounds of every order. These are Simon [37, Lecture 18, Theorem 3, and Lecture 12, Theorem 2]. The constants may depend on the fixed positive floor r.

Choose α smaller if necessary than the exponent just obtained. The linearization in u is uniformly elliptic and has strictly negative zero-order coefficient $-JF_x/\eta$; angle and gradient derivatives affect only its principal and first-order terms. Interpolate its coefficients with those of $\Delta_h - 1$, obtaining L_q, $0 \le q \le 1$. Uniform ellipticity and a strictly negative zero-order bound persist. The maximum principle and the finite-cover Schauder estimate [37, Lecture 12, Theorem 2] give

$$\|v\|_{C^{2,\alpha}(N)} \le C\|L_q v\|_{C^{0,\alpha}(N)} \qquad (0 \le q \le 1).$$

For smooth f, minimizing $\frac{1}{2}\int_N(|\mathrm{d}v|_h^2 + v^2) + \int_N fv$ over $H^1(N)$, with Riemannian volume understood, gives the unique weak solution of $(\Delta_h - 1)v = f$. It is smooth by Simon [37, Lecture 6, Theorem 1 and Corollary 1]. For $f \in C^{0,\alpha}$, approximate f uniformly by smooth functions with bounded $C^{0,\alpha}$ norms. The displayed estimate gives a subsequence of solutions converging in $C^{2,\alpha'}$, $\alpha' < \alpha$, and the α-Hölder bounds pass to the limit. Thus $\Delta_h - 1$ is an isomorphism $C^{2,\alpha}(N) \to C^{0,\alpha}(N)$. Along L_q, invertibility is open by a convergent power series for the inverse and closed because the same estimate and coefficient convergence make solutions for each fixed right side Cauchy in $C^{2,\alpha}$. Hence the linearization is an isomorphism.

At $\tau = 0$ the nonlinear equation has solution $u = 0$. The implicit function theorem gives openness in τ, and the a priori bounds of all orders give compactness in $C^{2,\alpha}$ and hence closedness. The height bound keeps the forcing function argument in its fixed compact interval. The soluble set is therefore all of $[0, 1]$, and the qualitative bootstrap makes its solutions smooth. □

4.4 Decay on the region of constant floor

Lemma 4.3 (Locality)**.** *Let u_r be the solutions from Lemma 4.2 at $\tau = 1$. For every compact subset of the fixed open constant-floor region $\psi = \eta_1 = 0$ and every integer $k \ge 0$, $u_r \to 0$ in C^k as $r \downarrow 0$.*

Proof. Here $\eta = r$ and $|u_r/r| \le S_1$. Expanding the left side of Equation (4.14) and using $F(0, t) = 0$ writes its equation as

$$a^{ij}(x)\partial_{ij}u_r + b^i(x)\partial_i u_r - \frac{c_r(x)}{r}u_r = 0, \qquad c_r = J\int_0^1 F_x(qu_r/r, t)\,\mathrm{d}q \ge as_0A_0T > 0. \tag{4.20}$$

The coefficients a^{ij} are uniformly elliptic and bounded, and b^i are bounded, independently of r. Indeed they are smooth functions of (x, Du_r) on the fixed compact jet range from Equation (4.16); the explicit x part of the flux vanishes at zero covector and is therefore a bounded first-order term. Also c_r is bounded above independently of r.

Take a coordinate box compactly contained in the constant-floor region. Let ℓ_f be its inward affine distance from face f in these coordinates and set

$$V_r = M_0 \sum_f \exp(-\gamma\ell_f/\sqrt{r}),$$

where M_0 dominates $|u_r|$ on the boundary for every r. Let Λ bound $a^{ij}(\partial_i \ell_f)(\partial_j \ell_f)$, let $c_* > 0$ be the lower bound for c_r, and let B_* bound the first-order contributions. The operator in Equation (4.20) applied to each summand of V_r is at most that summand times

$$\frac{\Lambda\gamma^2 - c_*}{r} + \frac{B_*\gamma}{\sqrt{r}}.$$

Choose $\gamma > 0$ with $\Lambda\gamma^2 < c_*/2$. For small r this quantity is negative. Comparison with both u_r and $-u_r$ proves, on each smaller box, $|u_r| \le C\exp(-c/\sqrt{r})$. A finite covering gives this estimate on any fixed compact subset.

To upgrade the estimate to derivatives, rescale a coordinate ball of radius comparable to r by

$$x = x_0 + ry, \qquad u_r(x) = rv(y).$$

The equation becomes

$$\partial_{y_i}\mathfrak{A}^i(x_0 + ry, Dv) = r\mathfrak{B}_0(x_0 + ry, v, Dv), \tag{4.21}$$

where $\mathfrak{B}_0$ includes the term involving t and $JF(v,t)$ and has no negative power of r. We have $|v| \le S_1$ and $|Dv| \le G$. The principal ellipticity is uniform. Explicit y derivatives of $\mathfrak{A}$ have a factor r, explicit y derivatives of $r\mathfrak{B}_0$ have a factor r^2, and its v, p derivatives have a factor r. Thus all explicit coefficient derivatives are bounded uniformly on this fixed jet range. Apply the Hölder estimate to the differentiated Equation (4.19), followed by the Schauder bootstrap on fixed balls in y. For every k this gives

$$|D_y^k v| \le C_k, \qquad |D_x^k u_r| \le C_k r^{1-k},$$

on smaller balls, with constants uniform in x_0 in a fixed compact set and in small r. To combine this polynomial bound with the exponentially small height, fix nested compact subsets $K \Subset K'$ of the constant-floor region. Taylor's formula on coordinate balls, applied to differences with step ℓ, gives, for $k \ge 1$ and small ℓ,

$$\|D^k u_r\|_{C^0(K)} \le C_k\big(\ell^{-k}\|u_r\|_{C^0(K')} + \ell\|D^{k+1}u_r\|_{C^0(K')}\big).$$

The two norms on the right are bounded by $A_r = Ce^{-c/\sqrt{r}}$ and $B_r = C_k r^{-k}$, respectively. Taking $\ell = (A_r/B_r)^{1/(k+1)}$, which fits inside the fixed coordinate balls for small r, bounds the left side by $C_k A_r^{1/(k+1)} B_r^{k/(k+1)}$. Hence $u_r \to 0$ in every fixed C^k norm. For $k = 3$ this is bounded by a constant times $r^{-9/4}\exp(-c/(4\sqrt{r})) + \exp(-c/\sqrt{r})$, which tends to zero. This last case ensures C^2 convergence of all increments of the metric and of t in Equation (4.7), which are smooth functions of Du_r. □

4.5 The scalar identity and its two slope regimes

All comparisons involving h_0 in this subsection are made on the original neighborhood $\{r_A < 1.1W\}$, where the auxiliary data agree with the given data.

Apply the construction in Equation (4.7) to the solution u_r from Lemma 4.2 at $\tau = 1$, and write $u = u_r$. We now prove its scalar estimate before applying the final spatial cutoff.

Lemma 4.4 (Raw scalar estimate)**.** *For the solution $u = u_r$ and the raw metric and function (h_u, t_u) in* (4.7)*, the following inequality holds on the original neighborhood $\{r_A < 1.1W\}$:*

$$S_1^{h_u}(t_u) \ge S_1^h(t) + (t_u^2 - t^2) - \delta(\mathrm{tr}_h\, h_0 - \mathrm{tr}_{h_u}\, h_0)t_u^2 - \frac{d}{2}t^2. \tag{4.22}$$

Proof. We derive the scalar comparison explicitly. Near a point with $\mathrm{d}u \neq 0$ use $q = u$ as the transverse coordinate and write

$$h = v^2 \mathrm{d}q^2 + \gamma_q, \qquad v = 1/s, \qquad e = v^{-1}\partial_q, \qquad \mathrm{J} = \frac{1}{2v}\partial_q \gamma_q, \qquad H = \operatorname{tr} \mathrm{J}.$$

Here J is the second fundamental form of the level sets, and all tangential norms and derivatives below use γ_q. Put $S = H - et$. The Gauss equation and normal variation of H give

$$\mathrm{Scal}_h = \mathrm{Scal}_\gamma - |\mathrm{J}|^2 - H^2 - 2eH - 2\frac{\Delta_\gamma v}{v}, \qquad \Delta_h t = \Delta_\gamma t + e(et) + Het + \langle \mathrm{d}_\gamma \log v, \mathrm{d}_\gamma t \rangle.$$

Consequently

$$\begin{aligned} S_1^h(t) = -|\mathrm{J}|^2 - S^2 - 2eS + \mathrm{Scal}_\gamma + 2\,\Delta_\gamma t - |\mathrm{d}_\gamma t|^2 \\ - 2\frac{\Delta_\gamma v}{v} + 2\langle \mathrm{d}_\gamma \log v, \mathrm{d}_\gamma t \rangle. \end{aligned} \tag{4.23}$$

The deformation replaces v by v/l and t by $t - \beta$. The change in the tangential terms on the right is

$$2\langle \mathrm{d}_\gamma \log v, \mathrm{d}_\gamma \beta \rangle - |\mathrm{d}_\gamma \beta|^2 = c_\beta(2 - c_\beta)|\mathrm{d}_\gamma \log v|^2 \geq 0, \qquad c_\beta = -s\beta_s \in [0, 2].$$

This computation follows by expanding $\Delta_\gamma(v/l)/(v/l)$ and $|\mathrm{d}_\gamma(t - \beta)|^2$; the $\Delta_\gamma \beta$ and the terms involving $\mathrm{d}_\gamma t$ cancel.

Put

$$\xi = (u - \psi)/\eta, \qquad M = \chi F(\xi, t), \qquad K = \frac{l^2}{\theta} e\beta.$$

Equation (4.15) at $\tau = 1$ gives $S = M - e \log \sqrt{\theta}$. Since $e \log \sqrt{\theta} = -K$, we have $S = M + K$. For the new metric, $e' = le$, $\mathrm{J}' = l\mathrm{J}$, and

$$S' = l(S + e\beta) = lM + K/l.$$

Substitution in the normal terms of Equation (4.23) gives

$$S_1^{h_u}(t_u) - S_1^h(t) \geq \theta E, \qquad E = |\mathrm{J}|^2 + M^2 + K^2/l^2 - 2MK + 2eM. \tag{4.24}$$

For completeness, the derivative terms before simplification are $-2le(lM + K/l) + 2e(M + K)$. Their eK terms cancel; using $e\beta = \theta K/l^2$ leaves $2\theta eM - 2\theta MK + 2\theta K^2/l^2$, which combines with the squared terms to yield Equation (4.24).

Since $e\chi = -m\chi K$ and $et = H - M - K$,

$$eM = -mKM + \chi(F_t et + F_x e\xi).$$

Completing the H and K squares gives $E \geq \mathcal{B}$, where

$$\mathcal{B} = \chi^2 \left\{ [1 - l^2(1 + m)^2]F^2 - 2[1 + l^2(1 + m)]FF_t - nF_t^2 \right\} + 2\chi F_x e\xi. \tag{4.25}$$

An exact verification, useful for tracking every sign, is

$$\begin{aligned} E - \mathcal{B} = \left(|\mathrm{J}|^2 - \frac{H^2}{n-1} \right) + \frac{(H + (n-1)\chi F_t)^2}{n-1} \\ + \left(\frac{K}{l} - l\chi[(1 + m)F + F_t] \right)^2 + \theta\chi^2 F_t^2 \geq 0. \end{aligned}$$

All forcing function expressions here are evaluated at (ξ, t).

Suppose first that $s \geq s_0$. Denote the expression in braces in Equation (4.25) by $\mathcal{C}$. Its F^2 coefficient is nonnegative, and $1 \leq 1 + l^2(1+m) \leq 2$. From Equation (3.1) we get

$$\mathcal{C} \geq -5\varepsilon F_x, \qquad -FF_t \geq P_0 F_x \implies \mathcal{C} \geq P_0 F_x. \tag{4.26}$$

Indeed, if $FF_t \geq 0$, then $F_t^2 \leq \varepsilon F F_t$ and $\mathcal{C} \geq -(4 + n\varepsilon)FF_t \geq -5\varepsilon F_x$. If $FF_t < 0$, the same absolute-value estimate gives $\mathcal{C} \geq (2 - n\varepsilon)|FF_t| \geq |FF_t|$.

Where $\eta \neq 1$ we have $\psi = 0$, and

$$e\xi = \frac{s - \xi e\eta}{\eta} \geq \frac{3s}{4\eta} \geq \frac{3s}{4}.$$

Using $1 \leq \chi \leq 2$, Equations (4.9) and (4.26) give $E \geq sF_x$ there. Where $\eta = 1$, we have $e\xi = s - e\psi \geq s - b$, with $b^2 = h_0(e, e)$ as in Equation (4.8), because $|\mathrm{d}\psi|_{h_0} \leq 1$. If $s > 8b$, the same estimates give

$$E \geq (7s/4 - 20\varepsilon)F_x \geq sF_x.$$

It remains to consider $s_0 \leq s \leq 8b$. The possible deficit from sF_x is at most $10F_x$: indeed $2\chi(s - b) - s \geq -4$ and $20\varepsilon < 1$. On the favorable forcing function branch, however,

$$E - sF_x \geq [20\chi^2 + 2\chi(s - b) - s]F_x \geq 0,$$

since $s \leq 8b \leq 8$. Thus a deficit is possible only when $F_x \leq b_0 t^2$. In that case $b \geq s_0/8$ and our choice of b_0 gives $10b_0 \leq \delta(s_0/8)^2$. We conclude throughout $s \geq s_0$ that

$$E \geq sF_x - \delta b^2 t^2 \geq 4(1 + s)t - \delta b^2 t^2. \tag{4.27}$$

The last inequality follows from $F_x \geq 1000t$ and $s \geq .01$. Since $t_u \geq t$, Equations (4.6), (4.8) and (4.24) now give

$$S_1^{h_u}(t_u) - S_1^h(t) \geq 4\theta(1 + s)t - \delta\theta b^2 t^2 \geq t_u^2 - t^2 - \delta(\mathrm{tr}_h\, h_0 - \mathrm{tr}_{h_u}\, h_0)t_u^2.$$

This proves the raw inequality in this region, even without its d term.

Now suppose $0 < s < s_0$. Since $m \leq -1/2$, the first coefficient in $\mathcal{C}$ is at least $3/4$. Using $|F_t| \leq \varepsilon|F|$ gives

$$\mathcal{C} \geq (3/4 - 4\varepsilon - n\varepsilon^2)F^2 \geq \tfrac{1}{2}F^2.$$

Where $\eta = 1$, $e\xi/\chi \geq -1$. Elsewhere, a negative $e\xi$ can occur only if $s \leq |\mathrm{d}\eta|_h$, and then

$$\frac{|e\xi|}{\chi} \leq \frac{|\mathrm{d}\eta|_h s}{s_0\eta} \leq \frac{|\mathrm{d}\eta|_h^2}{s_0\eta} \leq C_1.$$

By the quadratic forcing function upper bound,

$$E \geq \chi^2\{(1/2 - 2C_1\varepsilon)F^2 - 2C_1 b_0 t^2\} \geq -C_2\chi^2 t^2. \tag{4.28}$$

Since $\theta\chi^2 \leq 2a^2 s_0^2$, the loss in Equation (4.24) is at most $2C_2 a^2 t^2$. Also $-\beta \leq \theta \leq a^2 s_0^2$ and $t \geq 1$, so $t_u^2 - t^2 \leq 3a^2 t^2$. The choice in Equation (4.10) pays both terms with $(d/2)t^2$. The trace penalty is favorable and may be omitted in this regime. This proves Equation (4.22) wherever $s > 0$. At a zero of $\mathrm{d}u$ it follows by continuity unless $\mathrm{d}u$ vanishes on a neighborhood; on such a neighborhood the raw output equals the input and the inequality is immediate. □

4.6 Cutting off the deformation and obtaining distance

Define on the original manifold

$$h' = h + \zeta(h_u - h), \qquad t' = t + \zeta(t_u - t), \tag{4.29}$$

and extend by (h, t) outside the auxiliary neighborhood. The result is smooth, $h' \geq h$, $t' \geq t$, and its changes have compact support in $\{r_A < W\}$. We now choose r small enough to verify both required estimates.

First, the scalar inequality must use the trace of the actual cutoff metric. On the fixed compact annulus $\mathcal{A}_\zeta$, define the residual for any positive definite metric H and function z by

$$\mathfrak{E}(H, z) = S_1^H(z) - S_1^h(t) - (z^2 - t^2) + \delta(\mathrm{tr}_h\, h_0 - \mathrm{tr}_H\, h_0)z^2 + dz^2.$$

This is a continuous function of the C^2 jets of (H, z) on a compact set of positive definite metrics. By Equation (4.22),

$$\mathfrak{E}(h_u, t_u) \geq dt_u^2 - (d/2)t^2 \geq (d/2)T^2.$$

The annulus lies compactly in the fixed region $\psi = 0$, $\eta_1 = 0$. By Lemma 4.3, $u_r \to 0$ there in C^3; hence both the raw increments and their products with ζ tend to zero in C^2. Choose r so small that

$$|\mathfrak{E}(h', t') - \mathfrak{E}(h_u, t_u)| < dT^2/4$$

on this annulus. Then $\mathfrak{E}(h', t') > 0$ there. Off the annulus the output is either raw or unchanged, and the same desired inequality holds directly. This proves Equation (4.1).

It remains to prove the distance gain, including for curves that revisit the cutoff annulus or leave the auxiliary neighborhood. Introduce the globally defined effective height

$$v_{\mathrm{eff}} = \zeta u,$$

extended by zero outside the neighborhood. On A, where $\zeta = 1$, the height bound (4.16) with $S_1 < S_0 \leq 1$ and Equation (4.11) give $v_{\mathrm{eff}} > W/5 - 1$, whereas on B, where $\zeta = 0$, it is zero. Decrease r further, if necessary, so that

$$|\mathrm{d}v_{\mathrm{eff}}|_h \leq .01 \tag{4.30}$$

throughout $\mathcal{A}_\zeta$; locality justifies this choice. Consider a piecewise smooth curve from A to B of h' length at most L_*W. Its h length is at most L_*W. Partition its parameter interval according to $0 < \zeta < 1$, $\zeta = 1$, and $\zeta = 0$. At points where ζ equals either endpoint, $\mathrm{d}\zeta = 0$ because ζ is smooth and valued in $[0, 1]$. The total variation of v_{eff} on $\{0 < \zeta < 1\}$ is at most $.01L_*W$, counting all repeated visits. On the region where $\zeta = 1$ and $s \leq 5s_0 = .05$, its variation is at most $.05L_*W$. Where $\zeta = 0$ it has zero variation. Therefore the variation on

$$\Omega_{\mathrm{graph}} = \{\zeta = 1,\ s \geq .05\}$$

is greater than

$$(1/5 - .06L_*)W - 1 > .11W.$$

The strict last inequality holds for our fixed sufficiently large W. On this region $h' = h + \mathrm{d}u^2$, while everywhere $h' \geq h$. If γ is parametrized on an interval I, the integral triangle inequality in $\mathbb{R}^2$ yields

$$\begin{aligned}
\mathrm{Length}_{h'}(\gamma) &\geq \int_I \sqrt{|\dot\gamma|_h^2 + \mathbf{1}_{\Omega_{\mathrm{graph}}}|(u\circ\gamma)'|^2}\,\mathrm{d}q \\
&\geq \left\{\mathrm{Length}_h(\gamma)^2 + \left(\int_{\gamma^{-1}(\Omega_{\mathrm{graph}})} |(u\circ\gamma)'|\,\mathrm{d}q\right)^2\right\}^{1/2} \\
&> W\sqrt{1 + .11^2} > L_*W.
\end{aligned}$$

Here $\mathrm{Length}_h(\gamma) \geq W$ follows from $h \geq h_0$ and the original separation of A, B. This contradiction proves Equation (4.2); curves with infinite length are irrelevant. Finally, a smooth metric dominating a complete metric is complete: an h'-Cauchy sequence is h_0-Cauchy, and local equivalence of the two smooth metrics supplies convergence in h'. The proof of Theorem 4.1 is complete.

5 Wide separations at a fixed original scale

The local estimate pays for a metric change by a decrease in a trace. That decrease can be summed over arbitrarily many labels.

The estimates in this section apply in every dimension $n \geq 2$. For such n, fix $L_* = 1.004$ and choose $0 < \delta < 1/10$, depending only on n, so small that

$$2(n+1)\delta < 1 - L_*^{-2}. \tag{5.1}$$

Let $W, T \geq 1$ be the constants of Theorem 4.1 for these n and δ, and choose

$$R_0 \geq \sqrt{2}\,T, \qquad D = W R_0. \tag{5.2}$$

All of these constants depend only on n. In the proof of Theorem 5.1, the bound $R_0 \geq \sqrt{2}\,T$ initializes the positive quadratic term, since $R_0^2/2 \geq T^2$, while (5.1) restores it after each rescaled round through $L_*^2(1-(n+1)\delta) > 1$.

Theorem 5.1 (Prescribed separations for a locally finite family)**.** *Let (M, g) be a connected complete smooth boundaryless n-manifold, $n \geq 2$, with $\mathrm{Scal}_g \geq 1$, and let D be the constant in* (5.2)*. There is a countable set of centers (x_i) such that, with*

$$A_i = \overline{B}_g(x_i, D), \qquad B_i = M \setminus B_g(x_i, 2D),$$

the A_i cover M and the closed balls $\overline{B}_g(x_i, 2D)$ are locally finite. If some B_i is empty, then $\mathrm{diam}_g M \leq 4D$. Suppose that every B_i is nonempty. For every assignment of finite numbers $L_i > 0$, there exist a complete smooth metric $h \geq g$ and a countable family of smooth functions (p_α), whose supports form a locally finite family of compact sets, for which

$$\mathrm{dist}_h(A_i, B_i) > L_i \quad (i \geq 1), \tag{5.3}$$

$$\mathrm{Scal}_h - 2\Delta_h \sum_\alpha p_\alpha - \left| \mathrm{d} \sum_\alpha p_\alpha \right|_h^2 - \sum_\alpha |\mathrm{d}p_\alpha|_h^2 \geq \frac{2}{5}. \tag{5.4}$$

Every sum in (5.4) *is locally finite.*

The trace estimate in Lemma 5.2 permits each label to stop after its own finite number of rounds. The scalar identity in Lemma 5.3 then turns the locally finite increments of t into the finite circle factors required on each compact region. The proof of Theorem 5.1 follows these two inputs.

5.1 The trace estimate and the scalar identity

Lemma 5.2 (A complete round)**.** *Let M be a smooth boundaryless n-manifold, $n \geq 2$, with a complete smooth metric h_0 and a smooth function $t_0 \geq T$ satisfying*

$$S_1^{h_0}(t_0) \geq c + t_0^2$$

for a constant c. Let (A_i, B_i) be a finite or countable family of nonempty closed pairs, with each A_i compact and $\operatorname{dist}_{h_0}(A_i, B_i) \geq W$. Suppose there are locally finite compact sets K_i such that the open h_0-distance-W neighborhood of A_i is contained in $K_i \setminus B_i$. Then there are a complete smooth metric $h_1 \geq h_0$ and a smooth function $t_1 \geq t_0$, obtained by a locally finite sequence of compactly supported changes inside the K_i, such that

$$\operatorname{dist}_{h_1}(A_i, B_i) \geq L_* W \quad \text{for every } i, \tag{5.5}$$

$$S_1^{h_1}(t_1) \geq c + (1 - (n+1)\delta) t_1^2. \tag{5.6}$$

Proof. Enumerate the labels and choose $d_i > 0$ with $\sum_i d_i < \delta$. Let $(h^{(j)}, t^{(j)})$ denote the data after j changes, and apply Theorem 4.1 in this order, keeping h_0 as the background metric throughout and using the loss d_j at the jth application. Its hypotheses hold at every application: earlier changes only increase the metric and the function, so $h^{(j)} \geq h_0$ and $t^{(j)} \geq t_0 \geq T$, and its separation hypothesis $\operatorname{dist}_{h_0}(A_i, B_i) \geq W$ concerns the fixed metric h_0. Write

$$Q_j = \operatorname{tr}_{h^{(j)}} h_0, \qquad \kappa_j = 1 - \delta(n - Q_j) - \sum_{i \leq j} d_i.$$

Thus $Q_0 = n$, $0 < Q_j \leq n$, and $\kappa_j \leq 1$. We claim that

$$S_1^{h^{(j)}}(t^{(j)}) \geq c + \kappa_j (t^{(j)})^2. \tag{5.7}$$

It holds at $j = 0$. In a single step, abbreviate the old coefficient by κ, the new one by κ', and the old and new functions by t, t'. The local estimate gives

$$\begin{aligned} S_1^{h'}(t') &\geq c + \kappa t^2 + (t'^2 - t^2) - (\delta(Q - Q') + d) t'^2 \\ &= c + \kappa' t'^2 + (1 - \kappa)(t'^2 - t^2) \geq c + \kappa' t'^2. \end{aligned}$$

The last inequality uses both $\kappa \leq 1$ and $t' \geq t \geq T > 0$. This proves (5.7). Its coefficient is greater than $1 - (n+1)\delta$, which proves (5.6). Each processed separation satisfies (5.5), and later changes preserve it.

The supports lie in the locally finite family (K_i). Hence every point has a neighborhood on which the sequence is eventually constant; its limit is smooth and satisfies the same inequalities. Completeness follows from $h_1 \geq h_0$: an h_1-Cauchy sequence converges in h_0, and near its limit the two smooth metrics are equivalent. □

The passage from a positive scalar-curvature form to warped products with circle fibers uses torical symmetrization; see Gromov [21, Section 2]. The following identity fixes the finite-dimensional convention used here.

Lemma 5.3 (Scalar curvature of warped torus extensions)**.** *Let $p_1, \dots, p_k$ be smooth real functions on a Riemannian manifold (M, h), with $k \geq 0$ finite. The metric $\widehat{h} = h + \sum_{\alpha=1}^k e^{2p_\alpha} \mathrm{d}\theta_\alpha^2$ on $M \times \mathbb{T}^k$ has scalar curvature*

$$\operatorname{Scal}_{\widehat{h}} = \operatorname{Scal}_h - 2\Delta_h P - |\mathrm{d}P|_h^2 - \sum_{\alpha=1}^k |\mathrm{d}p_\alpha|_h^2, \qquad P = \sum_{\alpha=1}^k p_\alpha.$$

Proof. Let $E_\alpha = e^{-p_\alpha} \partial_{\theta_\alpha}$ be the unit vertical fields. For horizontal fields X, Y, metric compatibility and zero torsion give

$$\widehat{\nabla}_X Y = \nabla_X^h Y, \quad \widehat{\nabla}_X E_\alpha = 0, \quad \widehat{\nabla}_{E_\alpha} X = (X p_\alpha) E_\alpha, \quad \widehat{\nabla}_{E_\alpha} E_\alpha = -\operatorname{grad}_h p_\alpha,$$

and $\widehat{\nabla}_{E_\alpha} E_\beta = 0$ for $\alpha \neq \beta$. Therefore, for a horizontal unit vector X, the mixed sectional curvatures are

$$K(X, E_\alpha) = -\operatorname{Hess}_h p_\alpha(X, X) - (X p_\alpha)^2, \qquad K(E_\alpha, E_\beta) = -\langle \mathrm{d}p_\alpha, \mathrm{d}p_\beta \rangle_h.$$

Summing twice over all orthonormal two-planes gives the identity. □

Following Gromov's $\mathbb{T}^*$-stabilization [23, §1], we call $\operatorname{Scal}_{\widehat{h}}$, regarded as a function on M, the *stabilized scalar curvature* of h and the logarithms p_α of the warping functions e^{p_α}. Unit factors, with $p_\alpha = 0$, do not change it. By Lemma 5.3, the left side of (5.4) is the stabilized scalar curvature on every compact region, computed with the finitely many logarithms of the warping functions that are nonzero there.

5.2 Prescribed separations

Proof of Theorem 5.1. A complete connected Riemannian manifold is proper. Choose a maximal D-separated set of centers. Maximality gives the covering by closed D-balls. A compact set contains only finitely many centers; more generally, centers whose closed $2D$-balls meet a fixed compact set lie in a fixed compact neighborhood of that set. This proves local finiteness of the outer balls and countability of the net. The triangle inequality gives $\operatorname{dist}_g(A_i, B_i) \geq D$. If some B_i is empty, then $M = B_g(x_i, 2D)$, and $\operatorname{diam}_g M \leq 4D$ by the triangle inequality.

We keep a scalar reserve of $1/2$ in the original units. Put $R_j = L_*^j R_0$, start with $h^{[0]} = g$, $t^{[0]} = T$, and work in round j with the metric $h^{[j]}/R_j^2$. At the start of that round we require

$$S_1^{h^{[j]}/R_j^2}(t^{[j]}) \geq \frac{R_j^2}{2} + (t^{[j]})^2. \tag{5.8}$$

For $j = 0$, this follows from $\operatorname{Scal}_g \geq 1$ and $R_0^2/2 \geq T^2$.

A label remains active until its separation exceeds its prescribed L_i. At the start of round j, every active pair has separation at least WR_j in $h^{[j]}$; this is initially true and follows inductively from (5.5). Apply Lemma 5.2 to the active pairs in the working metric, with reserve $c = R_j^2/2$. The required support condition follows from $\operatorname{dist}_{h^{[j]}/R_j^2}(A_i, B_i) \geq W$: the open distance-$W$ neighborhood of A_i misses B_i, and is therefore contained in the original outer ball $B_g(x_i, 2D)$. We may take $K_i = \overline{B}_g(x_i, 2D)$.

Return the output metric to original units and denote it by $h^{[j+1]}$. It dominates $h^{[j]}$. Scalar curvature, the Laplacian and squared gradient norms all multiply by R^2 when a metric is divided by R^2. Consequently, on passing from working scale R_j to R_{j+1}, the round estimate becomes

$$S_1^{h^{[j+1]}/R_{j+1}^2}(t^{[j+1]}) \geq \frac{R_{j+1}^2}{2} + L_*^2(1 - (n+1)\delta)(t^{[j+1]})^2.$$

By (5.1), the coefficient of the final square is greater than one. Thus (5.8) is restored. In original units the reserve remains exactly $1/2$; it has not been spent by the change of scale.

Each fixed label becomes inactive after finitely many rounds, since its separation is at least WR_j as long as it remains active and $R_j \to \infty$. A compact set meets only finitely many of the original outer balls. Each of their labels is processed only finitely many times. Thus there are only finitely many changes on a neighborhood of any compact set over the *entire* construction, not merely in each round. The limiting data (h, t) are smooth, satisfy

$$h \geq g, \qquad t \geq T, \qquad S_1^h(t) \geq \frac{1}{2},$$

and satisfy (5.3). The same comparison argument as in Lemma 5.2 proves completeness of h.

It remains to realize the term involving t by ordinary finite-dimensional warped products on each compact region. Enumerate the nonzero compactly supported increments of t as $\tau_\nu \geq 0$, so that $t - T = \sum_\nu \tau_\nu$. The supports are locally finite. For each ν, choose an integer $m_\nu \geq 1$ such that

$$\frac{1}{m_\nu} \sup_M |\mathrm{d}\tau_\nu|_h^2 \leq \frac{2^{-\nu}}{10}.$$

The supremum is finite because the support is compact. Use m_ν fresh logarithms of the warping functions, all equal to $-\tau_\nu/m_\nu$. Listing these functions as (p_α), we obtain

$$\sum_\alpha p_\alpha = -(t - T), \qquad \sum_\alpha |\mathrm{d}p_\alpha|_h^2 \leq \frac{1}{10}.$$

On a neighborhood meeting only finitely many supports, equip a finite product of circles with metric $h + \sum_\alpha e^{2p_\alpha} \mathrm{d}\theta_\alpha^2$, omitting all unused unit factors. Its scalar curvature is the left side of (5.4); this is the identity in Lemma 5.3. Substitution gives $S_1^h(t) - \sum_\alpha |\mathrm{d}p_\alpha|_h^2 \geq 1/2 - 1/10 = 2/5$. This proves (5.4). □

6 Finite cuts with compatible ends

The stretching construction supplies separated sets and a scalar-curvature reserve. We now describe the second input: a cutting procedure that retains both sides of every cut and controls how later cuts meet them. Its output is a family of smooth boundaries and complete manifolds obtained by attaching cylinders to the closed sides. The definitions below make its hypotheses and output explicit; Appendices A–E give the proofs. Section 7 will separately justify its use with countably many labels.

Let $X = M$ denote the original complete manifold, let g be its original metric, and write $d_0 = d_g$ for its distance. The metric used in an analytic component can change; the map to X always records its position in the original manifold. Fix finitely many of the supplied ball pairs

$$A_i = \overline{B}_{d_0}(x_i, D), \qquad B_i = X \setminus B_{d_0}(x_i, 2D), \qquad 1 \leq i \leq \ell,$$

with A_i, B_i nonempty. There is one label for each ball pair. An additional first label is radial. Thus $N_* = \ell + 1$, and the order is radial, then $1, \ldots, \ell$. Properness of (X, g) makes each closed outer ball compact.

The analytic input at stage zero is a complete smooth metric $h \geq g$ on X and finitely many smooth positive circle warping functions for which the stabilized scalar curvature is at least .4. Each prescribed distance $\mathrm{dist}_h(A_i, B_i)$ can be required to exceed any given finite number. Theorem 5.1 supplies these data. For finitely many requested pairs its construction uses only finitely many compactly supported increments and hence finitely many circle factors. The countable version has a locally finite list of nonunit warping functions and is used in Proposition 7.2; the geometric statements below apply to each finite collection of the relevant factors.

For a single smooth cut, let u be a smooth function on an m-manifold E without boundary, with 0 a regular value, and put $L = u^{-1}(0)$. Attach $L \times [0, \infty)$ to each closed side $\{\pm u \geq 0\}$ along its boundary, and also retain L as a separate space. The two extended sides have dimension m, while the boundary has dimension $m - 1$. The cylinders replace the new boundaries by ends. The following definition records these spaces and their incidences after repeated cuts.

Definition 6.1 (Component diagram)**.** At stage r, an index is $\alpha \in \{+, -, *\}^r$. Its entries record the first r processed cuts. A sign $+$ or $-$ selects the corresponding retained side of a cut, and a star $*$ selects the zero set. Indices are ordered by replacing signs with stars: an index obtained from α by replacing $d \geq 1$ signs by stars lies below α. A stage-r component diagram consists of the following data, with the following properties.

(i) *Spaces.* For each index α, the space E_α is a smooth second countable manifold without boundary, possibly empty or disconnected; when nonempty its base dimension is $m(\alpha) = n - \#\{i : \alpha_i = *\}$. We consider its connected components.

(ii) *Product tails.* If B is a component of an index β and α is obtained from β by replacing d stars with signs, then in E_α the component B has exactly one incident component E and one corresponding occurrence of a full product tail

$$\iota_{BE} : B \times (T_1, \infty) \times \cdots \times (T_d, \infty) \longrightarrow E. \tag{6.1}$$

B is the base of this product end of E, with d normal factors. E is the corresponding incident higher-dimensional component. For $d = 1$, the product end has one normal coordinate. The normal coordinates are labelled by the changed index entries. Thresholds may depend on the incidence.

(iii) *Intersections and commutation.* For a single entry, tails from distinct end base components are disjoint. Intersections of sufficiently far tails for different entries are exactly the tails over their common further end bases. The identifications commute: composing end base maps in any order gives the same occurrence and the same individually labelled normal coordinates. After slightly raising the entry thresholds, each closed tail is properly embedded.

(iv) *Position maps.* Each component E is equipped with a continuous map $q_E : E \to X$.

(v) *After the radial stage*, in addition: every component has only finitely many incident end bases and incident higher-dimensional components, including through chains; the complement of its sufficiently far product tails with one normal factor is compact. On each sufficiently far end base product the maps q_E satisfy $q_E(\iota_{BE}(b, \mathbf{t})) = q_B(b)$.

A metric, scalar function, warping function, or set is *compatible* if, in every sufficiently far tail, it is exactly the product or pullback of the corresponding end base datum. For metrics the normal factors are $\mathrm{d}t_1^2 + \cdots + \mathrm{d}t_d^2$; all warping functions are independent of the normal coordinates there. The thresholds for different data may differ.

There is no boundary in an analytic component. Its end base is represented by an entire cylindrical end, not by a boundary subset. Finite boundaries appear only when we truncate these ends.

A *split along regular zero sets* applies the preceding construction using compatible smooth defining functions whose zero sets are regular on every index space. Each zero set gives the space indexed by a new star. In inherited ends the operation is the corresponding end base operation times the old normal factors. The proof that these operations preserve the component diagram is Lemma A.1.

For the diagrams constructed here, these maps are defined recursively, starting with $q_X = \mathrm{Id}_X$. In the split of Lemma A.1, let A be a new stage-$(r+1)$ component with stage-r parent E. Define $p_A : A \to E$ by inclusion on a zero set or the bulk of a joined side, and by $p_A(x, t) = x$ on each newly attached cylinder. Set $q_A = q_E \circ p_A$. Thus q_A is pulled back from the base on each fresh cylinder. In a sufficiently far inherited tail of A, let C be the corresponding new end base with old

parent B, and let I denote the old normal intervals. Under the identifications of these tails with $C \times I$ and $B \times I$, the map p_A is $p_C \times \mathrm{Id}_I$; explicitly,

$$p_A \circ \iota_{CA} = \iota_{BE} \circ (p_C \times \mathrm{Id}_I), \qquad q_A(\iota_{CA}(c, \mathbf{t})) = q_C(c).$$

Thus the recursion preserves compatibility. The proper maps q_r on finite reconstructions are constructed in Proposition A.5.

Figure 1 depicts one extended side and the separate zero set.

The label set and the index depth are finite in this section. The family of connected components may be countable; finiteness is required only for the directed chains through end bases and higher-dimensional components incident to each component. The downward diagram of a component consists of that component and all its further end bases. Numerical requests are allocated downward through these chains before actual data are constructed upward from end bases.

We now specify the analytic data that must survive each split. For the finite-label statements in this section and the appendices we use the following constants:

$$\begin{gathered} \lambda_r = \frac{1}{4} - \frac{r}{8N_*}, \qquad b_r = 1.12 - \frac{.05r}{N_*}, \qquad a_r = .60 + \frac{.03r}{N_*} \quad (0 \le r \le N_*), \\ \pi_0 = \frac{1}{10N_*}, \qquad k = 399, \qquad C = \frac{4k}{k+1} = 3.99, \qquad e_0 = \frac{\lambda_r - \lambda_{r+1}}{100} = \frac{1}{800N_*} \quad (r < N_*). \end{gathered} \tag{6.2}$$

Here k is the number of circle factors added at each split; the initial number may be different. The choice $k = 399$ realizes the coefficient $C = 4k/(k+1)$ in the scalar-curvature identity of Proposition E.1. With $C = 3.99$, the boundary and retained-side estimates leave gradient coefficients $C - 2 = 1.99$ and $C - 3.05 = .94$, respectively (Proposition B.10); these supply the reserve for enlargement of the prescribed open sets. In particular $\lambda_r \ge 1/8$, $a_r \le .63$, and $1.07 \le b_r \le 1.12$. The common floor π_0 and the equally spaced stage parameters are bookkeeping for a finite list, not uniform lower bounds for the countable construction. Section 7 and Appendix E write the same constants in terms of increasing budget parameters $t_r \in [0, 1]$, with increments $\Delta_r = t_{r+1} - t_r$, in (7.3). The equally spaced choice $t_r = r/N_*$ gives back (6.2), with $\Delta_r = 1/N_*$ and $e_0 = \Delta_r/800$, and gives the number $\eta_r = \Delta_r/200 = 1/(200N_*)$ that appears in Theorem 6.5.

Definition 6.2 (Analytic invariant)**.** At stage r, write J_r for the unprocessed labels. On a component E of base dimension m, retain its base metric and all circle warping functions. Write R_E for the scalar curvature of that full warped metric and ω_E for the product of its warping functions,

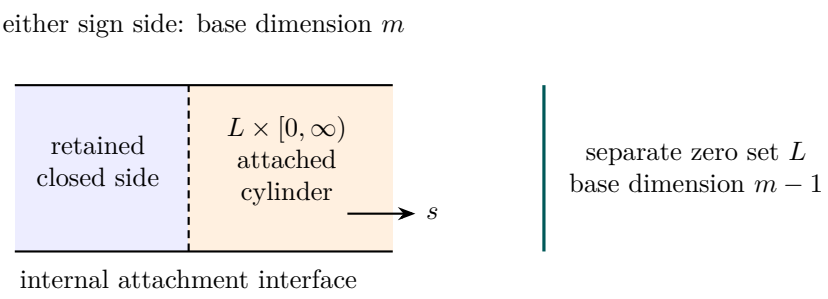


Figure 1: A regular cut retains each closed sign side with a full cylinder attached, and also retains the boundary as a separate zero set of one lower base dimension. The attachment interface is internal to the extended side. In old product ends this entire operation is a product. The raw metric and its later smoothing are defined in Appendices B and E.

suppressing fixed circle-volume constants. Gradients of invariant functions are base gradients; their integrals use $\omega_E \,\mathrm{d}\,\mathrm{vol}_E$, and their Laplacian is the corresponding weighted base Laplacian. For every $j \in J_r$ the data comprise a compatible continuous function $\mu_j : E \to [-\infty, \infty]$ whose finite truncations $\max\{-c, \min\{\mu_j, c\}\}$, $0 < c < \infty$, are locally Lipschitz, a compatible smooth potential $\Pi_j \geq \pi_0$, and compatible prescribed open sets U_j^+, U_j^- satisfying

$$\overline{U_j^{\pm}} \subset \mathrm{int}_E\{\mu_j = \pm\infty\}. \tag{6.3}$$

On the open finite-value locus $\{x \in E : |\mu_j(x)| < \infty\}$ the function obeys, almost everywhere,

$$2|\operatorname{grad} \mu_j| \leq a_r \mu_j^2 + \Pi_j. \tag{6.4}$$

Set

$$G_j = E \setminus (U_j^+ \cup U_j^-), \qquad D_E = \{x : \#\{j \in J_r : x \in G_j\} \geq m - 1\}. \tag{6.5}$$

The required scalar bound is

$$R_E \geq \lambda_r + b_r \sum_{j \in J_r} \Pi_j \quad \text{on } D_E. \tag{6.6}$$

For an unprocessed label i, the prescribed open sets contain $q_E^{-1}(A_i) \subset U_i^+$ and $q_E^{-1}(B_i) \subset U_i^-$. Enlarging prescribed open sets must preserve these conditions. The sets D_E are closed: they are finite unions of finite intersections of the closed sets G_j (and equal E when the count threshold is nonpositive). All data are exact products in the sense of Definition 6.1; at stage zero the single space is X, with no product-end requirement.

The threshold $m-1$ is preserved by the two kinds of output of a split. A side keeps its dimension and loses one unprocessed label; a zero set loses both one dimension and that label. The set H_E^* where at least $m-2$ remaining labels are undecided contains the projections of all new required sets, even after the remaining prescribed open sets have been enlarged.

Lemma 6.3 (The two count inclusions)**.** *Use the required sets and compatible prescribed open sets of Definition 6.2. If B is the base of a product end of E with d normal factors, the specified tail projection in* (6.1) *sends D_E in that tail into D_B. At a split of label j_0 on a parent of dimension m, suppose the remaining prescribed open sets have been compatibly enlarged, giving $G_j' \subset G_j$ for $j \neq j_0$, and define the closed set*

$$H_E^* = \{x : \#\{j \in J_r \setminus \{j_0\} : x \in G_j'\} \geq m - 2\}. \tag{6.7}$$

Suppose a regular zero set L lies in $\mathrm{int}\, G_{j_0}$*. Pull the remaining prescribed open sets to a new side or component of the zero set A by bulk inclusion, boundary restriction, and projection of the attached cylinder. Define D_A using these prescribed open sets, the remaining labels, and the dimension of A, as in* (6.5)*. Then the bulk inclusion, boundary restriction, or cylinder projection that defines D_A sends it into both H_E^* and the parent's old set D_E. The sets H_E^* likewise project into H_B^* on further end bases B.*

Proof. In an exact product tail all label memberships pull back unchanged, whereas the end base dimension is $m - d$. A count at least $m-1$ is at least $(m-d)-1$, proving the first assertion. The same argument with thresholds $m-2$ and $(m-d)-2$ proves the set H_E^* assertion. On a new side, whose dimension is m, a point of D_A has at least $m-1$ remaining undecided labels. This puts its projection in the set H_E^* and, because $G_j' \subset G_j$, in the old required set. On a component of the zero set, whose dimension is $m-1$, the required count is $m-2$. Its projection lies on $L \subset G_{j_0}$, so the processed label contributes one further old undecided label. The old count is therefore at least $m-1$. The cylinder projection has exactly the same remaining memberships as its base, so the side argument also covers all cylinder points. These are pointwise count statements. □

Lemma 6.4 (Initialization and radial confinement)**.** *For the stage-zero metric, warping functions, and ball pairs fixed above, assume that the stabilized scalar curvature is at least* .4 *and that each label pair satisfies*

$$\operatorname{dist}_h(A_i, B_i) > T_0 + 4, \qquad T_0 = 2\pi\sqrt{2/\pi_0}.$$

Then the given metric and warping functions admit the functions, prescribed open sets, and potentials of Definition 6.2 at stage zero with $\Pi_j = \pi_0$*. On a noncompact* X *the radial prescribed open sets can be chosen as separating inverse images of intervals of a smooth proper function, with alternating signs, with compact smooth barrier domains around their closures lying in the same pole plateaus. Every regular radial partition respecting these signs has locally finite compact boundary components and compact closed-side components before collar attachment. On a compact* X *a decided radial label with empty boundary has the same properties.*

Theorem 6.5 (Finite compatible cutting)**.** *Fix* N_* *and the stage constants in* (6.2)*. Let* $0 \le r < N_*$ *and let a stage-*r *diagram arise from the regular splits of Definition 6.1, with compatible smooth base metrics and a finite common list of positive circle warping functions satisfying Definition 6.2. After the radial stage, assume that each component is complete, has compact remainder outside its finitely many product ends, and has the exact metric and warping function identifications in Definition 6.1. Let* j_0 *be the next label, and assume that every nonempty parent has base dimension at least two.*

The diagram admits a compatible split along regular zero sets of j_0 *with the following properties.*

(i) On each parent E*, the smooth defining function has the prescribed signs on the current closures of the prescribed open sets, and its regular zero set lies in* $\operatorname{int} G_{j_0}$*. The new spaces are that boundary and the two closed sides with their full boundary cylinders attached. Their maps to* X *are the continuous maps* $q_A = q_E \circ p_A$ *specified above, including the commuting inherited tail projections.*

(ii) The remaining parent functions and potentials may change, with prescribed open sets only enlarged and potentials only increased. Their restrictions and cylinder pullbacks admit compatible smooth metrics and potentials and exactly 399 *new positive circle factors satisfying the full stage-*$(r+1)$ *invariant. Earlier forced signs persist. The new components are complete with compact remainders and exact product ends. This includes a possible one-dimensional output, whose scalar bound holds everywhere.*

(iii) Every old warping function identically equal to 1 *stays fixed. A parent's downward source diagram consists of that component and all its further end bases, together with every specified product occurrence* (6.1) *whose end base and incident higher-dimensional component both belong to this diagram. An end base copy means the data on the end base pulled back to such an occurrence. If a remaining label has the same constant pole function, full corresponding prescribed open set, and spatially constant potential* c_j *on every component of this finite diagram, and hence on all their end base copies, its pole and prescribed open set can stay fixed and its new potential can equal* $(1 + 3\eta_r/2)c_j$ *exactly, where* $\eta_r = 1/(200N_*)$*.*

At stage zero the parent may instead be an arbitrary complete smooth manifold. With the same scalar and separation assumptions and the radial initialization of Lemma 6.4, the first split has all the stated conclusions. Its boundary and closed-side components are compact before cylinder attachment and form locally finite families.

Proof. Realize the processed function by a smooth finite prescribed weighted mean curvature and take a local minimizer of weighted perimeter minus the weighted volume integral of that function, as in Lemmas B.1 and B.3. The Caccioppoli set has the forced signs of Lemma B.4; the support of

its perimeter measure has the regular-part geometry of Theorem B.6 but may still have singularities. After the radial cut, the generic prescribed weighted mean curvature choice in Proposition B.5 makes these boundary supports converge to the corresponding end base products at component ends (Corollary B.7). On the initial radial parent use shift zero and a Caccioppoli set from compactness; compact barriers replace this post-radial growth and uniqueness argument.

The boundary, and each retained side with its attached cylinder, carry the quadratic-form lower bounds of Proposition B.10. These bounds include a positive gradient reserve and, on a joined side, the scalar contribution of its internal attachment interface. They are not yet pointwise scalar bounds on smooth new components. Keep these boundary supports and their prescribed weighted mean curvatures fixed. The constructions of Lemma C.1 and Propositions C.5 and D.2 change only the remaining functions, enlarge their prescribed open sets, and add nonnegative potentials. Their form cost is paid by the gradient reserve and the gap $\lambda_r - \lambda_{r+1}$, and the updated set H_E^* of (6.7) misses the singular set. Thus the future required regions can be controlled through regular portions of the boundary support. When the parent dimension is at most seven the singular set is empty, and no singular forcing is needed; compactness on the resulting smooth regions is the smooth case of Lemma D.3. More generally this case with a smooth limiting boundary applies whenever the limiting boundary is smooth, including products of smooth end bases with Euclidean factors even when their total dimension is at least eight. At the first radial split the same arguments are applied on the locally finite compact transition blocks, as specified in Proposition D.2.

The approximation near H_E^* in Lemma E.3 now replaces the boundary support by a smooth regular boundary. Its defining functions are exact products in the old ends. The comparison with the boundary support on the set H_E^*, followed by the interface smoothing of Lemma E.5, transfers the form bound to the smooth split. The positive supersolutions of Lemmas E.8 and E.10 then supply 399 compatible new circle factors; the scalar identity in Proposition E.1 converts that form bound into the pointwise stage-$(r+1)$ scalar inequality. Throughout this transfer the parent scalar inequality is used only over D_E: Lemma 6.3 places every new required region inside that old region as well as inside the set H_E^*.

Lemmas A.1 and A.2 give the new geometry; Lemma 6.4 gives the radial confinement. Unit entries are fixed by restriction, product extension, and interface smoothing. For a remaining label with the same constant pole function, full corresponding prescribed open set, and constant potential throughout its source diagram, we omit its forcing updates, and Lemmas E.6 and E.7 reproduce its chosen constant potential exactly. The choice of η_r satisfies the strict stage margins in (E.3). □

7 Countably many compatible partitions

The separations in Theorem 5.1 may depend on the label. We now use this freedom to construct the partitions. The analytic construction is performed at finite stages. Its essential finiteness property concerns each *whole component*, including all its infinite product ends; local finiteness merely on compact subsets would not suffice. The finite-label procedure is stated in Theorem 6.5, and its detailed proof is in Appendices A to E. We adapt that proof using individual positive floors and finite exceptional data on each whole component.

7.1 Data and the countable construction

Let D be the length in (5.2), and let the centers x_i and the pairs (A_i, B_i) be those fixed in Section 2: a maximal D-separated set and its ball pairs, as in Theorem 5.1. We assume every B_i is nonempty, since otherwise $\operatorname{diam}_g M \le 4D$. There is one label for each center, and an additional label 0 used

first for radial confinement. The set of labels is finite or countable; write it as $I = \{0, 1, 2, \dots\}$, stopping the enumeration if necessary. Choose

$$p_i^* > 0, \qquad \sum_{i \in I} p_i^* = \frac{1}{20}. \tag{7.1}$$

Use Theorem 5.1 with

$$L_i = 10 + 2\pi\sqrt{2/p_i^*} \qquad (i \neq 0). \tag{7.2}$$

We henceforth use the resulting complete metric $h \geq g$, its locally finite list of logarithms of the warping functions p_α, and its stabilized scalar lower bound $2/5$.

The initial weighted base measure is globally well defined:

$$\omega \, \mathrm{d}\,\mathrm{vol}_h, \qquad \omega = \exp\Big(\sum_\alpha p_\alpha\Big) > 0.$$

The sum is locally finite. On a buffered compact set, only finitely many nonunit factors e^{p_α} occur, so ordinary finite warped-product formulas compute the stabilized scalar, weighted variations, and invariant test forms there. These formulas agree on overlaps: adjoining a unit factor changes none of them. In particular, the initial problem is a weighted problem on the n-dimensional base M.

Fix numbers

$$0 = t_0 < t_1 < t_2 < \cdots < 1, \qquad t_r \longrightarrow 1,$$

and use as many stages as there are labels. These are budget parameters, unrelated to a spatial coordinate. Put

$$\begin{gathered} a_r = .60 + .03 t_r, \qquad b_r = 1.12 - .05 t_r, \qquad \lambda_r = \frac{1}{4} - \frac{t_r}{8}, \\ \Delta_r = t_{r+1} - t_r, \qquad e_r = \frac{\Delta_r}{800}, \qquad \eta_r = \frac{\Delta_r}{200}, \qquad k = 399, \qquad C = \frac{4k}{k+1} = 3.99. \end{gathered} \tag{7.3}$$

With $t_r = r/N_*$, the entries shared with (6.2) agree with it, e_r being its e_0. Thus $\lambda_r \geq 1/8$, $.60 \leq a_r < .63$, and $1.07 < b_r \leq 1.12$. In a stagewise appendix lemma, its symbol e_0 denotes the current e_r.

We use the component notation and operation of splitting along regular zero sets introduced in Section 6. The countable invariant below replaces the common finite-label floor by the individual positive floors p_j^*. At stage r, the processed labels are $0, \dots, r-1$ and the remaining set is J_r. A component with s star entries has base dimension $m = n - s$. Its continuous collapse map to M is denoted by q_E. For each $j \in J_r$ retain an extended-real continuous function μ_j with locally Lipschitz finite truncations, a smooth potential $\Pi_j \geq p_j^*$, and disjoint prescribed open sets U_j^+, U_j^- with

$$\overline{U_j^\pm} \subset \operatorname{int}\{\mu_j = \pm\infty\}, \qquad 2|\operatorname{grad}\mu_j| \leq a_r \mu_j^2 + \Pi_j \quad \text{almost everywhere on the finite-value locus.} \tag{7.4}$$

For unprocessed center labels require $q_E^{-1}(A_j) \subset U_j^+$ and $q_E^{-1}(B_j) \subset U_j^-$. All data are compatible products in the component ends. With

$$G_j = E \setminus (U_j^+ \cup U_j^-), \qquad D_E = \{x : \#\{j \in J_r : x \in G_j\} \geq m - 1\}, \tag{7.5}$$

the scalar requirement is

$$R_E \geq \lambda_r + b_r \sum_{j \in J_r} \Pi_j \quad \text{on } D_E. \tag{7.6}$$

Here R_E denotes the stabilized scalar expression, and integrals and Laplacians use the product of the old warping functions as weight. The finiteness statement below makes these meanings unambiguous. A nonpositive counting threshold means the whole component.

Definition 7.1 (Finite exceptional data)**.** At stage $r \geq 1$, on each whole component E there are finite sets $I_E \subset J_r$ of label indices and F_E of old warping function indices such that, for $j \notin I_E$,

$$\mu_j \equiv -\infty, \qquad U_j^- = E, \qquad U_j^+ = \varnothing, \qquad \Pi_j \equiv c_r p_j^*. \tag{7.7}$$

Every old warping function indexed outside F_E is identically one. The number c_r is the same on all components and is prescribed by

$$c_0 = 1, \qquad c_{r+1} = (1 + 3\eta_r/2)c_r. \tag{7.8}$$

For any finite diagram used in a comparison, take the union of its finite exceptional sets, adding the current label explicitly. This one finite union then works simultaneously on all its whole components. At stage zero the analogous property is required on every compact radial block and buffer, with $c_0 = 1$.

The sets I_E need not consist exactly of the nonconstant entries: an entry may be retained in the exceptional list even if it happens to be constant. This convention avoids any need to discard previous exceptions. Only the existence of a finite list matters.

Proposition 7.2 (The countable compatible construction)**.** *For the initialized data above, there is a successive compatible split along regular zero sets for every label. At every finite stage the component diagram has the invariant above and no nonempty component of dimension at most one. After the radial stage it has finite exceptional data on whole components; at stage zero this assertion holds on buffered compact radial blocks. The finite reconstruction X_r, obtained by truncating the product ends and gluing the corresponding boundary faces selected by stars, has a proper collapse $q_r : X_r \to M$ and closed signed sides $P_i^{(r)}, N_i^{(r)}$ for each processed center label, with*

$$P_i^{(r)} \cup N_i^{(r)} = X_r, \qquad P_i^{(r)} \cap N_i^{(r)} = W_i^{(r)},$$

where $W_i^{(r)}$ is its boundary. The forced preimages satisfy

$$q_r^{-1}(A_i) \subset P_i^{(r)} \setminus W_i^{(r)}, \qquad q_r^{-1}(B_i) \subset N_i^{(r)} \setminus W_i^{(r)}.$$

These inclusions hold in open sides and persist at every subsequent finite stage. No $n-1$ distinct boundaries, even including the radial boundary, have a common point. Each center boundary lies over its original open outer ball, and its components are compact and locally finite.

Old finite truncation levels of the product ends can be enlarged by identifications preserving collapse, all boundaries, and all signed indices. For any prescribed sequence $\varepsilon_r > 0$ with $\sum_r \varepsilon_r < 1$, the next finite collar compression can be chosen with collapse displacement less than ε_r, preserving the previously decided boundary strata and signed sides. These assertions concern each finite stage.

Proof. Initialization is Lemma 7.3. If the construction has reached stage r, its parents have dimension at least two: a first nonempty output of smaller dimension would have dimension exactly one, since a zero set decreases dimension by one and a sign preserves it. Proposition 7.5 produces the next invariant, and Lemma 7.6 excludes that first output. Thus the induction continues through every finite stage.

Apply Proposition A.5 to each finite initial segment. Its hypotheses are the compatible splits along regular zero sets and the compact radial components already established. Its index pieces are exactly the corresponding compact component truncations. If $n-1$ boundaries met, assign those labels star entries and choose a closed sign index for each other processed label containing the point.

The corresponding nonempty truncation would have dimension one, a contradiction. Forced signs persist under the continuous collapse recursion and enlargement of the prescribed open sets. In particular

$$q_r(W_i^{(r)}) \subset M \setminus (A_i \cup B_i) \subset B_g(x_i, 2D).$$

Properness of q_r and compactness of the original closed outer ball make each boundary component compact; regular local boundary charts give local finiteness, as in Proposition A.5.

The truncation level identifications in Proposition A.5 preserve collapse, boundaries, and signed indices. Its collar compressions preserve the old strata and their sides, and Remark A.6 allows their collapse oscillations to be bounded by the prescribed ε_r. These choices are made separately on the locally finite compact component family. Section 8 chooses these homeomorphisms coherently on one fixed copy of M, proving Proposition 2.1. □

7.2 Initial functions and radial confinement

Lemma 7.3 (Initial functions and radial blocks)**.** *The fixed stretched metric and its initial warping functions admit stage-zero functions and prescribed open sets satisfying* (7.4)–(7.6) *with the prescribed potentials* $\Pi_i = p_i^*$*. The radial forced signs can be chosen so that every regular radial split preserving them has locally finite compact boundary and closed-side components before cylinder attachment. On every compact radial block and buffer, only finitely many center labels fail to be identically the negative pole with full negative prescribed open set, and only finitely many initial warping functions fail to be one.*

Proof. For a positive number p, the decreasing crossing

$$2|y'| = .5y^2 + p$$

from $+\infty$ to $-\infty$ has length

$$T(p) = \int_{-\infty}^{\infty} \frac{2\,\mathrm{d}y}{.5y^2 + p} = 2\pi\sqrt{2/p}.$$

Extend it by its poles. For center i, begin this crossing at stretched distance 2 from A_i and compose it with $\mathrm{dist}_h(A_i, \cdot)$. The distance is 1-Lipschitz, so the resulting function obeys the inequality with .5 in place of $a_0 = .60$. Take prescribed open sets

$$U_i^+ = \{\mathrm{dist}_h(A_i, \cdot) < 1\}, \qquad U_i^- = \{\mathrm{dist}_h(A_i, \cdot) > T(p_i^*) + 3\}.$$

Their closures lie in the respective pole interiors. The stipulated separation puts B_i strictly inside the negative prescribed open set. In particular, on the complement of the original closed ball $\overline{B}_g(x_i, 2D)$, the function is the negative pole and the negative prescribed open set is full.

The radial label is constructed as in the proof of Lemma 6.4 in Appendix A, with the potential π_0 replaced by p_0^* and T_0 by $T(p_0^*)$. The chartwise smooth-radius construction there, with its additive constant 2 subtracted, gives a smooth proper function $\rho : M \to \mathbb{R}$, bounded below, such that $|\rho - \mathrm{dist}_h(o, \cdot)| < 1$ and $|\operatorname{grad}\rho|_h < 2$. It uses only local smoothness and completeness, so applies to the present stretched metric. The same proof builds alternating positive and negative plateaus of ρ, separated by transition intervals wider than $2T(p_0^*)$, on which the rescaled crossing satisfies the function inequality with coefficient .5 and potential p_0^*; closures of the prescribed open sets strictly inside the plateaus; and compact smooth barrier domains, bounded by regular levels, whose closures stay in the same pole plateaus and whose value intervals tend to infinity. The same proof shows that a boundary respecting these signs has compact components, each lying in one block between consecutive separating plateaus, that every component of either closed side is trapped between two

separating regions of the opposite sign or lies in the initial compact block, and that both component families are locally finite. When M is compact, it uses a radial label with constant positive pole and full positive prescribed open set; its split has no boundary and its only side is compact.

The scalar requirement initially holds everywhere because

$$\lambda_0 + b_0 \sum_i p_i^* = .25 + 1.12/20 = .306 < .4.$$

Finally a compact block with a compact buffer meets only finitely many original closed outer balls and finitely many initial supports of the logarithms of warping functions. Every other center label and warping function has the asserted constant data there. □

7.3 Finite exceptional data on entire components

A *downward diagram* of a component consists of it and all its further end bases. An *upward diagram* consists of it and all its incident higher-dimensional components. Each is finite at a fixed stage, by Lemma A.1. A finite union of either type is finite. The constructions below make finitely many directed passes through such diagrams.

Radial confinement gives the geometric reason to expect finite data on a whole component. At the first split, a compact radial side or boundary component is prolonged by cylinders; its entire collapse image stays in that original compact component. At every later split a child A of E has

$$q_A = q_E \circ p_A, \qquad p_A = \begin{cases} \text{inclusion}, & \text{on the boundary or bulk}, \\ \text{projection to the boundary}, & \text{on a fresh cylinder}. \end{cases} \tag{7.9}$$

Hence $q_A(A) \subset q_E(E)$, including in all simultaneous normal directions. A finite incidence diagram therefore has image in a finite union of compact original radial blocks. Only finitely many original outer balls and supports of the logarithms of the initial nonunit warping functions meet that image. Enlargement of the prescribed open sets changes no collapse map, and regularization chooses new boundaries inside old parents, so both preserve this confinement.

This argument controls the initial supports. Later smooth data need not factor through q_A: they also use data copied from end bases. We must therefore retain finite exceptional lists through those directed dependencies, as well as preserve every other entry exactly.

Lemma 7.4 (Consequences and directed dependencies)**.** *At a finite stage after the radial split, assume the exact component geometry of Definition 6.1 and the finite exceptional data of Definition 7.1. Then:*

(i) The sum of potentials on a whole component is a smooth function with bounded derivatives of every fixed order. The undecided counts, required sets, and sets H_E^* use only a finite list and are closed.

(ii) On each whole component and each finite comparison diagram, the old weight, stabilized scalar, stability forms, and weighted Laplacian are computed with one finite list of warping functions, after padding with unit factors. The conclusions of a positive injectivity radius, uniform bounds on curvature and all its covariant derivatives, and polynomial weighted growth in Lemma A.2 apply, including on any finite Euclidean product.

(iii) Compatible choices with positive componentwise tolerances can be made in the following order: fix individual requests, propagate numerical requests down the finite chains through higher-dimensional components, and then construct data upward from deeper end bases. Enlarging

an incident higher-dimensional component's exceptional list by the union over its downward source diagram suffices for the last pass. No data are propagated back from a newly constructed incident higher-dimensional component to an already constructed end base.

Proof. On E, the sum of potentials is exactly

$$\sum_{j\in J_r} \Pi_j = \sum_{j\in I_E} \Pi_j + c_r \sum_{j\in J_r\setminus I_E} p_j^*. \tag{7.10}$$

The second term is a finite spatial constant. Indeed

$$1 \le c_r \le \exp\Big(\frac{3}{2}\sum_{s<r}\eta_s\Big) \le e^{3/400}.$$

The first term is a finite sum; its derivative bounds follow from compatible smoothness and the compact-part/product-end decomposition in Lemma A.2. On a finite comparison diagram choose common product thresholds for its finite exceptional list. Every other entry is already a spatial constant with the same value on each copy. The full sum is therefore an exact end base pullback beyond those thresholds. Entries outside the finite exceptional list have empty undecided set. Thus a set requiring at least $a \ge 1$ undecided labels is the finite union, over a-element subsets of I_E, of the intersections of their closed G_j's. This proves closedness, including for the sets H_E^* with threshold $m-2$. Threshold zero gives E.

For (ii), take the union of the finitely many relevant old warping function indices on the finite diagram and set every missing entry to one. Writing $u_\alpha = \log f_\alpha$ and $U = \sum_\alpha u_\alpha$, the scalar expression is

$$R_E = \mathrm{Scal}_{h_E} - 2\Delta_{h_E}U - |\mathrm{d}U|^2_{h_E} - \sum_\alpha |\mathrm{d}u_\alpha|^2_{h_E}.$$

A padded logarithm is zero; the product weight is unchanged. The weighted Laplacian is $\Delta_{h_E} + \langle \mathrm{d}U, \mathrm{d}\,\cdot\,\rangle_{h_E}$ and is unchanged as well. Weighted perimeter and its first and second variations depend on the same weight. Equivalently, a flat unit factor adds zero second fundamental form and zero curvature to the invariant variation calculation. Thus the boundary and interface forms also agree. The finite-diagram geometric lemma now gives completeness, a positive injectivity radius and uniform bounds on curvature and all its covariant derivatives, positive lower and finite upper bounds for the weight, and center-uniform polynomial volume growth. It gives the same statements after taking a finite Euclidean product. The regularity problem for a minimizing Caccioppoli set remains on the base, with singular dimension at most $m-8$ by Theorem B.6; padding does not alter that dimension.

For (iii), a particular end base receives requests from only its finitely many incident higher-dimensional components and chains through higher-dimensional components. Allowing for the finitely many fixed product and cutoff bounds, the minimum of the requested positive errors is positive. Settle all these numbers before constructing the approximations. A request only decreases an error tolerance: it does not copy any incident higher-dimensional component value into the end base. During the subsequent upward pass, the inputs on a component are its own data and already constructed copies from its finite downward diagram. Their exceptional sets have a finite union. Outside that union every copy in (7.7) agrees with every other copy. A newly constructed incident higher-dimensional component is not used to reconstruct its sources. The same argument applies to finitely many successive directed passes, and proves the stated dependency rule. □

7.4 One countable invariant step

Proposition 7.5 (Preservation of the invariant)**.** *Suppose that all nonempty parents at stage r have base dimension at least two and satisfy* (7.4)–(7.6)*. At $r = 0$ use the compact radial blocks of Lemma 7.3; at $r \geq 1$ suppose also that the component geometry and Definition 7.1 hold. There is a compatible split along regular zero sets of label r preserving the forced signs and satisfying* (7.4)–(7.6)*, the component geometry, and Definition 7.1 at stage $r + 1$. Exactly* 399 *new warping function factors, with common indices, are appended at this stage. Each whole resulting component has a finite relevant old warping function list.*

Proof. The finite exceptional lists let each geometric operation use only finitely many nonconstant functions on a whole component. The remaining functions and potentials pass through the operation as their prescribed constants. We first choose the current cut using its own positive floor, then carry these two types of data through enlargement of the prescribed open sets and regularization.

The current prescribed weighted mean curvature and minimizing Caccioppoli sets. The prescribed weighted mean curvature construction of Lemma B.1 uses a positive floor only for the label being processed. Replace its finite-family floor by p_r^*. Compatible clipping heights can first be chosen above the local barrier bounds, with a uniform extra barrier margin of one. Then use local smoothing errors to obtain, for the smoothed prescribed weighted mean curvature p_0 of Lemma B.1 (its subscript does not refer to the radial label 0),

$$2|\operatorname{grad} p_0| \leq .65p_0^2 + 1.01\Pi_r.$$

All smoothing requests involve only this one function and finite incidence diagrams. The elementary inequality

$$.67(y + c)^2 - .65y^2 \geq -21.775c^2$$

shows that one common nonempty shift interval, for example

$$|c| < \min\{.01\sqrt{p_r^*}, 1/2\}, \tag{7.11}$$

preserves both $2|\operatorname{grad}(p_0 + c)| \leq .67(p_0 + c)^2 + 1.02\Pi_r$ and the barrier inequalities. Floors for other labels do not enter this interval.

After the radial stage, Lemma 7.4 supplies the positive injectivity radius, uniform bounds on curvature and all its covariant derivatives, and weighted polynomial-growth hypotheses of Proposition B.5. There are countably many components, Euclidean product dimensions, and tests in that lemma's exhaustion. Remove the countable exceptional shift sets and choose a single c in (7.11). The resulting minimizers have the common product limits of Corollary B.7; choose them before any enlargement of the prescribed open sets. The smooth global positive weight and locally bounded smooth prescribed weighted mean curvature give local BV existence and strict compactness by Lemma B.3. The relatively compact radial barriers force the Caccioppoli sets by compact comparisons. All variations and tests lie in buffered compact regions with finite warping function lists, so the formulas in Proposition B.10 apply. The GMT and singular-set dimensions are the base dimensions at this first step as well.

Smooth approximations, sets H_E^*, and enlargement of the prescribed open sets. On each post-radial component retain its finite exceptional list, enlarged by its finite downward source diagram and by the current label. Labels outside it are already forced everywhere and require no shell addition. Choose compatible smooth functions h_j within .1 of $\max(-1, \min(1, \mu_j))$ as in Lemma C.1, retaining the exact value -1 when $\mu_j = -\infty$ identically. For each component of dimension $m \geq 8$ and each set of $m - 7$ remaining labels, the image of its singular set under the corresponding map

of smooth approximations has $(m-7)$-dimensional Lebesgue measure zero. There are countably many such conditions, even if all finite subsets of the countable label set are included. In the countable product of uniform probability intervals $(-.3, .3)$, each bad cylinder has measure zero by its finite-dimensional marginal. A threshold vector (t_j) outside their union therefore exists. At most $m-8$ approximations h_j equal their thresholds at any singular point. The sign exclusions at poles follow from the .1 approximation to the truncated function, exactly as in Lemma C.1.

Process the finitely many end base depths from deepest upward, keeping the boundary supports fixed. Suppose every strict end base B of E has H_B^* disjoint from its singular set. The remaining intersection of its singular set with H_E^* is

$$K_E = \Sigma_E \cap \{x : \#\{j \in J_r \setminus \{r\} : x \in G_j'\} \geq m-2\}$$

is closed. It is compact: an escaping sequence would, by Lemma A.3, have a product limit over a further end base with compact convergent base coordinates. The finite exceptional list and closed complements of the prescribed open sets pass the undecided count to that limit. The end base B has a smaller dimension, so the limit lies in H_B^*. The product convergence of boundary supports forces the limiting point to be singular, since regular-point graphical convergence excludes nearby singular points. This contradicts the completed enlargement of the prescribed open sets of that end base. This argument also treats escape in several normal coordinates.

Apply Lemmas C.2 and C.3 and Proposition C.5 on this compact set K_E, only for the finite list which can still be undecided. Enumerate the signed compact subsets F with the following property: in Euclidean chart coordinates, for some $r_F > 0$, each $F \cap B_r(x)$, $x \in F$, $0 < r < r_F$, lies within ηr of an affine $(m-8)$-plane, with η as small as required. Choose each shell scale by Lemma D.4, hold it fixed, and then send its incident higher-dimensional component starting levels far enough out by Lemma D.5. In the normal-coordinate function construction of Lemma D.6, the crossing for label j satisfies

$$2|w'| \leq .58w^2 + p_j^*.$$

Its length is finite; the inequality is paid by $\Pi_j \geq p_j^*$, and the rest of that lemma applies unchanged.

Only finitely many additions are required on E. Indeed, every point of the initial compact K_E eventually belongs to a prescribed open set for each of its strict inequalities $h_j > t_j$ or $h_j < t_j$ among the finite exceptional list. At most $m-8 < m-2$ labels can then remain undecided there. The remaining subsets of the initial K_E are nested and closed, and their intersection is empty. Compactness gives a finite index at which the residual is empty. Stop then. Each specified incident higher-dimensional component receives additions from finitely many source end bases through finitely many chains. Thus only finitely many additions and finite minima or maxima affect any whole component.

The local shell estimate (C.13) charges $1.16D_\varepsilon$ at gradient coefficient .78. The factor 1.16 is chosen to exceed $(1+3\eta_r)b_{r+1}$, the coefficient with which the additions D_j enter the next potentials in the later scalar-form transfer; see (E.3). To control all the countably many components at once, enumerate the pairs consisting of a component and a possible local addition number by distinct integers $\nu \geq 1$. As in the proof of Proposition D.2, assign $\gamma_\nu = .1\,2^{-\nu}$ to that pair, allocating at most half to its shell-scale choice and half to its simultaneous finite incident higher-dimensional component extensions. Starting the bookkeeping form bound at .78, the maximum and extension estimates give $\theta_* = .78 + \sum_{\nu \geq 1} \gamma_\nu = .88 < .9$ and hence

$$\int 1.16 \sum_{j \neq r} D_j v^2 \leq \theta_* (\mathcal{N}(v) + e_r \|v\|_2^2). \tag{7.12}$$

This is the bound (D.4) of Proposition D.2. Here $\mathcal{N}(v)$ is the sum of weighted integrals of $|\nabla v|^2$, with coefficient 1.7 on a boundary, or coefficient .8 on the bulk and 1.7 on its attached cylinder. The measure on each piece is its base volume times the product of the retained warping functions; $\|v\|_2^2$ is the sum of its weighted squared norms. Tests are compactly supported away from singularities, and their two traces match at a joined interface, as in Definition B.11. These are the same coefficients on every component and product. Only a finite list of additions occurs on each post-radial component, so the bound persists after a whole depth has been treated.

Initially, instead work in separated compact transition shells with compact buffers between radial barriers. Each has finitely many exceptional labels and warping functions. Its intersection of the singular set with H_E^* is compact, and the same finite termination applies; there is no incident higher-dimensional component extension yet. Choose supports within the shell buffers. Their local finiteness makes all updates locally finite on M, and a compact test meets only finitely many of them. The summable allowances prove (7.12) for every such test. Every compact radial side block meets only finitely many shell buffers. This supplies the whole-component finite lists after the first attachment.

Counts and the quantitative transfer. Enlargement of the prescribed open sets gives $G_j' \subset G_j$. Only the finite exceptional list can contribute to the counts on a post-radial component; the analogous statement holds on buffered compact blocks at the radial step. Thus Lemma 6.3 places every new required set in both the old required set and the set H_E^* disjoint from the singular set under projection, including along attached cylinders. The finite lists and closed complements of the prescribed open sets also pass these count inclusions to all further-end base product limits.

The normal tube near H_E^* and comparison constructions of Lemmas E.2 and E.3 use these closed sets H_E^*, regular graphical convergence on product limits, and finite componentwise derivative bounds. All are now available. Numerical errors are allocated down chains through higher-dimensional components before the defining functions are constructed upward.

Apply the quantitative transfer of Lemma E.4 with the current t_r, $\Delta_r = t_{r+1} - t_r > 0$, and $e_0 = e_r = \Delta_r/800$. Its stage margins (E.3) were proved for these general adjacent parameters. The potential calculation also applies: on each finite comparison diagram, (7.10) expresses the old potential sum as finitely many smooth functions plus one spatial constant. Only finitely many potential additions near singularities occur there. Thus, with $\widetilde{\Pi}_j = \Pi_j + D_j$, the same raw form and comparison estimates control

$$\lambda_{r+1} + (1 + 3\eta_r) b_{r+1} \sum_{j \neq r} \widetilde{\Pi}_j \tag{7.13}$$

near the new required set, with the positive mass and gradient reserves proved there. The constant tail has no derivative error; potential approximation errors concern only the finite exceptional list. At the radial step these estimates are made on buffered compact blocks, as in that proof. The interface and potential smoothing can consequently retain an e_r local mass gap. This positive number is fixed for the current stage; all geometric tolerances may depend on the component.

Exact constants outside the finite exceptional lists in regularization. Restrictions, bulk inclusions, and cylinder projections preserve identically constant functions. Interface smoothing of an old identically unit warping function leaves it one. For the remaining potentials, smooth $(1 + 3\eta_r/2)\widetilde{\Pi}_j$ as in Lemmas E.6 and E.7, so that

$$(1 + \eta_r)\widetilde{\Pi}_j \leq \Pi_j^{\text{new}} \leq (1 + 2\eta_r)\widetilde{\Pi}_j. \tag{7.14}$$

On a finite exceptional list, allocate the required relative errors separately, using $p_j^* > 0$ for its jth entry. If an aggregate error is prescribed, subdivide it among that finite list. If an entry lies outside every finite exceptional list in the source diagram, set its output *exactly* to $c_{r+1}p_j^*$. This is compatible with the smoothing procedure, which reproduces constants exactly by Lemma E.6.

Before blending on an incident higher-dimensional component, enlarge its exceptional list by the union over its entire finite downward source diagram. Thus an entry on an end base's exceptional list is recorded even if it was outside the incident higher-dimensional component's own exceptional list. The remaining copies agree identically. Higher-dimensional component requests merely adjust numeric tolerances; they do not introduce new source values on end bases. This proves exact reproduction of the specified constant data on the completed diagram and precludes a cascade through infinitely many same-depth neighbors. At the radial step choose a compact buffer for each compact radial block, enlarged to contain the finitely many chart closures and shell supports used by its approximations. Only finitely many original outer balls and supports of the logarithms of the initial nonunit warping functions meet this buffer. Enlarge the exceptional list by precisely these indices. Every other entry is the same constant on every averaging neighborhood and is therefore reproduced exactly before cylinder attachment.

The restriction argument of Lemma E.7, with the first inequality in (E.3) and (7.14), gives the next function inequality. Continuous pullback and enlargement of the prescribed open sets preserve all pole-neighborhood and forced-preimage conditions.

Corrections, supersolutions, and the new warping function group. The sum in (7.10) is now smooth, bounded, and exactly compatible, as are its finite-order derivatives. On every required finite diagram, pad the old warping functions by units. In a product end over B the operator of Lemma E.8 therefore satisfies exactly

$$L_A = L_B - C\Delta_{\mathrm{normal}}, \qquad L_A = -C\Delta_{\mathrm{old}} + R_{\mathrm{old}} - V_A^* + K_A, \qquad V_A^* = \lambda_{r+1} + b_{r+1} \sum_{j \neq r} \Pi_j^{\mathrm{new}}.$$

In particular the potential formula does not depend on the base dimension or on how many unit factors were padded. The correction is zero near the new required set and produces a positive form gap on each whole component. The proof of Lemma E.8 uses finite minima of the gaps on the component's end base diagram. Its bounded solution of the elliptic equation and the compatible positive supersolutions in Lemmas E.9 and E.10 apply with the positive injectivity radius, uniform bounds on curvature and all its covariant derivatives, and polynomial weighted volume already proved on each component. Their tolerances are chosen after the operators and their individual exact solutions are fixed, and before the final upward blending pass. Thus their finite-dependency hypothesis is precisely (iii) of Lemma 7.4.

Let $v_A > 0$ be the resulting supersolution with $L_A v_A \geq 1/2$. Append the common group of $k = 399$ warping functions $f = v_A^{2/(k+1)} = v_A^{1/200}$. The scalar identity in Proposition E.1 gives

$$R_{\mathrm{new}} = R_{\mathrm{old}} - C\frac{\Delta_{\mathrm{old}} v_A}{v_A} = V_A^* - K_A + \frac{L_A v_A}{v_A} \geq V_A^* \quad \text{on } D_A.$$

Only the number of *new* factors enters the coefficient C. The old finite list may vary from component to component; unit padding has no effect. Each stage adds just this one finite group, with the same factor indices for every component index. All its warping functions are positive and bounded above and away from zero on a whole component, and exactly compatible. The initial lists and their finite propagated exceptions remain finite, so the complete old list at stage $r + 1$ is still finite on each whole component and each finite comparison diagram.

Finally the split along regular zero sets preserves compact truncations and finite incidence by Lemma A.1. Initially the radial separators make all components compact before prolongation. Compatible smooth metrics and finite relevant warping functions then give the complete component geometry by Lemma A.2. The recursion (7.9) preserves confinement in original radial blocks. All exceptional-list enlargements just described are finite, and all constants outside the exceptional lists have multiplier c_{r+1}. This completes the induction step, including the first radial step. □

7.5 The obstruction in dimension one

Lemma 7.6 (The one-dimensional obstruction)**.** *At any finite stage of the construction, the scalar inequality* (7.6) *excludes a nonempty one-dimensional component with the complete geometry and finite warping function list stipulated above.*

Proof. Such a component is complete by Lemma A.2 and has only finitely many relevant positive smooth warping functions $f_1, \dots, f_q$ on its whole base. Its arclength universal cover is $\mathbb{R}$, including when the component itself is a circle. On this line put $u_\alpha = \log f_\alpha$ and $P = \sum_{\alpha=1}^{q} u_\alpha$. The warped-product scalar formula reduces to

$$R = -2P'' - (P')^2 - \sum_{\alpha=1}^{q} (u'_\alpha)^2.$$

The required count in dimension one is zero, so the invariant holds everywhere and gives $R \geq \lambda_r > 0$. Consequently

$$-2P'' \geq \lambda_r + (P')^2, \qquad \frac{\mathrm{d}}{\mathrm{d}s} \arctan \frac{P'}{\sqrt{\lambda_r}} \leq -\frac{\sqrt{\lambda_r}}{2}.$$

Integration over a line interval longer than $2\pi/\sqrt{\lambda_r}$ contradicts the range of the arctangent. If $q = 0$, the scalar formula is simply $R = 0$. The argument takes place at one finite stage and uses an ordinary finite sum. □

8 Reconstructing the partitions on the original manifold

We prove Proposition 2.1, the partition statement used in Section 2. Keep the original metric g, its distance d_0, the centers, and the ball pairs fixed there. The finite-stage separating sets and proper collapses are supplied by Proposition 7.2. Only finite truncations of the analytic product ends enter the reconstruction. The choice of homeomorphisms below makes every change for a center label lie in its original ball of radius $2D + 2$; local finiteness of those balls then gives the limit on one copy of M.

Proof of Proposition 2.1. Choose positive numbers $\varepsilon_0, \varepsilon_1, \dots$ with $\sum_{r \geq 0} \varepsilon_r < 1$, and use ε_j as the error of Remark A.6 for the split of label j, the transition from analytic stage j to $j + 1$ in Proposition 7.2. In this proof only, X_r denotes the reconstruction after the radial label 0 and the center labels $1, \dots, r$ have been split; it is X_{r+1} in the indexing of Propositions 7.2 and A.5. Thus the passage from X_r to X_{r+1} splits the center label $i = r + 1$, with error $\varepsilon_i = \varepsilon_{r+1}$. For a finite reconstruction write

$$q_r : X_r \longrightarrow M, \qquad H_r : X_r \longrightarrow M, \qquad Q_r = q_r \circ H_r^{-1}.$$

The map q_r is the proper collapse of Proposition A.5, whereas H_r is a homeomorphism to a fixed copy of the original manifold. We maintain

$$d_0(Q_r(x), x) \leq E_r, \qquad E_r \leq \sum_{j=0}^{r} \varepsilon_j < 1. \tag{8.1}$$

Separating sets and their two closed sides will also be viewed on this fixed copy using H_r.

For the radial stage, the cut has a locally finite family of compact components. Choose pairwise disjoint, locally finite collars of these components on which the collapse varies by less than ε_0 in

each normal segment. Compactness of each component and continuity of the collapse permit the collars to be this small, independently from one component to another. Compress the finite inserted cylinders into these collars. Since their supports are disjoint, the displacement bound is ε_0, rather than a sum over components. This constructs H_0 and proves (8.1) initially. The finite length of an inserted cylinder may be arbitrarily large; its length does not impose a lower bound on the width of the collar into which it is compressed.

Fix $r \geq 0$ and put $i = r + 1$. First raise any old finite truncation levels needed for the new compatible defining functions. Denote the resulting reconstruction by $\widehat{X}_r$. The truncation level reparameterization

$$\Phi_r : \widehat{X}_r \longrightarrow X_r$$

preserves collapse and every old separating set and signed side. Indeed, on an old product end the collapse is independent of the normal coordinate, and an increasing change of that coordinate preserves its sign and its zero section. Compatible products of these changes agree at corners. Thus we may set

$$\widehat{q}_r = q_r \circ \Phi_r, \qquad \widehat{H}_r = H_r \circ \Phi_r, \qquad \widehat{q}_r \circ \widehat{H}_r^{-1} = Q_r. \tag{8.2}$$

In particular, increasing a truncation level contributes no displacement on the fixed copy.

The new separating set lies over $B_0(x_i, 2D)$. Choose its collar strip S_i in $\widehat{X}_r$ so that

$$S_i \subset \widehat{q}_r^{-1}\big(B_0(x_i, 2D + 1)\big), \tag{8.3}$$

and so that collapse oscillation along every segment used by the compression is less than the error assigned to this stage. These choices are possible because the relevant separating set components in a finite reconstruction are compact and locally finite. More specifically, their union lies in the inverse image of the compact ball $\overline{B}_0(x_i, 2D)$ under the proper finite collapse; hence only finitely many components are involved in this label stage.

Choose these strips using the collars in the proof of Proposition A.5. That construction applies after the truncation levels have been raised beyond the product thresholds: the new defining function is regular on each old end base and independent of its old signed normal coordinates. Its collar flow and the resulting compression preserve every old separating set and signed side and their intersections, including the radial ones. Decrease the componentwise strip widths to satisfy (8.3) and the prescribed collapse oscillation, as permitted by Remark A.6. Let

$$c_r : X_{r+1} \longrightarrow \widehat{X}_r \quad \text{and} \quad h_r : X_{r+1} \longrightarrow \widehat{X}_r$$

be, respectively, the new cylinder collapse and the homeomorphism that compresses those cylinders into the chosen strips. They agree off the inverse image of S_i. Figure 2 shows the two maps schematically in a normal coordinate: c_r collapses the inserted interval, while h_r compresses it into the chosen old collar. Define

$$q_{r+1} = \widehat{q}_r \circ c_r, \qquad H_{r+1} = \widehat{H}_r \circ h_r, \qquad Q_{r+1} = \widehat{q}_r \circ c_r \circ h_r^{-1} \circ \widehat{H}_r^{-1}. \tag{8.4}$$

The oscillation choice gives $d_0(Q_{r+1}(x), Q_r(x)) < \varepsilon_{r+1}$ wherever they differ, proving (8.1) at the next stage. Their difference is supported in $\widehat{H}_r(S_i)$. If $x = \widehat{H}_r(z)$ belongs to this set, then (8.2), (8.3), and (8.1) imply

$$d_0(x, x_i) \leq d_0(x, Q_r(x)) + d_0(\widehat{q}_r(z), x_i) < 1 + (2D + 1) = 2D + 2. \tag{8.5}$$

This support radius is independent of the stage.

The balls in (8.5) form a locally finite family. Consequently, on every relatively compact neighborhood in the fixed copy, the maps Q_r are exactly unchanged after finitely many stages. Their locally stationary limit q is continuous and satisfies (2.1). It is proper: the inverse image of a compact set is a closed subset of its compact closed 1-neighborhood in (M, d_0).

At each finite stage the signed sides are the closed reconstructed sides, with their intersection equal to the central separating set. The stratum-preserving identifications just constructed preserve every previously installed side and separating set exactly. Newly installed label separating sets are confined to the fixed balls in (8.5). Thus all the partition data are locally eventually unchanged. Their final closed sets satisfy (2.2) and (2.3), since these are the finite-stage forced-sign properties of Proposition 7.2, and both collapse and indices are stationary near each point. The separating set family is locally finite by the same support bound. Finally, a point on $n-1$ distinct final label separating sets would, on a neighborhood where these data have stabilized, lie on the same $n-1$ finite-stage separating sets. This contradicts Proposition 7.2. □

9 Consequences for closed manifolds and their covers

The uniform simplex metric in the definition of continuous macroscopic dimension was fixed in the introduction. We first show that the coordinates of Section 2 retain their fibers in this metric, then prove Corollary 1.3. The filling-radius and aspherical consequences follow at the end of this section.

Lemma 9.1 (Changing the simplex metric without changing fibers)**.** *Let Y be a topological space and $k \geq 0$ an integer. Suppose a family of continuous functions $t_i : Y \to [0,1]$ has locally finite supports, at least one equals 1 at each point, and at most k are strictly between 0 and 1. Their coordinate map admits an injective conversion to a simplicial complex of dimension at most k with its uniform simplex metric. The converted map is continuous, and two of its values agree if and only if all the original coordinates agree.*

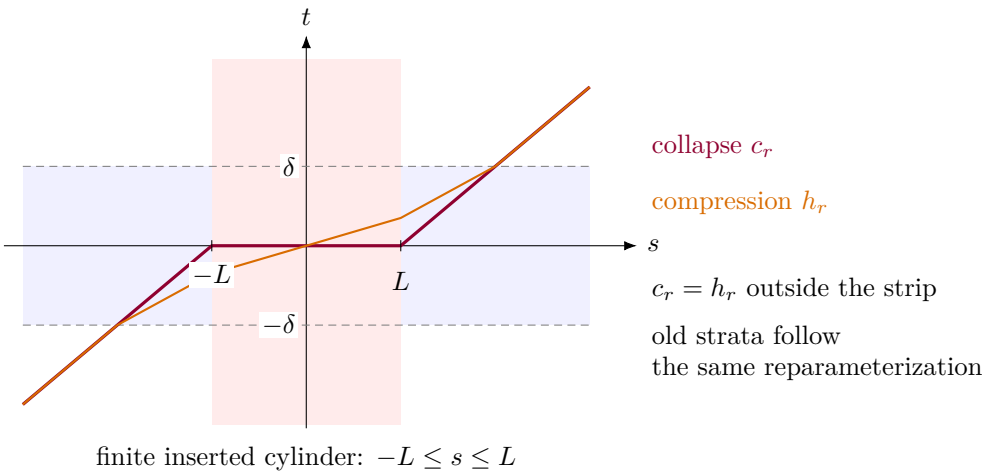


Figure 2: The two maps used in a finite reconstruction, shown schematically in local normal coordinates: s on the reconstructed domain X_{r+1} and t on the old target $\widehat{X}_r$. The target strip $|t| < \delta$ represents the chosen collar S_i. The collapse is constant on the inserted cylinder; the homeomorphism squeezes it into an arbitrarily narrow old collar. Where the maps differ, both images lie in the target strip. Infinite analytic ends are truncated before this topological operation.

Proof. Form the abstract simplicial complex of Section 2 for the indices of the family (t_i) and the present k: its simplices are those of (2.7), where S, A are disjoint finite sets of indices, $S \neq \varnothing$, and $|A| \leq k$. Realize each simplex as a standard simplex. The coordinate realization of this complex is injective, by the argument given in Section 2. Preserving barycentric coordinates therefore gives a well-defined injective conversion: two converted values agree exactly when the original coordinate vectors agree.

For $k \geq 1$, give each simplex edge length 1 and define d_K as the infimum of the total Euclidean lengths of finite piecewise straight paths in the simplices. This infimum is finite: vertices $\mathbf{1}_S$ and $\mathbf{1}_T$ are joined by adding the coordinates in $T \setminus S$ and then removing those in $S \setminus T$, always retaining a nonempty set of coordinates equal to 1. Every point is joined to a vertex in its simplex. For a vertex v, its barycentric coordinate b_v, set to zero on simplices not containing v, agrees on faces and is $\sqrt{2}$-Lipschitz on each unit-edge simplex. Indeed such a simplex is realized by the vectors $e_v/\sqrt{2}$ in a Euclidean coordinate space. Hence

$$|b_v(x) - b_v(y)| \leq \sqrt{2}\, d_K(x, y).$$

The barycentric coordinates distinguish points, so d_K is a genuine finite metric. For $k = 0$ use the discrete vertex metric.

Near a point of Y, only a finite set F of coordinates can be nonzero. The coordinate map there lands continuously in the finite cubical subcomplex with all other coordinates zero. The conversion is continuous on each of its closed simplices and hence on the finite complex. Inclusion into the full uniform realization is continuous on a simplex because distance there is at most the length of its straight segment. Finite pasting therefore proves continuity on the entire finite subcomplex, and thus near every point of Y. The same local finite argument applies for $k = 0$. □

Lemma 9.2 (Level components of a proper Morse function)**.** *Let $u : N \to \mathbb{R}$ be a proper Morse function on a connected smooth manifold without boundary. The connected components of the levels of u are the point fibers of a continuous map $q : N \to K$, where K is a locally finite simplicial graph with its uniform simplex metric.*

Proof. Choose an auxiliary smooth Riemannian metric on N to define gradients. The critical set is closed and its points are isolated. Properness therefore makes the number of critical points in any compact slab $u^{-1}([a, b])$ finite. Thus the critical values form a locally finite subset of $\mathbb{R}$, even if several critical points have the same value. Every level is compact and locally path connected: regular charts give a hypersurface, while the Morse normal form at a critical point gives the zero set of a nondegenerate quadratic form, which is a cone or an isolated point. A compact locally path connected space has finitely many components, since its components are open and form an open cover.

Let S consist of the critical values and the integers. It is locally finite. At each $a \in S$ put a vertex for every component of $u^{-1}(a)$. Between consecutive values $a < b$ of S, the flow of $\nabla u/|\nabla u|^2$ identifies $u^{-1}((a, b))$ with a regular level times (a, b). The flow exists for each compact subinterval because its trajectory stays in a compact slab. Each of the finitely many connected product components defines an open edge, parametrized by its u-value. The image of a proper map to $\mathbb{R}$ is closed; any finite endpoint of $u(N)$ is an attained extremum and hence a critical value. Thus no further finite endpoints are missing.

An edge component E has a unique attachment at each end. At its lower end, the nonempty compact connected sets

$$\overline{E \cap u^{-1}((a, a + \epsilon])}, \qquad 0 < \epsilon < b - a,$$

are nested as $\epsilon \downarrow 0$. Their intersection equals $\overline{E} \cap u^{-1}(a)$: a point of this latter set is approached by points of E whose heights tend to a, and therefore belongs to every displayed closure. The intersection is nonempty, connected and contained in $u^{-1}(a)$, hence in exactly one of that level's components. Attach the edge to its vertex. The same argument applies at b. There are finitely many edges in either adjacent slab, so every vertex has finite valence. The resulting graph is countable and locally finite. Its points correspond exactly to the connected components of all levels, using full level components as vertices when critical values coincide.

Define q by this correspondence. On an open edge it is continuous by the product description. At a vertex corresponding to a compact level component $C \subset u^{-1}(a)$, an adjacent edge not attached to C has no sequence approaching C: such a sequence would put a point of C into its endpoint intersection, contrary to its unique attachment. The finitely many other components of $u^{-1}(a)$ are disjoint compact sets. A neighborhood of C can therefore exclude them and the finitely many nonincident edges. Requiring also that $|u - a|$ be sufficiently small places its remaining points in any prescribed short tails of the finitely many incident edges. This proves continuity at the vertex and shows that its whole inverse image is C. The graph is connected, since q is a continuous surjection and N is connected.

Subdivide every edge once with a different midpoint vertex for each edge. This removes parallel edges from the abstract realization; loops do not occur because the endpoint heights differ. The result is a simplicial graph. Give each new edge the same length. Finite valence implies that its ordinary graph topology equals the topology of this uniform simplex metric: vertex neighborhoods contain short segments on each of finitely many incident edges. The subdivision and change of edge lengths preserve points and fibers. Thus q has the asserted continuity and exact fibers. □

Proof of Corollary 1.3. Put $s = \min_M \mathrm{Scal}_h > 0$ and $X = \widetilde{M}$. The lifted metric is complete. If X is compact, use a constant map to a vertex, whose only nonempty fiber has finite diameter.

For $n \geq 4$, the complete metric $s\widetilde{h}$ has scalar curvature at least 1. Apply Theorem 1.1 using the coordinates of (2.6), then Lemma 9.1. Local finiteness of their supports was proved in Section 2, and the conversion preserves their entire fibers. Scaling back gives the bound $C_n/\sqrt{s}$ in the original metric.

For $n = 2$, the Gaussian curvature of the cover is at least $s/2$. If a unit-speed minimizing geodesic had length $L > \pi\sqrt{2/s}$, take a parallel unit normal E along it and the endpoint-zero variation field $V(t) = \sin(\pi t/L)E(t)$. Its index form obeys

$$I(V,V) = \int_0^L \left(|\nabla_t V|^2 - K|V|^2\right) dt \leq \frac{\pi^2}{2L} - \frac{sL}{4} < 0,$$

contradicting minimality. Completeness supplies minimizing geodesics, so the cover has diameter at most $\pi\sqrt{2/s}$ and is compact by properness of a complete Riemannian manifold. The constant map therefore proves this case.

Finally let $n = 3$ and X be noncompact. Apply Liokumovich and Wang [26, Theorem 1.1, arXiv version 1] to the complete metric $(s/6)\widetilde{h}$, whose scalar curvature is at least 6. It gives a proper Morse function such that every connected component of every level, including critical levels, has ambient diameter less than 18π in that metric. Apply Lemma 9.2. Each whole fiber of its graph map is one such level component, so in the original lifted metric it has diameter less than $18\pi\sqrt{6/s}$. This is a continuous map to a one-dimensional uniform simplicial complex, as required. □

All map statements here are continuous; changing the simplex metric does not supply a Lipschitz bound.

9.1 Filling radius and the aspherical obstruction

Proof of Corollary 1.2. Gromov's comparison, in the form recorded by Nabutovsky, Rotman, and Sabourau [32, Definition 6.1 and equation (6.1)], uses the same continuous-map and entire-fiber width convention as above. A target of dimension at most $n-2$ is also admissible for UW_{n-1}. Since M is closed, the comparison and the scaled form of Theorem 1.1 give

$$\mathrm{FillRad}(M,g) \le \tfrac{1}{2}\,\mathrm{UW}_{n-1}(M,g) \le \tfrac{1}{2}\,\mathrm{UW}_{n-2}(M,g) \le \frac{C_n}{2\sigma}.$$

□

Proof of Corollary 1.4. It suffices to consider a connected manifold M. Suppose that it carries a metric h of positive scalar curvature, put $s = \min_M \mathrm{Scal}_h > 0$, and let X be its universal cover with the lifted metric $\widetilde{h}$, which is complete. Since M is aspherical, X is contractible, and the deck group acts on X properly, isometrically and with compact quotient. Hence X is uniformly contractible [18, §$2\frac{1}{2}$]. Gromov's Proposition-Example in the same section shows that, for every $0 < \varepsilon < \infty$, no continuous map from a complete uniformly contractible Riemannian n-manifold to a polyhedron of dimension less than n has all its fibers of diameter at most ε. On the other hand, Theorem 1.1 applied to the complete metric $s\widetilde{h}$, whose scalar curvature is at least 1, gives a continuous map from X to a simplicial complex of dimension at most $n-2<n$ whose fibers have $\widetilde{h}$-diameter at most $C_n/\sqrt{s}$. This contradiction proves the corollary. □

A Components, finite truncation levels, and the induction invariant

This appendix gives the geometry and bookkeeping for a finite number of partitions. The minimization, enlargement of the prescribed open sets, and regularization arguments in the following appendices construct the regular cuts to which this geometry applies. The passage to countably many labels is proved in Proposition 7.2.

The component diagrams, finite stage constants, analytic invariant, and radial initialization are stated in Section 6. We prove here their geometric properties and the compatibility of finite reconstructions.

A.1 Product ends and splitting along regular zero sets

Lemma A.1 (Compatible splitting and finite truncations)**.** *Let a post-radial diagram satisfy Definition 6.1, and let g_E be compatible smooth defining functions with 0 a regular value on every component and end base. Its split along regular zero sets retains $L_E = g_E^{-1}(0)$ and both closed sides with their full cylinders, using normal attachment collars that are products in the old tails. The connected components, with their appended indices, form a new component diagram.*

It has finite end base and incident higher-dimensional component incidence, properly embedded closed tails, and compact remainders. Each component admits a compact truncation which is a smooth manifold with corners and is a strong deformation retract of the full component. Truncations can be chosen simultaneously on every finite downward diagram, with their boundary faces equal to the corresponding truncated components. If thresholds are in the product range of g_E, the old retractions preserve its zero set and both closed sides. These conclusions also hold at the first radial stage provided that, before attachment, its boundary and closed-side components are compact and form a locally finite family in X.

Proof. First work in the finite downward diagram of one old component. Raise all entry thresholds until every overlap is in a commuting quadrant, and until all the functions in this diagram are products there. For a normal coordinate in a one-factor product end t_i choose a larger truncation level h_i. On its tail use $t_i \mapsto \min(t_i, h_i)$ and the homotopy $t_i \mapsto (1-s)t_i + s\min(t_i, h_i)$; use the identity elsewhere. The formulas agree continuously at $t_i = h_i$. In an overlap the other normal coordinates are unchanged and the formulas commute. One may therefore perform these clamps in any order. Their joint image is compact: remove the far product tails with one normal factor, and then inductively truncate each of the finitely many end base tails. This gives a finite union of compact products. Its boundary strata have precisely the claimed product descriptions. Each clamp fixes the cutting function, since that function is an end base pullback throughout the moving region.

The regular zero set in this compact manifold with corners is transverse to every corner stratum. Indeed on a far old boundary face the derivative in end base directions is the derivative of the regular end base function. Consequently the compact zero set and the two compact closed sides are manifolds with corners. Each has finitely many components: every point has a connected neighborhood in the relevant manifold with corners, and compactness supplies a finite such cover. Clamping shows that a full zero or side component has a nonempty connected intersection with the truncation. Conversely a connection in the full set clamps to a connection in the truncation. Thus no additional components or connections occur only at infinity.

On an old tail the whole operation is literally the same operation on its end base times the old normal intervals. In particular the cylinder attached to an end base zero set is the same subset in every order of passing to that end base. For the new sign-to-star relation the tail is exactly the newly attached cylinder. These observations prove all product and commutation rules, including the unique occurrence for a prescribed index obtained by replacing stars with signs: the outgoing product of a connected end base is connected and belongs to a unique component of that index. A new component has finitely many end bases by the compact truncation just constructed. An end base has at most one incident component for each index obtained by replacing stars with signs, of which there are finitely many. Applying these assertions through the finite index depth proves finiteness in both directions.

Fresh closed cylinder tails are proper by the attachment topology. The continuous collapse of a new space to its old parent sends a compact set to a compact set. On an inherited tail, old properness then bounds its old parameters in a compact subset of the old tail; the fresh cylinder length is also bounded on the given compact set. The remaining zero and side conditions are closed. Thus the inverse image of that compact set in the inherited closed tail is compact. Equivalently, it is contained in the union of closed old pieces and bounded fresh cylinders over compact zero pieces. This also shows that the complement of all new far tails is compact. Clamp the fresh cylinder coordinate as well to obtain the asserted new compact truncations and retractions.

At the radial stage there are initially no old tails. The hypotheses give compact boundary and side pieces, with finite incidence on each compact piece, and the fresh cylinders start the preceding induction. All diagrams remain countable, because all manifolds involved are second countable and the radial family is locally finite.

For later metric uses, normal collars can be obtained from any compatible smooth parent metric. In an old product tail, both the regular defining function and the metric are pulled back from the end base, so the unit normal and its normal geodesics split off the old normal coordinates. On a compact truncation of the zero set choose a positive tubular radius by finitely many normal-coordinate charts. Induction through its finite end base diagram extends this radius to a positive uniform radius on that parent component: any failure would be in a product over an end base where the radius was already chosen. Thus these collars are products on all old ends; at the radial step the same choice is made separately on each incident compact block. □

A.2 Completeness, growth, and product limits

Lemma A.2 (Metric geometry of a component)**.** *Let a post-radial component diagram carry compatible smooth base metrics and a finite list of compatible smooth positive warping functions, with the exact product identifications of Definition 6.1. Each connected component E is complete and has a positive injectivity radius and uniform bounds on curvature and every covariant derivative. Every warping function and all its covariant derivatives are bounded on E, and every warping function has a positive lower bound. For ω_E the product of the warping functions there are constants $A_E < \infty$ and an integer $d_E \le \dim E$ such that*

$$\int_{B_E(x,R)} \omega_E \,\mathrm{d}\,\mathrm{vol}_E \le A_E(1+R)^{d_E} \qquad (x \in E,\ R \ge 0). \tag{A.1}$$

The constant is uniform over centers in this component. The same assertions hold on $E \times \mathbb{R}^a$, with the product metric, pulled-back warping functions, and growth exponent at most $d_E + a$, for every finite a.

Proof. Induct upward from components with no end bases, which are compact. A full product tail with one normal factor over a complete end base has no finite-distance exit through its end base coordinate. More precisely, a finite-length curve in that product whose normal coordinate stays above the entry level has Cauchy end base and normal coordinates. Completeness of the end base and properness of the closed tail keep its limit in the product. It follows that, after fixing a closed-tail entry level T_i, the function

$$\rho_i = \max(t_i - T_i, 0) \quad \text{on that tail}, \qquad \rho_i = 0 \quad \text{outside it}$$

is continuous and globally 1-Lipschitz. The only exit is through $t_i = T_i$, and its differential has norm at most one wherever defined. A Cauchy sequence has all the finitely many ρ_i bounded. The compact-truncation assertion of Lemma A.1 puts such a sequence in a compact set. Hence the component is complete.

Recursively subdividing each coordinate into a bounded interval and its far range gives a finite cover of E by compact sets and sectors

$$K_B \times [T_1, \infty) \times \cdots \times [T_d, \infty), \tag{A.2}$$

where K_B is a compact subset of a further end base and the metric and warping functions are products. The recursion terminates because an end base decreases base dimension. On such a sector each displayed normal coordinate is also the corresponding coordinate in a one-factor product end on E, by commutation. In an R-ball its range therefore has length at most $2R$, by the Lipschitz functions ρ_i (enlarge by a fixed bounded interval at an entry if necessary). The sector's weighted volume in that ball is bounded by its compact base volume times a constant multiple of $(1+R)^d$. Summing the finitely many bounds proves Equation (A.1), for all centers, including centers outside the sector. Notice that $d \le \dim E$.

Bounds on smooth data follow from the same finite sector decomposition: on compact pieces they are finite, and on a product they are precisely end base bounds. For a positive injectivity radius and uniform bounds on curvature and all its covariant derivatives one can equivalently use uniform charts. Any failure of a uniform chart radius or of a fixed-order chart bound has an escaping sequence; a subsequence lies in a far product over a compact part of a further end base, as detailed in Lemma A.3. Fixed-radius neighborhoods there are product neighborhoods with uniform smooth chart bounds and a positive injectivity radius. Compact remaining regions have the same properties

by a finite chart cover. This proves a positive injectivity radius and uniform bounds on curvature and all its covariant derivatives and all warping function bounds. The product assertions follow either by the same proof or from $B_{E\times\mathbb{R}^a}((x,z),R)\subset B_E(x,R)\times B_{\mathbb{R}^a}(z,R)$. □

Lemma A.3 (Escaping sequences and product regions)**.** *Let E carry the compatible geometry of Lemma A.2. If a sequence escapes all compact subsets of E, a subsequence has coordinates $(x_\nu, t_{1,\nu},\dots,t_{d,\nu})$ in one further-end base product, where $d\geq 1$, x_ν converges in a compact subset of the end base B, and every $t_{i,\nu}\to\infty$. After translating these normal coordinates, the product charts contain each fixed metric neighborhood of the sequence for all sufficiently large ν. They exhaust $B\times\mathbb{R}^d$ locally. All compatible smooth data equal the corresponding product data on these regions.*

Proof. Choose a sector from the finite cover in Equation (A.2) containing a subsequence. Pass to a subsequence so each of its normal coordinates either converges finitely or tends to infinity. The compact base coordinate also converges. Absorb the bounded normal coordinates into the base of the intermediate end base specified by the remaining, unbounded coordinates. Commutation identifies this with the corresponding full product occurrence. At least one unbounded coordinate remains, since otherwise the sequence lies in a compact truncation. A path of length at most R changes each normal coordinate by at most R. It cannot escape through an end base coordinate, by completeness of the end base as in Lemma A.2. Thus a fixed-radius neighborhood lies in the product once all its normal coordinates exceed the entry levels by more than R. Translating yields the stated exhaustion. This argument uses only the completeness part of Lemma A.2, which was established before its conclusion of a positive injectivity radius and uniform bounds on curvature and all its covariant derivatives. □

At the radial split each bulk piece and each boundary piece is compact before cylinder attachment. The smooth data have finite derivative bounds there, the positive warping functions have positive minima, and the cylinders freeze those data. This initializes Lemma A.2. For a locally finite initial warping function list, a compact radial block and its compact buffer meet only finitely many nonunit factors; the same argument applies after omitting the unit factors. Polynomial growth is a conclusion for a single post-radial component.

A.3 Compatible choices

Lemma A.4 (Compatible choices and finite tolerance allocation)**.** *On a post-radial component diagram, fix a finite derivative order and, on each component, positive error tolerances for any finite collection of smooth data or approximations. A datum on an end base may additionally be required to meet the tolerances of all its incident higher-dimensional components, including through chains. There are strictly positive end base tolerances meeting all these requests and compatible smooth cutoffs and approximations with exact products beyond finite thresholds, provided the individual local approximations and their required convergence to the end base data are available. In particular there exist compatible smooth positive functions arbitrarily small in any prescribed finite derivative norm on each component. Also, if compatible closed sets F_E and compatible open sets O_E satisfy $F_E\subset O_E$ on every component, then* $\operatorname{dist}_E(F_E, E\setminus O_E)>0$ *on each fixed component whenever both sets are nonempty.*

Proof. There are only finitely many requests on a particular end base, because there are only finitely many chains through higher-dimensional components. Starting with the given requests, propagate stricter tolerances down these chains, allowing for the finitely many product, derivative, and cutoff constants. Take finite minima. This finishes all tolerance choices before construction begins. A

countable collection of components causes no problem, since each one has finitely many end bases and incident higher-dimensional components and the depth is finite.

For the construction on one component, enumerate its product tails with one normal factor. Choose smooth cutoffs $f_i(t_i)$ rising from zero to one sufficiently far out, with bounded derivatives through the chosen order, and extend them by zero off their tails. Write u_0 for an individual approximation and u_i for the already chosen end base copy. The weights

$$w_0 = \prod_i (1 - f_i), \qquad w_i = f_i \prod_{a<i} (1 - f_a), \qquad w_0 + \sum_i w_i = 1 \tag{A.3}$$

produce $w_0 u_0 + \sum_i w_i u_i$. Put the transitions beyond all relevant intersection thresholds. On an overlap the participating end base copies then agree by their own product rules. In particular, wherever some $f_i = 1$, every nonzero term is the same end base copy. The result is an exact copy there. Leibniz' rule bounds the derivative error by a finite multiple of the preassigned errors. A perturbation on the remaining compact region changes none of these end identities. This proves the conditional approximation assertion. It does not assert convergence for a temporary nonproduct object; whenever such an object is used below, its convergence is proved separately.

For positive small functions start with positive constants on the compact deepest components, take positive small individual choices on compact parts, and use the positive weights above. Decrease the preassigned tolerances to cover all incident higher-dimensional component derivative requests. This gives strict positivity at every point and any stipulated finite norm bounds, including the C^2 bounds used for interface widths. For the separation assertion, a sequence of distances tending to zero either stays in a compact region, where closedness and inclusion in the open set give a contradiction, or escapes. In the latter case Lemma A.3 supplies a common product region over a compact part of a deeper end base. The two sequences project to the same limit point, lying both in F_B and in $B \setminus O_B$, a contradiction. □

A.4 Finite reconstruction

Proposition A.5 (Reconstruction by finite truncation)**.** *Consider finitely many compatible splits along regular zero sets of X as in Lemma A.1. Assume that the first split has locally finite compact radial boundary and side components, and that at each later label split i the defining functions are positive on the inverse image of A_i and negative on that of B_i, in open regions of those signs. After any $r \geq 1$ of these splits, the diagram has a reconstruction X_r, obtained by finite collar truncations and gluing at the corresponding boundary faces, with the following properties:*

(i) X_r is a smooth manifold homeomorphic to X. Its r labelled boundaries are closed smooth hypersurfaces with simultaneous transverse intersection charts. Each boundary component is compact, the component families are locally finite, and the closed sign/zero-set pieces are exactly the compact cornered truncations of the analytic components, by components.

(ii) The component maps glue to a proper continuous map $q_r : X_r \to X$. There is a homeomorphism $H_r : X_r \to X$ such that

$$d_0\big(H_r(z), q_r(z)\big) \leq r \qquad (z \in X_r). \tag{A.4}$$

(iii) Truncation levels can be enlarged to exceed any already specified compatible product thresholds on the full analytic components. A truncation level is constant for all occurrences of a normal coordinate belonging to one connected global boundary component. Such enlargement has an identification with the previous reconstruction which preserves q_r, every old boundary, and every old side index.

(iv) Once label i has been processed, $q_r^{-1}(A_i)$ remains in its open positive side and $q_r^{-1}(B_i)$ remains in its open negative side at every subsequent stage. In particular its boundary misses $q_r^{-1}(A_i \cup B_i)$.

Proof. We prove the assertions in their order of dependence. At stage zero use $X_0 = X$ and $q_0 = H_0 = \mathrm{Id}$. At the radial split, the assumed compact pieces give the first finite reconstruction by inserting cylinders of finite length on the two sides of each boundary component. The properness and homeomorphism argument below applies to this locally finite compact family. Its compact pieces also initialize the analytic diagram by Lemma A.1.

Suppose the reconstruction through stage r has been made. Fix all product thresholds for the next cutting functions and their collar metrics in the *full analytic coordinates* first. Consider a connected component W of an existing global boundary. A component with the corresponding star entry has a nonempty connected truncation, by the commuting retractions of Lemma A.1; it therefore lies in a single such W. The same assignment holds at all incidences through further end bases, since the corresponding truncations glue along those boundary faces. Compactness of W and local finiteness of the reconstructed pieces imply that only finitely many such pieces meet W. Thus there are only finitely many threshold requests for its normal coordinate. Choose one truncation level exceeding all of them, with a positive buffer. Do this for each existing boundary component and coordinate.

This is a direct choice from previously fixed finite lists. Increasing a truncation level cannot introduce a new analytic threshold: the functions and the incidence diagram were fixed on the full infinite tails. On a tail where topology and q_r are already products, choose an increasing smooth reparameterization from the enlarged finite normal interval to the old one, equal to the identity before the old product range and sending the new endpoint to the old endpoint. Choose the same reparameterization at all occurrences of W, on both sides of its central zero set. It preserves those sides and the central zero set. At corners the separately labelled normal reparameterizations commute. Their product gives an identification $\Phi_r : \widehat{X}_r \to X_r$ preserving every boundary and sign index. The glued position maps satisfy

$$\widehat{q}_r = q_r \circ \Phi_r, \qquad \widehat{H}_r = H_r \circ \Phi_r, \qquad d_0(\widehat{H}_r, \widehat{q}_r) \le r,$$

because each position map is independent of the moved normal variables. Properness is preserved by this homeomorphism. Thus truncation level enlargement introduces no error or change of position data. For the rest of the step, relabel these enlarged representatives as X_r, q_r, H_r.

The compatible cutting functions now glue to a single smooth function on this enlarged X_r. Near any old boundary stratum the function is a pullback from the zero-set piece, independent of all its signed normal coordinates. Its zero set L is therefore regular and transverse to every old boundary stratum. More explicitly, in a chart $(b, s_1, \dots, s_d)$ at an old stratum, the old boundaries are $s_j = 0$ and the new function is $g_B(b)$. The end base hypothesis says $\mathrm{d}g_B \neq 0$ at its zeros. Hence the old normal covectors $\mathrm{d}s_1, \dots, \mathrm{d}s_d$ together with $\mathrm{d}g_B$ are independent. This also covers a stratum involving the radial boundary. At this point L has been formed without choosing a new truncation level. For a label split its forced signs give

$$q_r(L) \subset X \setminus (A_i \cup B_i).$$

In the present one-ball-per-label setup, $L \subset q_r^{-1}(B_{d_0}(x_i, 2D))$. The latter set has compact closure because q_r is proper and the original closed outer ball is compact. Since L is closed in X_r, the whole label boundary L is compact. In particular each component is compact, and there are only finitely many components. More generally, a closed regular hypersurface with compact components has a

locally finite component family: a regular hypersurface chart meets only its own component, and compact subsets of the hypersurface have finite such covers. The first radial boundary need not be compact as a whole; its stated locally finite compact-component hypothesis gives exactly what is needed for the same construction.

Only now assign a finite new truncation level to each component of L. Cut along L, attach a cylinder on each side, and identify the two outer cylinder boundaries as the new central zero set. Near an old boundary this is the product of the operation on its end base selected by a star with the old signed normal coordinate. The same statement holds at multiple old intersections. Thus old boundaries extend smoothly through the new cylinders, and all boundaries have transverse intersection charts. The resulting pieces are exactly the truncations of the newly appended indices: the old clamps preserve L and its signs and extend across fresh cylinders keeping their new coordinate fixed; the fresh-coordinate clamp commutes with all of them. In particular every full new component retracts onto one nonempty connected piece in the reconstruction. This verifies the component assertion needed to assign its coordinate at the next stage.

Let $c : X_{r+1} \to X_r$ collapse the inserted cylinders to L and be the identity on the bulk. This map is proper. A compact subset of X_r meets only finitely many new boundary components; its inverse image is contained in its compact bulk copy together with finitely many finite cylinders over compact subsets of these compact components. The glued map is $q_{r+1} = q_r \circ c$, and so is proper. Every old compact boundary meets finitely many new components, and its transverse intersection with them is compact. Its extension adds only finitely many compact finite cylinders; the old boundary therefore remains compact. Local finiteness of all pieces follows locally from the finite collection of transverse boundary charts, together with the compact truncation descriptions. This proves (i) and the properness assertion.

We next choose collars that preserve all old strata. The compatible component metrics glue on the enlarged reconstruction: near an old zero-set stratum they have the form $h_B + \sum_j \mathrm{d}s_j^2$. On a neighborhood of L, write u for the glued cutting function and use the transverse field

$$V = \frac{\operatorname{grad} u}{|\operatorname{grad} u|^2}, \qquad V(u) = 1.$$

In each old product chart, $u = g_B(b)$ and this field has no old normal component. Thus it is tangent to every old boundary meeting that chart and is the pullback of the corresponding end base field. Equivalently, one can patch such local fields with a compatible partition of unity: $V(u) = 1$ and tangency are linear conditions, so they survive patching. The flow supplies product collars at old intersections. It preserves every old boundary and each side, by uniqueness of the flow and continuity from the identity. This explains why a transverse ordinary collar chosen without the product condition would not suffice here.

Choose pairwise disjoint locally finite strips around the compact components of L using this flow. A positive strip width exists on each component by compactness; it may be decreased to keep strips disjoint. For each component, continuity of q_r and compactness permit the width to be so small that q_r-images of any two points on the same flow segment have d_0-distance less than one. A monotone normal reparameterization compresses the two finite inserted intervals and the old strip into that old strip, fixing its outer boundary. It produces a homeomorphism $h : X_{r+1} \to X_r$, equal to the identity off the strips, with

$$d_0\big(q_r(h(z)), q_r(c(z))\big) < 1.$$

Different strip widths and cylinder lengths are allowed on different components. Local finiteness makes these definitions a global homeomorphism. Set $H_{r+1} = H_r \circ h$. The triangle inequality and

Equation (A.4) give the new bound $r+1$. The compression follows the stratum-preserving collar, so it also preserves every old boundary and its two side indices. The old signs are pulled back under c, including across the cylinders, and the newly imposed signs hold on the prescribed inverse images. Pullbacks of their open sides stay open. This proves (iv) and finishes the induction. □

Remark A.6 (Prescribed reconstruction errors). The number 1 in the last collar comparison can be replaced at stage r by any prescribed $\varepsilon_r > 0$. Each compact boundary component has its strip chosen separately, with the same error bound ε_r on every component. Disjoint strips give a supremum bound, rather than a sum over components. The resulting error after finitely many stages is at most $\sum_{j<r} \varepsilon_j$. This freedom concerns the original position map and is independent of the analytic lengths of the inserted finite collars. The spatial localization and limit of such reconstructions are treated in the main reconstruction argument.

A.5 Initialization

Proof of Lemma 6.4. The decreasing solution of $2|y'| = .5y^2 + \pi_0$, from $+\infty$ to $-\infty$, has finite crossing length

$$\int_{-\infty}^{\infty} \frac{2\,\mathrm{d}y}{.5y^2 + \pi_0} = 2\pi\sqrt{2/\pi_0} = T_0.$$

Extend it by $+\infty$ before the crossing and by $-\infty$ after it. This is continuous as an extended-real-valued function, and all its finite truncations are locally Lipschitz. Composing the transition, begun at distance 2 from A_i, with the 1-Lipschitz function $d_h(A_i, \cdot)$ gives

$$2|\operatorname{grad}_h \mu_i| \le .5\mu_i^2 + \pi_0 \le a_0\mu_i^2 + \pi_0$$

almost everywhere on the finite-value locus. Take

$$U_i^+ = \{d_h(A_i, \cdot) < 1\}, \qquad U_i^- = \{d_h(A_i, \cdot) > T_0 + 3\}.$$

Their closures lie strictly inside the respective pole interiors; they contain A_i and B_i. The requested finite separations are available from Theorem 5.1 after N_*, and hence T_0, are fixed. Their size does not change the original radius D.

Assume first that X is noncompact, fix $o \in X$, and put $d(x) = d_h(o, x)$. Choose a countable locally finite cover by relatively compact coordinate neighborhoods on which the metric oscillation is small, and a subordinate smooth partition of unity $\{\chi_j\}_{j\ge1}$ whose supports are compact in those neighborhoods. Convolution in a slightly larger coordinate neighborhood gives smooth d_j with

$$|d_j - d| < \epsilon_j, \qquad |\mathrm{d}d_j|_h \le \tfrac{3}{2} \quad \text{near } \operatorname{supp}\chi_j.$$

The derivative estimate follows from the local Lipschitz bound for d and the small metric oscillation; the convolution radius can then be reduced to make its value error arbitrarily small. Let $C_j = \sup|\mathrm{d}\chi_j|_h < \infty$, and choose $0 < \epsilon_j < \min\{1/2, 2^{-j-2}/(1 + C_j)\}$. The locally finite sum $\rho = 2 + \sum_j \chi_j d_j$ is smooth and $|\rho - (2 + d)| < 1/2$. Moreover, since $\sum_j \mathrm{d}\chi_j = 0$,

$$\mathrm{d}\rho = \sum_j \chi_j \mathrm{d}d_j + \sum_j (d_j - d)\mathrm{d}\chi_j, \qquad |\mathrm{d}\rho|_h \le \tfrac{3}{2} + \sum_j 2^{-j-2} < 2.$$

Completeness makes d proper, so ρ is proper and bounded below. Thus a smooth proper radius with bounded gradient has been constructed using only local smoothness and completeness.

Choose successive plateau intervals of the values of ρ, with alternating signs and prescribed open subintervals of positive length strictly inside each plateau. Separate successive plateaus by transition

intervals of width greater than $2T_0$. On such an interval, a copy of the crossing y, parametrized by $(\rho - a)/L$ with $L > 2$ and with sign reversed on alternate intervals, obeys the same function inequality because $|\mathrm{d}\rho|_h/L \leq 1$. Include an initial plateau extending below the minimum of ρ. The closures of these subintervals have positive value margins to both ends of their plateaus, except at the harmless lower end of the initial plateau. Since ρ is smooth, Sard's theorem supplies regular values in those margins [36, Chapter 2, Remark 6.5]. The sublevel domain at the initial plateau, and the domains between two such regular levels at later plateaus, are compact smooth barriers enclosing the corresponding closures of the prescribed open sets and lying in the same pole interiors. These barriers and the transition blocks form locally finite families, since their value intervals tend to infinity. Connectedness and properness imply that the image of ρ contains all sufficiently large values, so the separating prescribed open sets are nonempty.

A zero set respecting the signs misses every separating prescribed open set. On any connected boundary component, the continuous function ρ therefore takes values in one bounded intervening block. The component is closed in X, and properness of ρ makes it compact. For a fixed sign, successive prescribed open sets of the opposite sign bound the ρ-values of every connected component of the closed side; the central component is bounded by the first opposite separator. These closed-side components are compact as well. In a compact block, regularity gives a finite cover by charts in which the boundary is a single sheet and each closed side is a half-space or the full chart. Only finitely many components can meet that block. The blocks are locally finite, which proves the required local finiteness of boundary and side components.

If X is compact, set the radial function identically $+\infty$, with $U^{+}_{\mathrm{rad}} = X$ and $U^{-}_{\mathrm{rad}} = \varnothing$. The constant positive defining function realizes its empty boundary; the single nonempty closed side is X, which is compact. All radial function and confinement assertions then hold directly.

Finally, the initial scalar requirement holds everywhere, since

$$\lambda_0 + b_0 \sum_{j \in J_0} \Pi_j = \frac{1}{4} + 1.12 N_* \frac{1}{10 N_*} = .362 < .4 \leq R_X.$$

The function, prescribed open set, and forced-preimage conditions have already been verified. This proves the invariant at stage zero. □

The geometric lemmas require compatible regular defining functions. The minimizing boundaries used to obtain them may initially be singular. The remaining appendices first construct these minimizing partitions, enlarge the remaining prescribed open sets until the required sets H^*_E avoid the singular sets, and then produce the regular compatible functions to which Lemma A.1 applies. The curvature restriction to the required sets is respected precisely by Lemma 6.3.

B Weighted minimizing cuts and the forms on retained sides

Prescribed-curvature perimeter minimization and convergence of its boundaries have classical precedents in Massari and Massari–Pepe [28, 29]. For scalar-curvature applications, prescribed-mean-curvature minimization is the μ-bubble method described by Gromov [18, §5$\frac{5}{6}$]. We prove here the weighted, varying-metric and product-end versions required by the cutting construction, including the forms on both retained sides.

Fix a finite stage r and the label j_0 to be processed. The component and function hypotheses are those of Definitions 6.1 and 6.2. For the formulation with finite truncations in this appendix, only finitely many labels and finitely many nonunit old warping functions occur in each incident diagram under consideration. The processed potential has a positive lower bound

$$\Pi_{j_0} \geq \pi_{\mathrm{cut}} > 0.$$

The constants satisfy $a_r \le .63$, $b_r \ge 1.07$, and $\lambda_r > \lambda_{r+1}$. The extension to a countable collection of labels is proved in the main text; every argument below uses only the current prescribed weighted mean curvature and the indicated finite lists.

Throughout, E denotes a parent component of base dimension $m \ge 2$. After the radial split it has the compact truncations, finite incidence, and exact product ends of Definition 6.1. At the initial split we instead work on the complete original base, and use the locally finite compact radial barriers of Lemma 6.4.

On a finite incident diagram, insert unit warping functions where necessary so that the old warping function lists agree. Their metric is

$$\widehat{g} = g + \sum_{i=1}^{q} f_i^2 \, \mathrm{d}\theta_i^2, \qquad \omega = \prod_{i=1}^{q} f_i, \qquad \mathrm{d}\mu = \omega \, \mathrm{d}\,\mathrm{vol}_g \,.$$

Empty products are one. Fiber volume constants are suppressed throughout. Adding a unit warping function changes neither ω nor any formula below. At the initial radial stage the list need only be finite on each compact region; the compatible local expressions then agree on overlaps. All minimizing sets, BV compactness, tangent cones, and singular-dimension bounds concern the m-dimensional base, regardless of the length of the warping function list. Write $R = \mathrm{Scal}_{\widehat{g}}$ and $\mathrm{div}_\omega Y = \omega^{-1} \mathrm{div}_g(\omega Y)$. For a set A of locally finite perimeter define its weighted perimeter by

$$P_\omega(A; U) = \int_{U \cap \partial^* A} \omega \, \mathrm{d}\mathcal{H}_g^{m-1}.$$

The reduced-boundary representation is the framework developed by De Giorgi [11, 12]; the current formulation and slicing originate in Federer and Fleming [15]. We use the precise representation and slicing statements in Simon [36, Chapter 3, Section 4, and Chapter 6, Section 4]. All set equalities and inclusions involving measurable Caccioppoli sets are understood up to μ-null sets. The Caccioppoli set representative used for geometry will be specified below. Constants in local estimates may depend on a compact region; after the radial split they can be chosen uniformly on each fixed component, without any uniformity between components. The parent scalar bound enters only at the end of this appendix, and only on the required region D_E of Equation (6.5).

B.1 Smooth finite prescribed weighted mean curvatures and local minimizers

Lemma B.1 (Realizing the processed function)**.** *The processed function μ_{j_0} admits a smooth finite compatible prescribed weighted mean curvature p_0 and a number $c_* > 0$ such that every $p = p_0 + c$, $|c| < c_*$, satisfies*

$$2|\operatorname{grad} p| \le .67p^2 + 1.02\Pi_{j_0}. \tag{B.1}$$

There are compatible smooth barrier domains D^+, D^- whose interiors contain the corresponding closures of the prescribed open sets and whose closures lie in the respective pole interiors. Each admits a smooth vector field $Y^\pm$ with $|Y^\pm| \le 1$, equal to the outward unit normal on its boundary, such that, for all these shifts,

$$p \ge 1 + \mathrm{div}_\omega Y^+ \quad \text{on } D^+, \qquad -p \ge 1 + \mathrm{div}_\omega Y^- \quad \text{on } D^-. \tag{B.2}$$

After the radial split all this data has bounded derivatives on each component. At the initial radial split the barriers are the separate relatively compact domains in the radial regions where the function is infinite, and the choices are locally finite with locally prescribed bounds.

Proof. The nesting of the closures of the prescribed open sets inside the pole interiors and the compatible approximation construction of Lemma A.4 give smooth separating cutoff functions. A level which is regular on every component gives the required domains: there are only countably many components, so Sard's theorem permits a common level. Their far collars are exact copies of the end base domains. On a fixed component, compact parts and the finite end base diagram give a positive tubular radius and a positive injectivity radius and uniform bounds on curvature and all its covariant derivatives for the resulting boundary. Extend its outward normal in a smaller tube and multiply by a cutoff of absolute value at most one. This supplies $Y^{\pm}$ and a bound for its weighted divergence. The radial case uses the relatively compact pole barriers individually.

Fix the barriers and their vector fields first. By Lemma A.4, choose a compatible positive smooth function h with sufficiently small value and first derivative, and put $M = h^{-1}$. Require

$$M \geq 3 + |\operatorname{div}_\omega Y^{\pm}| \quad \text{on } D^{\pm}, \qquad 2|\operatorname{grad} h| \leq a_r.$$

Only finitely many incident higher-dimensional component requests reach a given component; at the radial step the requests are locally finite on compact sets. The choices therefore have the stated compatibility. The derivative inequality implies $2|\operatorname{grad} M| \leq a_r M^2$. The clipped function

$$z = \max(-M, \min(\mu_{j_0}, M))$$

is finite and locally Lipschitz. At almost every point its derivative is that of one active branch, and consequently

$$2|\operatorname{grad} z| \leq a_r z^2 + \Pi_{j_0}.$$

This assertion also holds near the original poles, since the clipping levels are locally bounded and finite truncations of μ_{j_0} are locally Lipschitz. On $D^{\pm}$ one has $z = \pm M$.

Here is the quantitative smoothing fact being used. If a locally Lipschitz function z obeys $|\operatorname{grad} z| \leq F$ almost everywhere for a continuous positive F, then, for arbitrary positive continuous error targets, local convolution in sufficiently small metric charts followed by a partition of unity gives a smooth z_1 with prescribed value error and $|\operatorname{grad} z_1| \leq F$ plus the prescribed error. Indeed convolution averages the gradient inequality, the oscillation of F and the metric on its support tends to zero, and differentiating the partition of unity produces only the value errors times its fixed derivatives. A locally finite cover permits the errors to be assigned summably at each point. For compatible smoothing, start on deepest end bases. On overlaps in far collars the clipped functions and all end base copies agree. Blend with the cutoffs in Lemma A.4; the same value-error estimate pays the additional cutoff derivatives. Preallocate smaller errors on end bases to meet their finitely many incident higher-dimensional component requests. Thus the gradient estimate is always compared with the single continuous function $(a_r z^2 + \Pi_{j_0})/2$, rather than with unrelated bounds for the pieces.

Choose the value error at most $\min\{1/4, 10^{-3}\sqrt{\Pi_{j_0}}\}$ and the gradient error so that

$$2|\operatorname{grad} p_0| \leq a_r z^2 + 1.005\Pi_{j_0}.$$

These are positive continuous error targets. For $e = p_0 - z$,

$$.65(z+e)^2 - .63z^2 \geq -20.475e^2,$$

so $a_r \leq .63$ gives

$$2|\operatorname{grad} p_0| \leq .65p_0^2 + 1.01\Pi_{j_0}.$$

On $D^{\pm}$ the value error and the earlier choice of M give $\pm p_0 \geq 11/4 + |\operatorname{div}_\omega Y^{\pm}|$. Finally set

$$c_* = \min\{1/2, .01\sqrt{\pi_{\mathrm{cut}}}\}.$$

The elementary identity

$$.67(y+c)^2 - .65y^2 \geq -21.775c^2 \qquad (y, c \in \mathbb{R})$$

and $\Pi_{j_0} \geq \pi_{\mathrm{cut}}$ prove (B.1) for every $|c| < c_*$; the barrier margin also persists. Thus the domains, clipping height, smoothing, and open shift interval have all been fixed before selecting a minimizing Caccioppoli set. On a post-radial component every smooth datum just constructed has bounded derivatives, by its compact parts and exact products. At the radial step the construction gives the corresponding bounds on each compact region, which is all that is used. □

Definition B.2 (Local minimizer)**.** For smooth p a Caccioppoli set A is a local minimizer if for every relatively compact open $U \subset E$ and every locally finite-perimeter set B with $A \triangle B \Subset U$,

$$P_\omega(A; U) - \int_{A \cap U} p \, \mathrm{d}\mu \leq P_\omega(B; U) - \int_{B \cap U} p \, \mathrm{d}\mu. \tag{B.3}$$

The notation $A \triangle B \Subset U$ means that the discrepancy is contained, modulo null sets, in a compact subset of U.

Lemma B.3 (Existence and strict local compactness)**.** *On any smooth Riemannian manifold, a smooth positive weight and a smooth locally bounded prescribed weighted mean curvature admit a Caccioppoli set satisfying* (B.3)*. Suppose on exhausting coordinate regions that metrics and positive weights converge smoothly and prescribed weighted mean curvatures converge locally uniformly. Any sequence of corresponding local minimizers has a subsequence whose indicators converge in L^1_{loc} to a local minimizer of the limit data. Along that subsequence the weighted perimeter measures converge weakly locally, as do the unweighted perimeter measures and their associated unoriented perimeter varifolds. The assertion includes smooth pointed rescalings when the rescaled data have the stated convergence.*

Proof. We first establish the comparison formulas that will show both minimality of a limiting Caccioppoli set and absence of perimeter loss. We then construct exhaustion minimizers and apply these formulas to locally convergent sequences.

Comparison formulas. Local L^1 compactness under local perimeter bounds is Simon [36, Chapter 2, Theorem 2.6]. To pass from its Euclidean variation to the present perimeter, fix a relatively compact coordinate chart and write

$$\Phi_{g,\omega}(x, \xi) = \omega(x)\sqrt{\det g(x)}\sqrt{g^{ij}(x)\xi_i\xi_j}.$$

The weighted variation of a BV function w is $\int \Phi_{g,\omega}(x, \sigma_w)\, d|Dw|$, where $Dw = \sigma_w |Dw|$ is its Euclidean distributional derivative. For $w = \mathbf{1}_A$ this is $P_\omega(A; \cdot)$; the sign of σ_w does not matter. On this chart $\Phi_{g,\omega}$ is uniformly comparable to $|\xi|$, so the imported compactness theorem applies. Lower semicontinuity for the weighted perimeter follows from the dual formula

$$P_\omega(A; U) = \sup_{\substack{X \in C^1_c(U; TE) \\ |X|_g \leq 1}} \int_U \mathbf{1}_A \operatorname{div}_g(\omega X)\, d \operatorname{vol}_g :$$

each fixed test integral is continuous under local L^1 convergence. If $(g_i, \omega_i) \to (g, \omega)$ smoothly, uniform convergence on the compact unit cotangent bundle gives numbers $\epsilon_i \to 0$ with

$$(1 - \epsilon_i)\Phi_{g,\omega}(x, \xi) \le \Phi_{g_i,\omega_i}(x, \xi) \le (1 + \epsilon_i)\Phi_{g,\omega}(x, \xi).$$

These inequalities hold for every covector ξ and hence for the variation of every Caccioppoli set. They transfer the same lower semicontinuity to the varying metrics and weights.

We also use strict local smoothing in this weighted variation. In the mollification argument of Simon [36, Chapter 2, Lemma 2.5], replace the Euclidean norm by $\Phi_{g,\omega}(x, \cdot)$. Convexity gives the upper bound for the convolved derivative, and, on a fixed compact chart,

$$\Phi_{g,\omega}(x, \xi) \le (1 + o(1))\Phi_{g,\omega}(y, \xi) \quad \text{when } |x - y| \le \varepsilon \longrightarrow 0,$$

uniformly in ξ. Integrating against a compactly supported continuous nonnegative cutoff therefore gives the upper limit bound; the dual formula gives the lower limit bound. Thus mollification converges locally in L^1 and in weighted variation measure. The further BV inputs are the perimeter lattice inequality and the trace and gluing formulas, both across a fixed smooth hypersurface and across almost every admissible level of a Lipschitz slicing function. The lattice inequality is

$$P_\omega(A \cap B; U) + P_\omega(A \cup B; U) \le P_\omega(A; U) + P_\omega(B; U). \tag{B.4}$$

For smooth functions u, v, the gradients of $\min(u, v)$ and $\max(u, v)$ are the two original gradients, possibly interchanged, almost everywhere. Strictly approximate the two characteristic functions, apply this identity, and use lower semicontinuity on the minimum and maximum. This gives (B.4), with its smooth metric and weight.

For a fixed smooth comparison hypersurface, flatten a buffered collar to $Q \times (-a, a)$, with $T = Q \times \{0\}$. A BV function w on the plus half-collar must have finite variation on smaller collars reaching T, as do restrictions of our ambient finite-perimeter Caccioppoli sets. For $K \Subset K' \Subset Q$ and $a_0 < a$, mollification and the fundamental theorem of calculus give

$$\int_K |w(y, t) - w(y, s)| \, \mathrm{d}y \le |D_t w|\big(K' \times (s, t)\big), \qquad 0 < s < t < a_0, \tag{B.5}$$

on one full-measure set of heights. To pass to the limit, choose mollifications with summable local L^1 errors on a countable exhaustion of buffered boxes. Fubini gives slice convergence on a common full-measure set; omit also the countably many heights charged by $|D_t w|$. The enlarged convolution strips then give the displayed bound.

Since $|D_t w|(K' \times (0, \delta)) \to 0$ as $\delta \downarrow 0$, the good slices are Cauchy in $L^1(K)$. Their limits agree on overlapping boxes and define $\operatorname{Tr}_+ w \in L^1_{\mathrm{loc}}(T)$. Reflection gives $\operatorname{Tr}_- w$. For a Caccioppoli set these traces take values in $\{0, 1\}$ almost everywhere, by subsequential pointwise convergence of its characteristic slices.

For a smooth vector field supported tangentially inside the collar and away from its outer end, apply the distributional derivative identity after multiplying the field by a cutoff that rises from zero to one between heights ε and 2ε. Its derivative has integral one, so uniform L^1 convergence of the good slices to the trace gives the boundary term as $\varepsilon \downarrow 0$. The other terms converge by finite variation and integrability. Undoing the coordinate change incorporates the smooth volume density and gives

$$\int_{U_\pm} w \operatorname{div}_g X \, \mathrm{d}\operatorname{vol}_g = -\int_{U_\pm} \langle X, \mathrm{d}D_g w \rangle + \int_T \operatorname{Tr}_\pm w \, \langle X, \nu_{\mathrm{out},\pm} \rangle \, \mathrm{d}\mathcal{H}^{m-1}_g.$$

Here $D_g w$ is the intrinsic distributional gradient, defined by this identity without the boundary term for tests in the open side. The signs come from the outward normals $-\partial_t$ and ∂_t in the flattened Euclidean calculation.

Orient ν_T from minus to plus. Adding these formulas for a function u on the minus side and v on the plus side shows that their gluing has gradient measure

$$(D_g u)\lfloor U_- + (D_g v)\lfloor U_+ + (\operatorname{Tr}_+ v - \operatorname{Tr}_- u)\nu_T \, \mathcal{H}_g^{m-1}\lfloor T.$$

The three measures are concentrated on the disjoint sets U_-, U_+, and T, so their variations add. Multiplication by the smooth positive weight gives the interface cost

$$\int_T \omega \, |\operatorname{Tr}_+ v - \operatorname{Tr}_- u| \, \mathrm{d}\mathcal{H}_g^{m-1}.$$

Thus for Caccioppoli sets the additional weighted perimeter is the weighted area of trace disagreement. The weight multiplies the variation measure; it is not differentiated.

We will also glue across almost every level of a locally Lipschitz slicing function f, whose level sets need not be smooth. For this distinct use, the reduced-boundary representation identifies a Caccioppoli set locally with its integer-multiplicity integration current and identifies boundary mass with perimeter [36, Chapter 3, Section 4, Equation (4.1) and Theorem 4.3]. In a locally oriented chart the integration current of the set has locally finite mass and locally finite boundary mass. Lipschitz coarea and the current slicing and restriction formulas therefore apply [36, Chapter 3, Section 2, Equations (2.8)–(2.9); Chapter 6, Section 4, Lemmas 4.1 and 4.5 and Definition 4.4]. In particular, writing J for either integration current of the set, for almost every level τ one has

$$\langle J, f, \tau \rangle = \partial(J\lfloor\{f < \tau\}) - (\partial J)\lfloor\{f < \tau\}.$$

Apply this identity to both Caccioppoli sets, excluding also the null set of levels charged by their local volume or perimeter measures. For a top-dimensional integration current of the set the slice has multiplicity one on the sliced Caccioppoli set and zero elsewhere. Indeed the slicing definition restricts its set indicator to the noncritical part of the level; coarea shows that critical points and any changes of the ambient almost-everywhere representative affect almost no level in positive hypersurface measure. The two Caccioppoli set slices have the same orientation induced by f in the locally oriented chart, so their difference has multiplicity equal to the difference of the two sliced indicators. The boundary of their gluing has the two restricted boundary currents and the difference of their level slices as its interface term. Its extra weighted mass is bounded by the weighted hypersurface measure of the set where the sliced indicator values differ. This is the almost-everywhere interface-cost bound used in the annular comparisons, distance-ball comparisons, and level-set deletions below. A countable collection of Caccioppoli sets, including a converging sequence and its limit, can use one common set of admissible levels by intersecting their full-measure sets. If the set indicators converge locally in L^1, coarea on a fixed buffered slicing annulus makes the integrated sliced disagreement tend to zero. Pass to a subsequence for which these integrals are summable. Fubini then makes the disagreement tend to zero at almost every level, still within the common admissible set. This supplies a fixed gluing level with vanishing interface cost in the compactness argument below. All orientations in this argument are local.

Exhaustion minimizers and local compactness. Choose a smooth relatively compact exhaustion $U_k \Subset E$. Minimize over Caccioppoli sets empty outside $\overline{U}_k$, taking the total perimeter, including the interface on ∂U_k, and the prescribed weighted mean curvature integral inside U_k. The empty Caccioppoli set is admissible. Because the prescribed weighted mean curvature is bounded on $\overline{U}_k$, an energy bound gives a perimeter bound. The class is closed under L^1 convergence, the volume term is continuous, and perimeter is lower semicontinuous. The direct method thus gives a minimizer, locally absolute minimizing in U_k.

The same local bound applies to arbitrary local minimizers. In a coordinate ball B_t compactly contained in their comparison domain, comparison with the Caccioppoli set cleared inside B_t gives, for almost every t,

$$P_\omega(A; B_t) \leq \int_{\partial B_t} \omega \, \mathrm{d}\mathcal{H}_g^{m-1} + \int_{B_t} |p| \, \mathrm{d}\mu.$$

Choose t in a fixed annulus around the smaller region to be bounded. Coarea bounds the first term for a suitable slice. Smoothly convergent metrics and positive weights and locally uniformly bounded prescribed weighted mean curvatures make this a uniform local bound. Diagonal BV compactness therefore produces a locally L^1 convergent subsequence, both for the exhaustion minimizers and for any sequence in the statement.

Minimality and perimeter convergence. The gluing formulas now apply to the locally convergent subsequence. Choose nested relatively compact regions $U \Subset V$ with a Lipschitz foliation of the intervening annulus. The integral over slicing parameters of the trace discrepancy between A_i and A tends to zero by L^1 convergence and coarea. One can therefore choose a slicing hypersurface between U and V whose discrepancy tends to zero along a subsequence. Glue A inside this surface to A_i outside. Minimality of A_i bounds its interior perimeter from above by that of A, the vanishing trace cost, and a vanishing difference of volume terms. The last difference vanishes by the convergence of the data and indicators. First use regions whose limiting perimeter gives zero mass to their boundaries; then shrink the annulus. Together with lower semicontinuity this proves weak local convergence of the perimeter measures. Replacing the interior copy of A by any compact competitor B gives the same upper comparison with B, since $B = A$ on the gluing annulus. Lower semicontinuity on the left proves (B.3) for A. Smooth metric and weight changes make only errors tending uniformly to zero on each fixed region, so the same argument proves the convergent-data assertion.

Changing metrics and varifold convergence. For the varifold assertion, account explicitly for the changing metric. In a coordinate chart let $D_0\chi_{A_i}$ be the Euclidean distributional derivative and let (g_i) denote the positive metric matrices. The vector measures

$$\mathbf{v}_i = \sqrt{\det g_i}\, g_i^{-1/2} D_0\chi_{A_i}$$

have Euclidean total variation equal to the g_i-perimeter measures. The matrices and their derivatives converge smoothly, so distributional convergence and the local mass bounds give $\mathbf{v}_i \rightharpoonup \mathbf{v}$. The perimeter conclusion gives $|\mathbf{v}_i| \rightharpoonup |\mathbf{v}|$ as well.

This strict convergence also controls the unit polar vectors. To see this, on a compact region approximate the limiting polar vector in $L^2(|\mathbf{v}|)$ by a continuous bounded vector field e. Expand

$$\int |\mathbf{v}_i/|\mathbf{v}_i| - e|^2 \, \mathrm{d}|\mathbf{v}_i| = \int (1 + |e|^2) \, \mathrm{d}|\mathbf{v}_i| - 2 \int e \cdot \mathrm{d}\mathbf{v}_i.$$

Weak convergence gives its limit, and then the approximation makes that limit arbitrarily small. Uniform continuity consequently gives convergence for every continuous compactly supported integrand of position and polar vector. The unoriented g_i-orthogonal tangent hyperplane is a continuous function of this vector and g_i. This proves varifold convergence. Multiplication or division by the smooth positive weight on a fixed compact region gives the asserted weighted and unweighted versions. □

B.2 Barriers, ordering, and exact product limits

Lemma B.4 (Barriers hold for every minimizer)**.** *For any prescribed weighted mean curvature $p = p_0 + c$, $|c| < c_*$, and the corresponding fixed barriers in Lemma B.1, every local minimizer*

contains D^+ and excludes D^- modulo null sets. Its boundary support misses a neighborhood of each current closure of the prescribed open set. After the radial split this neighborhood can be chosen with a positive uniform radius on each component. At the radial split the assertion is local on the corresponding compact radial blocks.

Proof. We first prove the positive assertion. Put $D = D^+$, $Y = Y^+$ and $G = D \setminus A$. For almost every distance radius R about a fixed point, $U = B_R$ is a finite-perimeter set, its boundary has zero mass for the relevant perimeter measures, and BV traces are available. Write

$$V(R) = \mu(G \cap B_R), \qquad S_R = \int_{\partial^* B_R} \operatorname{Tr}(\chi_G)\omega \, \mathrm{d}\mathcal{H}^{m-1}.$$

Distance coarea, with $|\operatorname{grad} d| = 1$ almost everywhere off the center, gives local absolute continuity and $V'(R) = S_R$ for almost every R. The weighted divergence theorem, $Y = \nu_D$ on ∂D, and $|Y| \le 1$ give

$$\begin{aligned} P_\omega(D; U) - P_\omega(A \cap D; U) \le &\int_{G \cap U} \operatorname{div}_\omega Y \, \mathrm{d}\mu \\ &- \int_{\partial^* U} \operatorname{Tr}(\chi_G)\, Y \cdot \nu_U \, \omega \, \mathrm{d}\mathcal{H}^{m-1}. \end{aligned}$$

By (B.4) this bounds the interior perimeter increase on replacing A by $A \cup D$ in U. Glue back to A outside U; the extra perimeter is at most S_R. This is a compact comparison in a slightly larger relatively compact open set. By (B.2) and minimality,

$$V(R) \le 2S_R = 2V'(R). \tag{B.6}$$

The weighted ball volume is at most polynomial by Lemma A.2, since ω is bounded on the component. If $V(R_0) > 0$, (B.6) would imply $V(R) \ge V(R_0)e^{(R-R_0)/2}$, a contradiction. Thus G is null. For a relatively compact radial barrier, simply add all of D; no slicing terms occur, and the same comparison gives $\mu(G) = 0$ directly. The complement of A minimizes the same functional with prescribed weighted mean curvature $-p$, proving the negative assertion. The domains contain the closures of the prescribed open sets in their interiors; the componentwise positive separation follows from exact products and the compact-part property. A Caccioppoli set whose characteristic function is one or zero on an open set has no perimeter support there. □

Proposition B.5 (Generic uniqueness on all required products)**.** *After the radial split there is a single $c \in (-c_*, c_*)$ such that $p_0 + c$ has exactly one local minimizer modulo null sets on every component E and every $E \times \mathbb{R}^d$, for each integer $d \ge 0$. On a product the metric is the specified product metric, and the weight and prescribed weighted mean curvature pull back along its projection to E. The unique product Caccioppoli set is exactly $A_E \times \mathbb{R}^d$ modulo null sets, where A_E is the unique Caccioppoli set on E.*

Proof. On any one such space, consider *arbitrary* minimizers A for $p_0 + c_1$ and B for $p_0 + c_2$, with $c_1 < c_2$. Put $G = A \setminus B$, $\delta = c_2 - c_1$, and use the same distance-ball slicing as in the preceding proof. Replace A by $A \cap B$ inside the ball and B by $A \cup B$ there, gluing each to its own exterior Caccioppoli set. The two gluing costs total at most $2S_R$. Their interior perimeter increments have nonpositive sum by (B.4); their prescribed weighted mean curvature increments sum to precisely $-\delta\mu(G \cap B_R)$. Adding the two minimality inequalities gives

$$\delta V(R) \le 2V'(R). \tag{B.7}$$

Polynomial weighted growth, also on $E \times \mathbb{R}^d$, forces $V = 0$. Thus *every* minimizer at the smaller shift is contained in *every* minimizer at the larger shift.

Choose countably many nonnegative compactly supported continuous test functions φ_a which separate locally integrable functions. For each a and shift c, let

$$I_a(c) = \left(\inf_{A \text{ minimizes at } c} \int \varphi_a \chi_A \, \mathrm{d}\mu, \quad \sup_{A \text{ minimizes at } c} \int \varphi_a \chi_A \, \mathrm{d}\mu \right).$$

Existence is supplied by Lemma B.3. The preceding ordering says that the nonempty open intervals $I_a(c)$ for distinct c are disjoint. Each contains a rational number, so only countably many can be nonempty. If minimizers differ, some φ_a distinguishes them and $I_a(c)$ is nonempty. There are therefore at most countably many exceptional shifts on this space. The component collection and the set of integers d are countable, so their exceptional shifts have a countable union. Choose c in the remaining part of $(-c_*, c_*)$.

A Euclidean translation of the unique minimizer on $E \times \mathbb{R}^d$ is again a minimizer. Uniqueness implies translation invariance of its indicator. It suffices to impose the countable dense group of rational translations: continuity of translations in L^1_{loc} then gives all translations, so its Euclidean distributional derivatives vanish. The indicator is therefore $\chi_F(x)$ for a base Caccioppoli set F. BV slicing shows that F has locally finite perimeter. If a compact base competitor improved its energy by $\eta > 0$, use that competitor over $(-L, L)^d$ and the Caccioppoli set F outside, retaining the original Caccioppoli set outside the compact base discrepancy. The energy gain is $\eta(2L)^d$. The interfaces at the Euclidean box boundary cost at most $O(L^{d-1})$ times the weighted volume of that compact discrepancy. For large L this contradicts minimality. Thus F minimizes on E, and base uniqueness identifies it with A_E. The case $d = 0$ needs no product argument. □

At the radial split set $c = 0$ and choose any minimizer supplied by Lemma B.3. Its compact barriers confine its boundary to the locally finite compact transition shells. The support of the selected minimizer's perimeter measure will later be replaced by a compatible regular boundary near the required region.

B.3 The precise local regularity input

Theorem B.6 (Local GMT properties used here)**.** *Let A be a Caccioppoli set of locally finite perimeter satisfying* (B.3) *on a smooth Riemannian m-manifold, $m \geq 2$, with a smooth positive weight ω and smooth prescribed weighted mean curvature p. Put $S = \operatorname{supp} |D\chi_A|$. Then $S = S_{\mathrm{reg}} \cup \Sigma$, where S_{reg} is a smooth embedded hypersurface, Σ is relatively closed, and*

$$\dim_{\mathcal{H}} \Sigma \leq m - 8. \tag{B.8}$$

A negative dimension bound means that Σ is empty. The outward measure-theoretic normal extends smoothly on S_{reg}. With $B(u, v) = \langle \nabla_u \nu, v \rangle$, its base mean curvature satisfies

$$H_g = p - \partial_\nu \log \omega. \tag{B.9}$$

The unweighted multiplicity-one perimeter varifold has locally bounded generalized mean curvature. Its density exists at every support point; monotonicity gives local uniform area upper bounds and positive area lower bounds at support points. Under the convergences of Lemma B.3, densities are upper semicontinuous at converging points, provided the local generalized-mean-curvature bounds are uniform. The supports converge locally, and on every compact subset of the regular limiting hypersurface convergence is smooth and single-sheeted when the data converge smoothly. On compact subsets of the interiors of the limiting Caccioppoli sets there is eventually no boundary.

Every tangent at a point of S is a multiplicity-one locally perimeter-minimizing boundary cone in the Euclidean tangent space. A flat tangent implies regularity. A tangent at a singular point is therefore nonflat, and its subspace of translation-invariance directions has dimension at most $m-8$.

Proof by first variation, strict blowups, and regularity. Put $d = m-1$. The classical inputs are the bounded-mean-curvature monotonicity theorem and Allard's graphical theorem [36, Chapter 4, Theorem 3.17; Chapter 5, hypotheses 5.1 and Theorem 5.2], and the singular-dimension theorem for *exactly* minimizing boundary currents [36, Chapter 7, Theorem 5.8]. The minimizing-boundary dimension bound follows from the work of Simons and Federer [14, 38]. The last theorem will be applied only to constant-metric blowup limits. We verify the transfer to the present smooth weighted problem, including the singular-set compactness needed for the Hausdorff-dimension conclusion.

First variation and local area bounds. First variation is available before any regularity conclusion. The flow of a compactly supported smooth field Z is an admissible comparison in (B.3), so

$$\int_{\partial^* A} \omega(\operatorname{div}_S Z + \langle \operatorname{grad} \log \omega, Z\rangle)\, \mathrm{d}\mathcal{H}^d_g = \int_{\partial^* A} p\omega\langle Z, \nu\rangle\, \mathrm{d}\mathcal{H}^d_g.$$

Here tangent planes and ν are the measure-theoretic ones. Substituting $Z = W/\omega$ cancels the tangential weight derivatives and gives

$$\int_{\partial^* A} \operatorname{div}_S W\, \mathrm{d}\mathcal{H}^d_g = \int_{\partial^* A} (p - \partial_\nu \log \omega)\langle W, \nu\rangle\, \mathrm{d}\mathcal{H}^d_g.$$

Thus the unweighted perimeter varifold has intrinsic generalized mean-curvature vector $-(p - \partial_\nu \log\omega)\nu$, locally bounded by $|p| + |\operatorname{grad}\log\omega|$. Its multiplicity is one almost everywhere.

On a relatively compact coordinate patch, first pass to a smaller patch and extend its metric to a closed manifold. Explicitly, place its coordinate closure inside a coordinate ball of a flat torus and blend the given positive definite metric with the flat metric outside the smaller patch; the convex combination stays positive definite. The resulting metric is smooth and the ambient manifold is compact, so the smooth isometric embedding theorem applies [33, Theorem 2]. The preceding intrinsic first variation then gives Euclidean generalized mean curvature with the additional bounded term $\operatorname{tr}_{TS} \mathrm{II}$, where II is the second fundamental form of this fixed embedding. Indeed decompose a Euclidean test field into its ambient-manifold tangent and normal parts and differentiate that decomposition along TS. Its normal contribution is exactly the trace of II. Consequently Euclidean monotonicity applies locally. Intrinsic and embedded density limits agree, since the embedding is isometric to first order. Monotonicity gives density existence and upper semicontinuity, including at nonreduced support points; their densities are at least one, by approximation by reduced points. Together with the clearing-ball upper bound from Lemma B.3, it gives uniform upper and positive lower area bounds on compact patches.

A common ambient model for varying metrics. We record carefully the uniform version when the metric changes. Suppose $g_i \to g$ smoothly on a fixed patch, and all other data have the local bounds in the theorem. Any claimed failure of regular-point persistence or support convergence has a subsequence on which the failure persists. Pass further so that, on a slightly larger compact patch,

$$\|g_i - g\|_{C^i} \le 2^{-i^2}.$$

Set $t_i = 2^{-i}$ and take disjoint smooth bump functions $\beta_i(t)$, equal to one on a neighborhood of t_i, supported in intervals of width comparable to t_i, and satisfying $|\beta_i^{(b)}| \le C_b t_i^{-b}$. After omitting finitely many terms,

$$g_t = g + \sum_i \beta_i(t)(g_i - g), \qquad G = \mathrm{d}t^2 + g_t$$

is a smooth positive metric on a product neighborhood, with $g_0 = g$ and $g_{t_i} = g_i$. For each pair of fixed orders a, b, the summands after a spatial and b temporal derivatives are bounded by $C_b 2^{ib-i^2}$ for all sufficiently large i. These bounds, divided by any fixed power of t_i, still tend to zero. Thus all mixed derivatives extend at $t = 0$. The plateaus also give $\partial_t g_t = 0$ at t_i, so the slices carrying g_i are totally geodesic in G.

Extend this *one* smooth metric G across the edge of a smaller product patch to a closed manifold and isometrically embed it once. The images of the slices $t = t_i$ converge smoothly to the slice $t = 0$, and the second fundamental form of this single embedding is bounded on their common compact neighborhood. The preceding first-variation calculation therefore gives one Euclidean mean-curvature bound for all the embedded perimeter varifolds. Strict perimeter convergence from Lemma B.3 becomes varifold convergence in this same Euclidean space.

Exact minimizing tangent cones. Next consider a blowup of one Caccioppoli set at $x \in S$. In smooth coordinates, at scale ρ, the normalized data are

$$A_{x,\rho} = \rho^{-1}(A - x), \quad g_{x,\rho}(y) = g(x + \rho y), \quad \omega_{x,\rho}(y) = \omega(x + \rho y), \quad p_{x,\rho}(y) = \rho p(x + \rho y).$$

The metric here has been divided by the square of the physical scale: perimeter scales by ρ^d, volume by ρ^{d+1}, and this accounts for the factor ρ in the prescribed weighted mean curvature. On every fixed region the data converge smoothly to $g(x), \omega(x), 0$. The strict compactness lemma therefore identifies every subsequential tangent varifold with the unit-multiplicity perimeter varifold of a Caccioppoli set minimizing perimeter in the constant metric $g(x)$. A constant weight can be divided out. After one linear isometry this is an ordinary Euclidean perimeter-minimizing Caccioppoli set. It is also a minimizing boundary current. Indeed, orient this Euclidean chart. A compactly supported integral current filling a boundary competitor has top dimension and is given by an integer-valued BV function q. The competitor is the boundary of $\chi_A + q$. Clipping that integer-valued function to $[0, 1]$ leaves it unchanged outside the compact discrepancy and gives a characteristic function without increasing variation. Any current competitor with smaller mass would therefore give a smaller-perimeter set competitor. This verifies the precise minimizing-boundary hypothesis of Simon [36, Chapter 7, Theorem 5.8]. All orientations in this argument are local.

The support lower bound makes this limit nonzero at the origin. Monotonicity based at x makes its mass ratio constant on all radii; the equality case, or the stationary radial identity in Simon [36, Chapter 4, Equation (3.6)], makes it conical. Equivalently, after a fixed local isometric embedding, dilations of the embedded ambient manifold converge smoothly to its tangent space, its second fundamental form is multiplied by ρ, and the same blowups give this cone. Its multiplicity is one as a perimeter varifold, although its density at the vertex need not be one.

Regularity and persistence under convergence. We now apply Allard's graphical regularity theorem [2]. In its Euclidean form used here, fix an exponent $q > d$. For a rectifiable d-varifold of multiplicity at least one, on a ball centered at a support point and contained in the first-variation domain, sufficiently small excess of the normalized mass above one and sufficiently small

$$\rho^{1-d/q} \left(\int_{B_\rho} |\mathbf{H}|^q \, \mathrm{d}\mu_V \right)^{1/q}$$

give a single $C^{1,1-d/q}$ graph on a smaller ball [36, Chapter 5, Theorem 5.2]. A unit flat tangent supplies the small mass excess. The bounded mean curvature supplies the other small quantity, since it is $O(\rho)$ under the local area bound. Thus a flat tangent makes the original point regular as a $C^{1,\alpha}$ graph. The graph satisfies the weak prescribed-curvature equation $H_g = p - \partial_\nu \log \omega$, whose coefficients and right-hand side are smooth functions of position, graph height, and graph gradient.

In a buffered chart write it as

$$\partial_i A^i(x,u,Du) = B(x,u,Du).$$

The flux A is the slope gradient of the smooth positive weighted area integrand, so A_p is symmetric and uniformly positive on the bounded slope range. Testing translated differences with $\eta^2\delta_h u$, and moving the difference quotient of the bounded function B onto the test, gives by ellipticity and Cauchy's inequality

$$\int \eta^2 |D\delta_h u|^2 \le C_\eta$$

independently of small h. Here the difference of A is $a_h D\delta_h u + c_h\delta_h u + b_h$, with a_h uniformly elliptic and all three coefficients uniformly bounded, and $\delta_h u$ is bounded because u is Lipschitz. Hence $u \in H^2_{\mathrm{loc}}$. For $v = \partial_k u$, weak differentiation now gives

$$\partial_i(a^{ij}\partial_j v) = \partial_i F^i, \qquad a^{ij} = A^i_{p_j}, \qquad F^i = \delta_{ik}B - A^i_{x_k} - A^i_u v.$$

Both a and F are $C^{0,\alpha}$ since the graph is $C^{1,\alpha}$. To apply the a priori divergence Schauder estimate, mollify a, F and solve the uniformly strictly coercive Dirichlet problems on a smaller ball with trace v. Their solutions v_ϵ satisfy, by testing the difference of the equations,

$$\lambda\|D(v_\epsilon - v)\|_2 \le \|F_\epsilon - F\|_2 + \|a_\epsilon - a\|_\infty \|Dv\|_2,$$

where $\lambda > 0$ is a common ellipticity constant, so $v_\epsilon \to v$ in H^1. Smooth-coefficient interior regularity, the local L^2–L^∞ estimate, and then divergence Schauder give interior $C^{1,\alpha}$ bounds depending only on the common ellipticity and Hölder bounds [37, Lecture 6, Theorem 1 and Corollary 1; Lecture 18, Theorem 2; Lecture 12, Theorem 1]. Passing to the limit gives $v \in C^{1,\alpha}_{\mathrm{loc}}$ and hence $u \in C^{2,\alpha}_{\mathrm{loc}}$. Repeating this argument for differentiated equations gives smoothness, with uniform higher estimates for families having common $C^{1,\alpha}$ graph bounds and uniformly smooth data.

Likewise, near a regular point of a limiting Caccioppoli set choose a sufficiently small ball with mass ratio as close to one as needed, with strict slack in the Allard threshold, and with small scaled mean-curvature bound. Choose its radius also so that its sphere has zero limiting perimeter measure; almost every sufficiently small radius has this property. For smoothly convergent data, use the single ambient embedding construction above. Weak perimeter-measure convergence now passes the mass bound on this ball to the approximating Caccioppoli sets. To center at a nearby approximating support point, decrease the radius by a fixed arbitrarily small factor: the recentered ball lies in the chosen ball for all large indices, and the strict slack pays the change in the normalizing power of the radius. Allard therefore gives one graph in each smaller ball for all sufficiently large indices. The graph equation and smooth data convergence then give smooth convergence on smaller compact patches. This also proves

$$x_i \in \Sigma_i, \quad x_i \longrightarrow x \quad \Longrightarrow \quad x \in \Sigma$$

for interior limits: a regular limiting point has just been excluded, and a point outside the limiting support is excluded by the uniform support lower bound and perimeter convergence. The same lower bound and weak convergence prove local support convergence; they also exclude stray support in compact subsets of Caccioppoli set interiors. The regular set is open in the support, so its complement Σ is relatively closed.

Transfer of the singular-dimension bound. For the singular-dimension transfer we use the content argument in Simon [36, Chapter 1, Theorem 3.26(2), and Appendix A, proof of Theorem A.4, step (1)]. Its hypotheses in the present application have all been established: on every compact coordinate patch and at every center, each vanishing scale sequence has a strict tangent subsequence; these tangents are homogeneous minimizing boundaries; singular sets commute exactly with translations

and dilations; and, on compact regions of every such convergent sequence, singular sets eventually lie in every open neighborhood of the limiting singular set. For the last assertion, a contrary sequence of points has a convergent subsequence, contradicting regular-point persistence or support convergence. The data on the rescaled regions have uniform local derivative bounds, with derivatives of order j acquiring factors ρ^j and prescribed weighted mean curvature derivatives acquiring ρ^{j+1}. Thus the compactness and regularity argument applies to all these sequences. We use only the cited content-transfer step, and spell it out next.

Suppose $m \geq 8$ and, for some $s > m - 8$, the singular set in an interior region had positive $\mathcal{H}^s$ measure. Let $\mathcal{H}^s_\infty$ denote Hausdorff content, defined by covering with balls and summing their radii to the power s; normalization constants are immaterial. The Hausdorff-content density selection in the proof of Theorem A.4 provides an interior $x \in \Sigma$, scales $\rho_i \downarrow 0$, and $c > 0$ such that

$$\rho_i^{-s} \mathcal{H}^s_\infty(\Sigma \cap \overline{B}_{\rho_i}(x)) \geq c.$$

Pass to a strict tangent limit C along these scales. The exact minimizing-boundary theorem gives $\mathcal{H}^s(\Sigma_C) = 0$, because its hypersurface dimension is $d = m - 1$ and its bound is $d - 7 = m - 8$. Cover $\Sigma_C \cap \overline{B}_{3/2}$ by finitely many open balls whose sum of sth powers of radii is less than $c/2$; the cover can first be chosen countable with this bound and then reduced to a finite subcover by compactness. Singular-set persistence puts every $\rho_i^{-1}(\Sigma - x) \cap \overline{B}_1$ in that same union for large i: otherwise a convergent sequence in its compact complement would give a singular limiting point outside the cover. Scaling of Hausdorff content now contradicts the preceding lower bound. This proves $\mathcal{H}^s(\Sigma) = 0$ for every $s > m - 8$.

If $m \leq 7$, the exact minimizing-boundary theorem says that every tangent cone is regular at its vertex and hence is a hyperplane. Flat-tangent regularity then makes Σ empty. Finally a nonflat minimizing cone is singular at its vertex, and translation invariance carries that vertex to each point of its subspace of translation-invariance directions. The subspace of translation-invariance directions is therefore contained in its singular set, giving the asserted bound on its dimension. The preceding monotonicity argument, applied in the common embeddings for convergent data, also gives upper semicontinuity of densities at moving points by first passing to the limit at a fixed radius and then letting the radius decrease to zero. The smooth graph's coorientation comes from the Caccioppoli set, and its first-variation equation is (B.9). □

Corollary B.7 (Boundary supports converge at component ends)**.** *Fix the common shift of Proposition B.5, and let A_E, A_B be its unique Caccioppoli sets on a component and a specified end base. On translated regions escaping in a product collar over a compact part of B, the Caccioppoli sets A_E converge to $A_B \times \mathbb{R}^d$ in L^1_{loc}, their perimeter measures and varifolds converge, and their regular boundaries converge smoothly on compact sets off the limiting singular set. Their boundary supports converge locally as well. The same conclusions hold after passage to any deeper product furnished by Lemma A.3.*

Proof. The metric, weights, and shifted prescribed weighted mean curvature in the translated expanding regions are exactly the data pulled from B. Local compactness produces a subsequential limiting Caccioppoli set on $B \times \mathbb{R}^d$, and it is an absolute minimizer by Lemma B.3. The uniqueness and factorization of Proposition B.5 identify it with $A_B \times \mathbb{R}^d$. Every subsequence has this same possible limit, which proves convergence of the full sequence. Apply Theorem B.6 for the support and smooth assertions. The deeper-end base statement follows by using the commuting product coordinates in Lemma A.3. □

B.4 Stability on a boundary and the retained-side estimate

We now choose the representative of each Caccioppoli set which is open away from S and agrees near S_{reg} with the corresponding smooth closed side when a trace is required. Write $S = S_{\mathrm{reg}}$ in integrals below. Every test has compact support away from Σ; joined-side tests also avoid the cylinder over Σ. The old torus variables remain invariant, and $\mathrm{d}\mu_S = \omega\,\mathrm{d}\mathcal{H}^{m-1}_g$ is the boundary measure. The intrinsic scalar curvature R_S and the second fundamental form B refer to $S \times \mathbb{T}^q$ with its induced old metric on the warped torus extension. Its torus principal curvature in direction i is $\partial_\nu \log f_i$, so (B.9) gives

$$H = \operatorname{tr} B = H_g + \partial_\nu \log \omega = p$$

for the outward normal of the plus Caccioppoli set. The invariant boundary Laplacian is $\omega^{-1} \operatorname{div}_S(\omega \operatorname{grad}_S \cdot)$, so its integration by parts uses precisely $\mathrm{d}\mu_S$.

Lemma B.8 (The boundary stability form)**.** *The regular boundary $S = S_{\mathrm{reg}}$ of the fixed minimizing Caccioppoli set satisfies, for every compactly supported smooth invariant test v,*

$$\int_S (2|\operatorname{grad} v|^2 + R_S v^2)\,\mathrm{d}\mu_S \geq \int_S (R + p^2 + |B|^2 + 2\partial_\nu p) v^2\,\mathrm{d}\mu_S. \tag{B.10}$$

In particular its right side is at least $\int_S (R - 1.02\Pi_{j_0}) v^2\,\mathrm{d}\mu_S$.

Proof. Weighted perimeter is the area of the old torus product for invariant Caccioppoli sets. The invariant normal variation with speed v is compactly supported in the smooth regular part and comes from an admissible base variation. Thus its energy has a minimum at parameter zero. For completeness the variation identities hold in every dimension: differentiating the induced metric gives $\dot g_{ab} = 2vB_{ab}$, differentiating the unit normal gives $\dot\nu = -\operatorname{grad}_S v$, and commuting the two ambient covariant derivatives in the derivative of B_{ab} gives, after tracing and differentiating the inverse metric,

$$\dot H = -\Delta_S v - (\operatorname{Ric}_{\widehat g}(\nu,\nu) + |B|^2) v.$$

The area derivative is $Hv\,\mathrm{d}\mu_S$, while the volume variation with prescribed weighted mean curvature is $pv\,\mathrm{d}\mu_S$. The first energy derivative is therefore $\int_S (H-p) v\,\mathrm{d}\mu_S$. At $H = p$ the derivatives of the area measure and of the chosen speed multiply zero, and $\dot p = v\partial_\nu p$. Hence the nonnegative second derivative is

$$\int_S \left[-v\,\Delta_S v - (\operatorname{Ric}_{\widehat g}(\nu,\nu) + |B|^2 + \partial_\nu p) v^2 \right] \mathrm{d}\mu_S.$$

Weighted integration by parts gives

$$\int_S |\operatorname{grad} v|^2\,\mathrm{d}\mu_S \geq \int_S (\operatorname{Ric}_{\widehat g}(\nu,\nu) + |B|^2 + \partial_\nu p) v^2\,\mathrm{d}\mu_S.$$

Compare the first variation in Chodosh and Li [9, Lemma 13] and the second-variation calculation in the proof of its Lemma 14. The displayed calculation proves the identity in the present warped torus extension; the stability and Gauss-equation mechanism is that of Schoen and Yau [35, Section 1]. The Gauss identity

$$R_S = R - 2\operatorname{Ric}_{\widehat g}(\nu,\nu) + H^2 - |B|^2$$

gives (B.10). By (B.1),

$$p^2 + 2\partial_\nu p \geq p^2 - 2|\operatorname{grad} p| \geq .33p^2 - 1.02\Pi_{j_0},$$

which proves the final statement. □

For either Caccioppoli set side Ω, put $p_s = p$ on the plus side and $p_s = -p$ on the minus side. The side is itself an locally minimizing Caccioppoli set for the signed prescribed weighted mean curvature p_s, and its outward mean curvature in the warped torus extension is $H_{\mathrm{out}} = p_s$.

Lemma B.9 (Deletion and the interface estimate)**.** *On either fixed Caccioppoli set side Ω, every nonnegative compactly supported smooth function f on its closed regular side, supported away from the singular set, satisfies*

$$\int_S f \,\mathrm{d}\mu_S \le \int_\Omega (|\operatorname{grad} f| + p_s f)\,\mathrm{d}\mu. \tag{B.11}$$

For every smooth real test v of the same type,

$$3.05 \int_\Omega |\operatorname{grad} v|^2 \,\mathrm{d}\mu + 2\int_S H_{\mathrm{out}} v^2 \,\mathrm{d}\mu_S \ge -1.02 \int_\Omega \Pi_{j_0} v^2 \,\mathrm{d}\mu. \tag{B.12}$$

Both statements extend to compactly supported Lipschitz tests by approximation.

Proof. For almost every $t > 0$, delete $\Omega \cap \{f > t\}$ from the minimizing Caccioppoli set. The removed boundary area is the area on $S \cap \{f > t\}$; the newly introduced boundary is the level surface of f in the interior of Ω. The volume term with prescribed weighted mean curvature increases by $\int_{\Omega\cap\{f>t\}} p_s \,\mathrm{d}\mu$. Minimality gives the corresponding perimeter inequality. Integrating in t, the layer-cake formula and weighted coarea give (B.11). All comparisons are compact and occur on regular patches; the perimeter slicing formula justifies the same argument at exceptional critical levels by discarding a null set of levels.

Apply (B.11) to $f = 2(-p_s)_+ v^2$, initially by Lipschitz approximation. Its right side is bounded above by

$$\int_{\Omega\cap\{p_s<0\}} \left[(2|\operatorname{grad} p_s| - 2p_s^2)v^2 + 4|p_s|\,|v|\,|\operatorname{grad} v|\right] \mathrm{d}\mu.$$

Young's inequality gives

$$4|p_s|\,|v|\,|\operatorname{grad} v| \le 3.05|\operatorname{grad} v|^2 + \frac{4}{3.05} p_s^2 v^2.$$

The prescribed weighted mean curvature estimate holds unchanged for p_s, and

$$2 - \frac{4}{3.05} - .67 = \frac{113}{6100} > 0. \tag{B.13}$$

Discard the resulting nonpositive multiple of $p_s^2 v^2$. Thus

$$2\int_S (-p_s)_+ v^2 \,\mathrm{d}\mu_S \le 3.05 \int_\Omega |\operatorname{grad} v|^2 \,\mathrm{d}\mu + 1.02 \int_\Omega \Pi_{j_0} v^2 \,\mathrm{d}\mu.$$

Since $2p_s \ge -2(-p_s)_+$ on the boundary and $H_{\mathrm{out}} = p_s$, this is (B.12). □

B.5 Raw joined forms and the numerical reserve

Attach $S \times [0,\infty)_t$ to the regular boundary of either side, freezing all induced metric and old warping function data on the cylinder. This is a joined space with a metric corner: it is smooth on its two pieces, and the interface retains its scalar corner contribution. An admissible test is a pair $v = (v_b, v_c)$, smooth separately up to regular interface patches, compactly supported away from singularities and their cylinders, satisfying $v_b|_S = v_c|_{t=0}$. Compact Lipschitz versions are allowed by local approximation in regular interface charts. For the boundary alone the tests are compact smooth functions on S away from Σ.

Set $C = 3.99$. On a boundary define

$$Q_S(v) = \int_S (C|\operatorname{grad} v|^2 + R_S v^2)\,\mathrm{d}\mu_S.$$

On a joined side define

$$\begin{aligned} Q_\Omega(v) = {} & \int_\Omega (C|\operatorname{grad} v_b|^2 + R v_b^2)\,\mathrm{d}\mu + 2\int_S H_{\mathrm{out}} v_b^2\,\mathrm{d}\mu_S \\ & + \int_0^\infty \int_S (C(|\operatorname{grad}_S v_c|^2 + |\partial_t v_c|^2) + R_S v_c^2)\,\mathrm{d}\mu_S\,\mathrm{d}t. \end{aligned} \tag{B.14}$$

The sign in the interface term can also be checked directly: in outward arclength s the mean curvature changes from H_{out} on the bulk side to zero on the frozen cylinder, and the scalar expression contains $-2\partial_s H$. Its distributional contribution is $+2H_{\mathrm{out}}$ at the interface. The smooth realization and its form convergence will be established in Appendix E.

Proposition B.10 (Scalar domination before smoothing)**.** *For the forms before smoothing and admissible matching tests defined above, set $V = R - 1.02\Pi_{j_0}$ on the parent, restrict it to the boundary, and pull that restriction to the cylinder. Then*

$$Q_S(v) \geq \int_S V v^2\,\mathrm{d}\mu_S + 1.99\int_S |\operatorname{grad} v|^2\,\mathrm{d}\mu_S, \tag{B.15}$$

$$\begin{aligned} Q_\Omega(v) \geq {} & \int_\Omega V v_b^2\,\mathrm{d}\mu + \int_0^\infty \int_S V v_c^2\,\mathrm{d}\mu_S\,\mathrm{d}t \\ & + .94\int_\Omega |\operatorname{grad} v_b|^2\,\mathrm{d}\mu + 1.99\int_0^\infty \int_S |\operatorname{grad} v_c|^2\,\mathrm{d}\mu_S\,\mathrm{d}t. \end{aligned} \tag{B.16}$$

Here $|\operatorname{grad} v_c|^2 = |\operatorname{grad}_S v_c|^2 + |\partial_t v_c|^2$.

Proof. The boundary assertion follows by adding $(C-2)|\operatorname{grad} v|^2$ to Lemma B.8. For the bulk use Lemma B.9; the remaining gradient coefficient is $C - 3.05 = .94$. Apply the boundary assertion to every cylinder slice and integrate in t. The horizontal coefficient is $C - 2 = 1.99$, while the vertical coefficient is still $C = 3.99$, so lowering the latter to 1.99 gives (B.16). The interface trace in the bulk estimate is precisely the matching trace in (B.14). □

Definition B.11 (Reserved form and test completion)**.** For the boundary or a joined side define, respectively,

$$\begin{aligned} \mathcal{N}_S(v) &= 1.7\int_S |\operatorname{grad} v|^2\,\mathrm{d}\mu_S, \\ \mathcal{N}_\Omega(v) &= .8\int_\Omega |\operatorname{grad} v_b|^2\,\mathrm{d}\mu + 1.7\int_0^\infty \int_S |\operatorname{grad} v_c|^2\,\mathrm{d}\mu_S\,\mathrm{d}t. \end{aligned}$$

We write $\mathcal{N}$ when the tested space is understood. The squared norm $\|v\|_2^2$ is the sum of the corresponding piecewise weighted integrals. With

$$e_0 = (\lambda_r - \lambda_{r+1})/100 > 0, \qquad \mathfrak{h}(v) = \mathcal{N}(v) + e_0\|v\|_2^2,$$

the test completion means completion of the admissible compact tests just specified in the norm $\mathfrak{h}^{1/2}$. In particular its interface condition is matching traces, with no zero trace imposed there.

The coefficients have a reserve even after the later cost of the added potentials .9ħ. Specifically,

$$.94 - .9(.8) = .22, \qquad 1.99 - .9(1.7) = .46. \tag{B.17}$$

The potential term in Proposition B.10, combined with the parent inequality, satisfies

$$R - 1.02\Pi_{j_0} \geq \lambda_r + b_r \sum_{j \neq j_0} \Pi_j + (b_r - 1.02)\Pi_{j_0} \qquad \text{on } D_E. \tag{B.18}$$

Here $b_r \geq 1.07 > 1.02$. The form inequalities before smoothing themselves hold for all admissible compact tests; the replacement of their potential by the right side of (B.18) is permitted only where the test is supported over the old required region. The count inclusions of Lemma 6.3 supply exactly this localization for the later required regions. These are the forms and constants used in enlargement of the prescribed open sets and regularization. The minimizing cut, its prescribed weighted mean curvature, and the old weight remain fixed while the other functions and their prescribed open sets are changed.

C Local enlargement of the prescribed open sets near singularities

Enlargement of the prescribed open sets makes each singular point irrelevant to the next curvature requirement by enlarging enough remaining prescribed open sets near that point. The forcing costs a potential concentrated in a thin annulus. We first pay for this potential from the reserved gradient form; the next appendix makes the choices compatible throughout the diagram.

We work at one fixed stage with a finite set of remaining labels. The countable extension is proved in Proposition 7.5. The local estimates use only $.60 \leq a_r \leq .63$ and the positive reserve of Definition B.11. Smooth geometric bounds are taken on the specified compact regions.

Fix a boundary support S in a parent component E of base dimension m, as in Theorem B.6. Its singular set is Σ, and its regular part is S_{reg}. The old torus volume factor is ω. All measures below include this factor; on a cylinder it is pulled back from S_{reg}. For either Caccioppoli set Ω, write

$$J_\Omega = \overline{\Omega}_{\mathrm{reg}} \cup_{S_{\mathrm{reg}}} (S_{\mathrm{reg}} \times [0, \infty)).$$

Here the notation retains only regular boundary points when forming the join. Tests on J_Ω are compactly supported, piecewise smooth functions with matching values at the interface, whose supports avoid Σ and the cylinders above it. Lipschitz tests with the same properties follow by approximation in regular charts. Tests on the boundary have the analogous meaning. Use the reserved energy from Definition B.11, namely

$$\mathcal{N}(v) = \begin{cases} 1.7 \displaystyle\int_{S_{\mathrm{reg}}} |\nabla v|^2, & \text{on the boundary,} \\ .8 \displaystyle\int_\Omega |\nabla v|^2 + 1.7 \int_{S_{\mathrm{reg}} \times [0,\infty)} |\nabla v|^2, & \text{on a joined side.} \end{cases} \tag{C.1}$$

The notation $\|v\|_2^2$ denotes the sum of the corresponding weighted L^2 integrals. An ambient function is restricted to the boundary and frozen in the cylinder coordinate. In particular, these conventions specify every potential integral in this section. The boundary support remains fixed throughout: only functions of labels other than the processed label j_0 will change.

Lemma C.1 (Smooth approximations to truncated functions)**.** *For the fixed remaining functions, there are compatible smooth functions h_j, one for each remaining label, within* .1 *of*

$\max(-1, \min(1, \mu_j))$, *and common numbers* $t_j \in (-.3, .3)$ *such that, on the singular set in each parent of dimension* m, *at most* $m - 8$ *of the equalities* $h_j = t_j$ *hold. The assertion is vacuous in dimensions below eight. Moreover*

$$\{h_j > t_j\} \cap \{\mu_j = -\infty\} = \varnothing, \qquad \{h_j < t_j\} \cap \{\mu_j = +\infty\} = \varnothing.$$

Proof. Compatible smoothing by Lemma A.4 gives the stated approximations: smooth the continuous clipped functions on compact parts and blend with their already prepared exact end base copies. Allocate the error .1 beforehand over the finitely many requests on each end base, and reproduce constants exactly on copies with constant data. For a component of dimension $m \geq 8$ and a set I of $m-7$ remaining labels, the smooth map $h_I = (h_j)_{j \in I}$ is locally Lipschitz. Since $\dim_{\mathcal{H}} \Sigma \leq m-8$, its image of Σ has zero $(m-7)$-dimensional Lebesgue measure. This follows on each relatively compact chart from the defining Hausdorff covers and then on the component by a countable union. The set of threshold vectors for which $t_I \in h_I(\Sigma)$ is null by Fubini. For measurability, intersect Σ with a compact exhaustion first: each image is compact, and their countable union is measurable. Give each interval $(-.3, .3)$ its normalized Lebesgue measure. There are countably many components and finitely many relevant subsets per component, so choose a threshold vector outside the union of all exclusions. The same argument also works in a countable product of the threshold intervals: each exclusion is a null cylinder set depending on only $m-7$ coordinates. If fewer than $m-7$ labels remain, the asserted count holds without a condition. At a negative pole $h_j \leq -.9 < t_j$, and at a positive pole $h_j \geq .9 > t_j$, proving the last assertions. □

The set H_E^* in (6.7) requires at least $m-2$ remaining undecided labels. At $x \in \Sigma$, enlarging the positive prescribed open set for every label with $h_j(x) > t_j$ and the negative prescribed open set for every label with $h_j(x) < t_j$ to contain x would leave at most $m - 8 < m - 2$ labels undecided. These pointwise decisions would therefore remove x from the set H_E^*.

We need quantitative pieces of Σ, rather than just its Hausdorff dimension. A coordinate chart in the following lemma is chosen with compact buffer and with its metric as close to Euclidean as required. Such charts can be chosen around every point. The plane condition is expressed in Euclidean chart coordinates. Replacing Euclidean balls by metric balls changes only fixed factors, which can be absorbed by requesting a smaller flatness constant.

Lemma C.2 (Compact pieces with one-sided flatness)**.** *For the fixed boundary support in a parent component of dimension* m, *let* O *be open in that parent and let* $\eta > 0$. *The set* $\Sigma \cap O$ *is covered by countably many compact sets* $F \subset \Sigma \cap O$, *each contained in a buffered coordinate chart, with the following property. There is* $r_F > 0$ *such that, for every* $x \in F$ *and* $0 < r < r_F$, *some affine* $(m-8)$*-plane* $L_{x,r}$ *satisfies*

$$F \cap B_r(x) \subset \{z : \operatorname{dist}_{\mathrm{Eucl}}(z, L_{x,r}) < \eta r\}. \tag{C.2}$$

The plane may depend on both the center and the radius.

Proof. Write $\Theta(x)$ for the unweighted boundary density. First consider a tangent minimizing boundary cone C at a singular point. It is nonflat, since a flat tangent gives regularity. For any $z \in C$, the density ratio centered at z has limit $\Theta_C(0)$ as its radius tends to infinity: balls centered at z are sandwiched between balls centered at zero whose radii differ by $|z|$. Consequently

$$\Theta_C(z) \geq \Theta_C(0) \quad \Longrightarrow \quad \text{every density ratio centered at } z \text{ equals } \Theta_C(0).$$

Equality in monotonicity says that C is also a cone centered at z. Composing a dilation about z with the inverse dilation about zero shows translation invariance along z. Conversely, translation

invariance along z gives equality of the two densities. The translation invariance directions form a vector subspace. Every point in this subspace is singular: otherwise translation gives regularity at the vertex, and the cone is a plane. The minimizing-cone singular dimension estimate therefore bounds the dimension of this subspace by $m-8$. These tangent-cone and regularity inputs are precisely those in Theorem B.6.

Fix a singular point x and a small number $\eta' > 0$. There are $d_x > 0$ and $r_x > 0$ such that, at every radius $r < r_x$, the points

$$\{z \in \Sigma \cap B_r(x) : \Theta(z) \geq \Theta(x) - d_x\} \tag{C.3}$$

lie within $\eta' r$ of some $(m-8)$-plane through x. If not, choose failing radii $r_i \downarrow 0$ with density tolerances tending to zero. Rescale at x and pass to a tangent cone C. Fix a plane containing its subspace of translation-invariance directions. Failure for this particular plane gives points z_i in the rescaled unit ball a fixed positive distance from it. A convergent subsequence has limit z. Upper semicontinuity of densities under these convergent blowups gives $\Theta_C(z) \geq \Theta(x) = \Theta_C(0)$. The preceding paragraph puts z in the subspace of translation-invariance directions, a contradiction. Smooth metric errors and bounded mean-curvature errors vanish in this rescaling.

Here is the countable decomposition, including its closure step. Group singular points according to lower bounds $r_x \geq 1/a$ and $d_x \geq 1/b$, with positive integers a, b, and according to a density interval of length less than $1/(2b)$. Any two points x, z in the same group satisfy $\Theta(z) > \Theta(x) - d_x$. Thus the group itself has the plane containment at every one of its centers and every radius less than $1/a$. Work within one buffered chart at a time. Intersect these groups with a countable family of compact subpatches $Q \Subset O$ exhausting that chart's portion of O, and take $F = \overline{Q \cap \text{group}}$. Relative closedness of Σ and the compact buffer ensure $F \subset \Sigma \cap O$. These compact sets still cover $\Sigma \cap O$.

To check the geometric condition at a closure center x, let x_i be group points tending to x and fix $r < 1/(3a)$. Apply the group estimate in $B_{2r}(x_i)$. The planes pass through x_i, so a subsequence converges to a plane through x. Every group point in $B_r(x)$ is in these larger balls eventually. Passing first to the limit of the planes and then to closure points gives containment with width $2\eta' r$. Taking, for example, $\eta' < \eta/3$ proves (C.2); choose r_F below $1/(3a)$ and the buffered chart radius. □

The next construction fixes the added function and potential to be estimated. Distances denoted by δ from now on are ambient base distances.

Lemma C.3 (A shell forcing function)**.** *Fix a remaining label j and a compact set $F \subset \Sigma \cap \{h_j > t_j\}$. For every sufficiently small $\varepsilon > 0$ there is an extended-real function u_ε, equal to $+\infty$ for $\delta \leq \varepsilon$ and $-\infty$ for $\delta \geq R_\varepsilon$, where*

$$\delta(x) = \operatorname{dist}(x, F), \qquad R_\varepsilon = e^{5\pi}\varepsilon,$$

whose finite truncations are locally Lipschitz and which satisfies

$$2|\nabla u_\varepsilon| \leq .58 u_\varepsilon^2 + 2\delta^{-2} \quad (\varepsilon < \delta < R_\varepsilon). \tag{C.4}$$

There is a smooth nonnegative ambient function D_ε with

$$\begin{aligned} D_\varepsilon &\geq 2\delta^{-2} && \text{on } \{\varepsilon \leq \delta \leq R_\varepsilon\}, \\ D_\varepsilon &\leq 2.12\delta^{-2} && \text{everywhere off } F, \\ \operatorname{supp} D_\varepsilon &\subset \{\varepsilon/2 < \delta < 2R_\varepsilon\} \Subset \{h_j > t_j\}. \end{aligned} \tag{C.5}$$

Replacing μ_j by $\max(\mu_j, u_\varepsilon)$, adding D_ε to Π_j, and adding $\{\delta < \varepsilon/2\}$ to U_j^+ preserves the function inequality with coefficient a_r. There is an identical negative construction in $\{h_j < t_j\}$, using

$\min(\mu_j, -u_\varepsilon)$ *and enlarging* U_j^-*. All supports and newly forced regions are compactly contained in the stated set* $\{h_j > t_j\}$ *or* $\{h_j < t_j\}$*.*

Proof. If F is empty, use $\delta = +\infty$, $u_\varepsilon = -\infty$, and $D_\varepsilon = 0$; the update is empty. Assume henceforth that F is nonempty. The polynomial $q(y) = .29y^2 - y + 1$ is strictly positive, and

$$\int_{-\infty}^{\infty} \frac{dy}{q(y)} = 5\pi.$$

Let $y : (0, 5\pi) \longrightarrow \mathbb{R}$ solve $y' = q(y)$, with the two infinite endpoint limits, and put

$$u_\varepsilon(x) = \frac{y(\log(R_\varepsilon/\delta(x)))}{\delta(x)}.$$

At differentiability points of δ, its gradient has norm at most one. Differentiating in δ gives

$$\left| \frac{d}{d\delta} \frac{y(\log(R_\varepsilon/\delta))}{\delta} \right| = \frac{y' + y}{\delta^2} = .29u_\varepsilon^2 + \delta^{-2},$$

which proves (C.4). Extending by the stated poles is continuous in the extended real line. Each finite clipping has a finite Lipschitz constant locally, including across either clipping interface.

Choose a continuous cutoff of δ, equal to one on $[\varepsilon, R_\varepsilon]$ and supported strictly in $(\varepsilon/2, 2R_\varepsilon)$. Multiplying it by $2.06\delta^{-2}$ gives a continuous nonnegative function supported away from F. Smooth it by nonnegative local convolutions and a partition of unity, with sufficiently small value error on this compact annulus and with support retained in a slightly larger annulus of the indicated size. The positive margins .06 on the central annulus and .06 below the upper bound give (C.5). The annuli lie in the corresponding set $\{h_j > t_j\}$ or $\{h_j < t_j\}$ for all small ε, by compactness of F.

At almost every point where the new maximum is finite, its gradient is the gradient of an active branch. The old branch obeys the old inequality. The new one obeys it because $.58 < a_r$ and $D_\varepsilon \geq 2\delta^{-2}$ wherever that branch is finite. The same reasoning applies at branch equality, using the elementary weak derivative rule for maxima of locally Lipschitz functions. The new prescribed open set has closure inside the open positive pole region $\{\delta < \varepsilon\}$. By Lemma C.1 no old negative pole occurs in the support of the positive forcing. Positive and negative additions for the same label have disjoint sets $\{h_j > t_j\}$ and $\{h_j < t_j\}$, so repetitions introduce no conflict. Indeed any opposite pole introduced later lies in the opposite set $\{h_j > t_j\}$ or $\{h_j < t_j\}$ and therefore cannot meet this support. Original poles and previously introduced opposite prescribed open sets consequently remain protected throughout. Replacing the function by its negative proves the other assertion. □

The localization in Proposition C.5 will produce a partition error bounded by $C\xi^2\delta^{-2}v^2$, where ξ is a small localization parameter and C depends only on m and the chart distortion. We first prove a rough estimate for the inverse-square potential with a positive coefficient c_0 that depends only on the dimension and a fixed chart distortion, so that choosing ξ small in terms of c_0 makes this error consume only a prescribed fraction of $\mathcal{N}$. The box scale of that localization is a multiple of the shell scale fixed in terms of ξ; the stronger flatness used for the sharp estimate is chosen in terms of this multiple; and the pieces, with their constants d_F, C_F, are selected only for that flatness. Hence c_0 must be fixed before the flatness and may depend neither on it nor on the piece.

Lemma C.4 (Uniform rough coefficient for the inverse-square potential)**.** *For each* $m \geq 8$ *there are numbers* $\eta_0 > 0$ *and* $c_0 > 0$ *with the following property. Let* F *be a compact singular piece satisfying* (C.2) *with* $\eta \leq \eta_0$ *in a sufficiently accurate buffered chart. There are* $d_F > 0$ *and* $C_F < \infty$ *such that, on the boundary and on either joined side,*

$$c_0 \int_{\delta < d_F} \delta^{-2} v^2 \leq \mathcal{N}(v) + C_F \|v\|_2^2. \tag{C.6}$$

The constants η_0, c_0 depend only on m and a fixed permitted chart distortion. The constants d_F, C_F may depend on the compact piece, its scale threshold, and the local smooth metric, weight, and prescribed weighted mean curvature.

Proof. If F is empty, the estimate is vacuous. Otherwise, the idea is to use the flatness of F along $(m-8)$-planes to build a positive function U which, in a sufficiently small punctured neighborhood of F, is comparable to $\delta^{-3/2}$ and satisfies $-\Delta_\omega U/U \geq c\delta^{-2}$ in the bulk and on the regular boundary. The covering and tubular-volume estimates come first, since they make the kernel defining U integrable and give the bounds for U and its derivatives. The vector-field identity (C.12) below, obtained by completing a square, turns these bounds into an inequality for the inverse-square potential using the fields $-b\rho\nabla \log U$ for small $b > 0$. On a joined side a vertical component added on the cylinder makes the interface terms nonnegative for arbitrary matching interface values.

Put $s = m - 8$. At scale r, the one-sided plane condition covers $F \cap B_r(x)$ by at most $C_m\lambda^{-s}$ balls of radius λr, with centers in F, where $\lambda = 8\eta_0$ can be made as small as desired. To see the bound, take a maximal λr-separated family in the set. Projection to the approximating s-plane changes pairwise distances by at most $2\eta_0 r$; the projections are separated by a fixed multiple of λr in an s-ball of radius $2r$. Its elementary Euclidean packing bound gives the claim, with harmless changes in C_m and the covering radius. Absorb these fixed changes into the initial choice of λ. Choose η_0 so small that the resulting covering number is at most $\lambda^{-(s+1/2)}$.

Iterating the same covering at centers in F gives, for $0 < t \leq d$ below a fixed fraction of r_F, a covering by at most $C(d/t)^{m-7.5}$ balls of radius t of the part of F controlling a ball of radius d. Here one first centers a ball of radius a fixed multiple of d at a nearby point of F and then iterates; the constants use only the fixed rough choice. Enlarging the final balls by a fixed factor proves the tubular-volume estimate

$$|B_d(x) \cap \{\delta < t\}| \leq Cd^{m-7.5}t^{7.5}, \qquad 0 < t \leq d, \tag{C.7}$$

whenever $\operatorname{dist}(x, F) \leq d$ and the slightly enlarged ball is in the chart at allowed scales. Metric and Euclidean distances are uniformly comparable here. Dyadically splitting the regions $2^{-k-1}d < \delta \leq 2^{-k}d$, and treating $\delta > d$ separately, gives

$$\int_{B_d(x)} \delta^{-5.5}\, dx \leq Cd^{m-5.5}. \tag{C.8}$$

Indeed the sum of the bounds is a geometric series with ratio 2^{-2}. In particular this density is locally integrable.

Choose a nonnegative smooth chart cutoff χ, equal to one on a neighborhood of F, and, using Euclidean coordinates, define

$$U(x) = 1 + \int \chi(y)\delta(y)^{-5.5}|x-y|^{-(m-4)}\, dy. \tag{C.9}$$

For small $\delta(x) > 0$, the part $|x-y| < c\delta(x)$ has $\delta(y)$ comparable to $\delta(x)$; it supplies the lower bound for U. The same region supplies the upper bounds for U and its first derivatives. On the successive larger annuli use (C.8); their contributions have orders $(2^k\delta)^{-1.5}$ and $(2^k\delta)^{-2.5}$ respectively. The parts above the allowed local scale are bounded constants depending on F. Shrinking the final distance threshold absorbs them, and gives dimension-dependent constants $c, C > 0$ such that

$$c\delta^{-1.5} \leq U \leq C\delta^{-1.5}, \qquad |\nabla U| \leq C\delta^{-2.5}. \tag{C.10}$$

Set $a = m - 4$ temporarily. For any Euclidean d'-plane T, direct differentiation of $K(z) = |z|^{-a}$ gives

$$-\operatorname{tr}_T D^2K(z) = a|z|^{-a-2}\left(d' - (a+2)|\operatorname{proj}_T(z/|z|)|^2\right) \geq a(d' - a - 2)|z|^{-a-2}.$$

For $d' = m$ or $m-1$ the last coefficient is positive. A sufficiently small, dimension-dependent metric distortion preserves a fixed positive fraction of it for traces in metric orthonormal frames. The connection and weight terms in the weighted Laplacian involve only first derivatives of U. On S_{reg} the restriction formula adds its mean-curvature vector paired with ∇U. This vector is locally bounded, since $H_{\mathrm{base}} = p - \partial_\nu \log \omega$ by Theorem B.6. There is no dependence here on individual principal curvatures. Integrating the positive second-derivative kernel over $|x - y| < c\delta(x)$ gives $c\delta^{-3.5}$. Equations (C.10), followed by another reduction of the distance threshold, therefore imply, in the ambient space and on the regular boundary,

$$-\frac{\Delta_\omega U}{U} \geq c\delta^{-2}, \qquad |\nabla \log U| \leq C\delta^{-1}. \tag{C.11}$$

The leading constants again depend only on the fixed rough data. For example the first-order errors before division have size at most $C_F\delta^{-2.5}$, one power lower than the positive term. The differentiated kernels through order two are locally integrable off F: their strongest singularity is $|x-y|^{-(m-2)}$. Since the density is bounded near any such x and integrable elsewhere on its compact support, ordinary differentiation under the integral, with a vanishing small diagonal ball, proves that U is C^2 off F. No diagonal distributional term is present.

For a weighted domain Q, completing a square and integrating by parts gives the vector-field identity in inequality form

$$\int_Q |\nabla v|^2 \geq \int_Q (\operatorname{div}_\omega A - |A|^2)v^2 - \int_{\partial Q} (A \cdot \nu_Q)v^2. \tag{C.12}$$

Choose a smooth cutoff ρ, equal to one near F and supported in the chart, and set $A_b = -b\rho\nabla \log U$ in the bulk. On the boundary use $A_h = -b\rho\nabla_S \log U$. For $0 < b \leq 1$, where $\rho = 1$,

$$\operatorname{div}_\omega(-b\nabla \log U) - b^2|\nabla \log U|^2 = -b\Delta_\omega U/U + b(1-b)|\nabla \log U|^2 \geq bc\delta^{-2}.$$

Outside a smaller neighborhood of F all errors from these fields are bounded by a piece-dependent constant.

On the cylinder take the horizontal field A_h and add the vertical component $a(x)\partial_t$, where

$$a(x) = \frac{.8}{1.7}\, b\rho(x)\big(1 + |\nabla \log U|_g^2\big)^{1/2}\big|_{S_{\mathrm{reg}}}.$$

It has zero divergence in the vertical direction and satisfies $1.7a \geq .8|A_b|$ at the interface. The outward cylinder normal there is $-\partial_t$. Applying (C.12) with coefficients .8 and 1.7 on the two pieces, their boundary terms sum to

$$\int_{S_{\mathrm{reg}}} [-.8A_b \cdot \nu_\Omega + 1.7a]\, v(0)^2 \geq 0.$$

This explicitly permits arbitrary matching interface values. The vertical square costs at most $Cb^2\delta^{-2} + C_F$ in the cylinder interior. Choose $b > 0$ sufficiently small, using the universal constants in (C.11), so that this is absorbed by half the positive term of order $b\delta^{-2}$. All remaining localization terms cost $C_F\|v\|_2^2$. This proves (C.6) on a joined side. The boundary proof is the same calculation without the interface or the vertical field. Taking the smaller leading coefficient makes one c_0 valid for the boundary and both sides. □

The coefficients 1.16 and .78 in the next estimate are set by its later use. The shell potentials enter the next scalar requirement with coefficient $(1 + 3\eta_r)b_{r+1}$, which is smaller than 1.16 by (E.3). The bound .78 leaves room for the summable allowances of the proof of Proposition D.2 below the total enlargement of the prescribed open sets charge $.9\mathfrak{h}$, which the gradient reserves of Proposition B.10 can pay by (B.17).

Proposition C.5 (The estimate for the added potential near singularities)**.** *There is a sufficiently small flatness number $\eta > 0$, depending only on m and the fixed numerical coefficients, with the following property. Let F be any piece from Lemma C.2 for this flatness number, in a sufficiently accurate chart. For the shell potential D_ε of Lemma C.3 on that same piece, there are $\varepsilon_F > 0$ and $C_F < \infty$ such that, for every $0 < \varepsilon < \varepsilon_F$,*

$$1.16 \int D_\varepsilon v^2 \le .78\mathcal{N}(v) + C_F\|v\|_2^2. \tag{C.13}$$

This holds on the boundary support and either joined side, and hence on translated compact tests in a pure boundary cylinder. The constants ε_F, C_F may depend on the local smooth data and F, but are independent of ε. No vanishing trace condition is imposed at a joined interface.

Proof. The proof localizes a test to boxes of scale $R_1 = K\varepsilon$ near F by a squared partition of unity. The rough estimate of Lemma C.4 pays for the partition gradients, so that these errors can be fixed before choosing the stronger flatness needed for the plane comparison. In each box the regularized distance ζ to an approximating $(m-8)$-plane gives, through the vector-field identity (C.12), the potentials $4.9\zeta^{-2}$ in the bulk and $5.9\zeta^{-2}$ on the boundary. As in the rough estimate, a vertical cylinder component controls the interface. The strong flatness makes $\zeta \le 1.03\delta$ on the shell, and the resulting bound for $\int \delta^{-2}v^2$ there controls $1.16\int D_\varepsilon v^2$ through the upper bound in (C.5).

We choose the parameters in this order. Fix the rough number η_0 and coefficient c_0 from Lemma C.4. Below, constants in partition gradient bounds depend only on dimension and the fixed chart distortion. Choose $\xi > 0$ so small that the rough estimate pays their total squared-gradient error with at most .04 of $\mathcal{N}$. Next choose a large fixed K, then a small fixed $\alpha > 0$, and finally a strong flatness $\eta \le \eta_0$, as specified below. Only after the piece F has been selected for this η do we choose its small distance threshold and then ε_F. In particular C_F is never used to choose ξ, K, or η.

Put $a_- = 1/2$, $a_+ = 2e^{5\pi}$ and $R_1 = K\varepsilon$. The shell supporting D_ε lies in $a_-\varepsilon < \delta < a_+\varepsilon$. Choose a smooth angle $\vartheta(\delta)$ equal to zero for $\delta \le a_+\varepsilon$ and equal to $\pi/2$ for $\delta \ge \xi R_1$, with

$$|\nabla\vartheta| \le \xi/\delta.$$

Such a function is obtained by a smooth transition in $\log(\delta/(a_+\varepsilon))$ of length at least a fixed multiple of $1/\xi$. Thus it suffices to choose $K \ge (a_+/\xi)\exp(C/\xi)$, independently of the piece. Set $q = \cos\vartheta$ and $q_\infty = \sin\vartheta$. They extend as constants across F, and $|\nabla q|^2 + |\nabla q_\infty|^2 \le \xi^2\delta^{-2}$.

Take a smooth normalized squared partition $\sum_i \phi_i^2 = 1$ on chart boxes of scale R_1, with uniformly bounded overlap and $\sum_i |\nabla\phi_i|^2 \le C_m R_1^{-2}$. One construction translates a fixed compactly supported bump on a Euclidean lattice and divides each bump by the square root of the sum of their squares. Restrict the same ambient functions to the boundary and freeze them on the cylinder. Only finitely many boxes meet $\operatorname{supp} q$, which is compact in the chart at the scales in use. The family $q\phi_i, q_\infty$ is a squared partition, and its total gradient square is at most $C_m\xi^2\delta^{-2}$ on $\{\delta < \xi R_1\}$ and zero elsewhere. The estimate follows from $\delta < \xi R_1$ wherever $q \ne 0$.

For a squared partition $\sum_i \chi_i^2 = 1$, direct expansion gives

$$\sum_i \mathcal{N}(\chi_i v) = \mathcal{N}(v) + \int a(x)\sum_i |\nabla\chi_i|^2 v^2, \tag{C.14}$$

where $a(x)$ is .8 on bulk and 1.7 on boundary or cylinder. The mixed terms vanish because $\sum_i \chi_i \nabla \chi_i = 0$. Choose ε_F small enough that $\xi R_1 < d_F$. Equation (C.6) and the choice of ξ now give

$$\sum_i \mathcal{N}(q\phi_i v) \le 1.04\mathcal{N}(v) + C_F\|v\|_2^2. \tag{C.15}$$

For example it is enough to impose $1.7C_m\xi^2/c_0 \le .04$; the other partition term is nonnegative and has been discarded. Lipschitz distance cutoffs are allowed by approximation on the regular test supports.

Consider one box meeting the support of q. Choose a nearby point of F and apply (C.2) at a radius $L_m R_1$, where a fixed L_m includes the box and every nearest point of F needed in it. Taking ε_F smaller ensures this is below r_F. Let L be the resulting $(m-8)$-plane and define

$$\zeta^2 = \mathrm{dist}_{\mathrm{Eucl}}(x, L)^2 + \alpha^2\varepsilon^2, \qquad P = \tfrac{1}{2}\zeta^2.$$

The Euclidean Hessian of P is an orthogonal projection of rank eight. Its trace on every $(m-1)$-dimensional subspace is at least seven. Its gradient has size at most ζ, and throughout the box $\zeta \le C_m R_1$ (after fixing α and K).

For either the ambient or the tangential weighted derivatives, the vector field $A = d_1 \nabla \log \zeta$ has potential

$$(\mathrm{div}_\omega A - |A|^2)\zeta^2 = d_1\Delta_\omega P - (2d_1 + d_1^2)|\nabla P|^2/\zeta^2. \tag{C.16}$$

Let τ bound the chart's metric distortion. The rank traces have error at most $C_m\tau$, and $|\nabla P|^2/\zeta^2 \le 1 + C_m\tau$. The connection, weight, and boundary mean-curvature contributions to $\Delta_\omega P$ have size at most $C_F R_1$. Thus one may arrange $C_m\tau \le .005$ and $C_F R_1 \le .025$, uniformly in the box and its plane. Equation (C.16) then gives

$$\begin{aligned} d_1 = 1 \text{ in bulk}: &\quad \mathrm{div}_\omega A - |A|^2 \ge 4.9\zeta^{-2}, \\ d_1 = 2 \text{ on boundary}: &\quad \mathrm{div}_\omega A - |A|^2 \ge 5.9\zeta^{-2}. \end{aligned}$$

Indeed the respective Euclidean lower bounds are $8 - 3 = 5$ and $2 \cdot 7 - 8 = 6$. Only the mean curvature of the boundary enters this calculation, so these bounds remain valid arbitrarily near its singular set on the regular test supports. The positive term $\alpha^2\varepsilon^2$ also makes the fields nonsingular along L; the estimates hold inside the inner shell as well.

In the bulk use $A_b = \nabla \log \zeta$. On a joined cylinder use $A_c = 2\nabla_S \log \zeta + \beta\zeta^{-1}\partial_t$, where $\beta = (.8/1.7)\sqrt{1 + C_m\tau}$ is chosen so that $\beta/\zeta \ge (.8/1.7)|A_b|$ at the interface. The metric distortion determines this fixed upper factor. The vertical component has zero divergence and is orthogonal to the horizontal component. Its only interior cost is $\beta^2\zeta^{-2}$. The boundary terms in (C.12) are nonnegative after summing, by exactly the cancellation used in Lemma C.4. After multiplication by the coefficients in (C.1), the limiting interior coefficients are

$$.8 \cdot 5 = 4 \quad \text{in bulk}, \qquad 1.7 \cdot 6 - .8^2/1.7 > 9.8 \quad \text{in the cylinder},$$

and $1.7 \cdot 6 = 10.2$ on the boundary alone. With the explicit error bounds just imposed, the coefficients are at least $.8 \cdot 4.9 = 3.92$ in bulk, $1.7 \cdot 5.9 - (.8^2/1.7)1.005 > 9.6$ in the cylinder, and $1.7 \cdot 5.9 > 10$ on the boundary. In particular all exceed 3.8. Applying the vector inequality to $q\phi_i v$ gives

$$\mathcal{N}(q\phi_i v) \ge 3.8 \int \zeta^{-2}(q\phi_i v)^2. \tag{C.17}$$

This is a bound throughout the localized support; its nonnegative right side may therefore be restricted to the shell.

We finally compare ζ with δ on that shell. A nearest point $z \in F$ lies in the ball used for the plane condition, so

$$\zeta(x) \le 1.005\delta(x) + L_m \eta K \varepsilon + \alpha\varepsilon.$$

The first factor is ensured by the fixed chart accuracy and its buffer, after shrinking the scale so that a minimizing short path to F stays in the chart. Since $\delta \ge a_-\varepsilon$ there, choose $\alpha/a_- \le .005$ and then $L_m\eta K/a_- \le .015$. These choices give $\zeta \le 1.03\delta$. Notice that K was already fixed using the rough coefficient, and that the strong flatness was chosen before selecting F. Since $3.8/(1.03)^2 > 3.5$, (C.17) controls $3.5\delta^{-2}$ on the shell. Summing over boxes uses $q = 1$ there and $\sum_i \phi_i^2 = 1$, and then (C.15) gives

$$3.5 \int_{\operatorname{supp} D_\varepsilon} \delta^{-2} v^2 \le 1.04\mathcal{N}(v) + C_F \|v\|_2^2.$$

Together with (C.5), this proves the stronger coefficient bound

$$1.16 \int D_\varepsilon v^2 \le \frac{1.16 \cdot 2.12 \cdot 1.04}{3.5} \mathcal{N}(v) + C_F \|v\|_2^2, \qquad \frac{1.16 \cdot 2.12 \cdot 1.04}{3.5} < .731 < .78.$$

This yields (C.13). A compact test on a pure cylinder can be translated to positive height in a joined cylinder and extended by zero onto the bulk, so it obeys the same estimate. □

D Global enlargement of the prescribed open sets and extension to incident higher-dimensional components

The local annular cost does not add once for every singular piece: a concentration argument bounds a new shrinking cost by the maximum of the old and local form coefficients. The finite-label hypotheses in this appendix are extended in Definition 7.1 and Proposition 7.5. Each label may already have its own positive potential floor p_j^*.

We keep the boundary supports fixed throughout this section. Only functions of remaining labels, their prescribed open sets, and their potentials are changed. In particular, none of the convergence assertions below concerns minimizers for a changing prescribed weighted mean curvature.

Definition D.1 (The matching form domain)**.** Let E be a parent component of dimension m, with boundary support S and singular set Σ. A raw piece Z means either S_{reg} or one of the two joined spaces

$$Z = \Omega \cup_{S_{\mathrm{reg}}} (S_{\mathrm{reg}} \times [0, \infty)).$$

The tests on a joined space are compactly supported, piecewise smooth functions whose two traces agree at the regular interface and whose support avoids Σ and the cylinders over Σ. The boundary has the corresponding compact tests away from Σ. All measures include the old warping function weight. Put

$$\mathfrak{h}_Z(v) = \mathcal{N}_Z(v) + e_0 \|v\|^2_{L^2(Z)}, \qquad e_0 = \frac{\lambda_r - \lambda_{r+1}}{100} > 0, \tag{D.1}$$

where $\mathcal{N}$ is the reserved gradient form in Definition B.11: its gradient coefficients are .8 on the bulk and 1.7 on the boundary or cylinder. Write $\mathcal{H}_Z$ for the completion of these tests in $\mathfrak{h}_Z^{1/2}$. Pure cylinders $S_{\mathrm{reg}} \times \mathbb{R}$ use the coefficient 1.7 throughout. The definitions also apply after adjoining finitely many Euclidean product coordinates, with the same coefficients and e_0. Such products are used only to identify limits on translated regions; they do not append further circle warping functions or change the weight. The compact regions used here consist of a fixed compact ambient base region together with a bounded cylinder interval, when a cylinder is present.

The trace matching in this definition is essential. There is no Dirichlet condition at the interface. A pure-cylinder test can be translated into the cylindrical part of a joined space, with zero bulk part. Consequently every form inequality on that joined space for potentials pulled back from its base also holds on the pure cylinder.

Proposition D.2 (Global enlargement of the prescribed open sets)**.** *At a fixed stage r, fix the current label j_0 and its minimizers, with the finite remaining data specified above. The remaining functions and prescribed open sets admit updates and compatible smooth potential additions $D_j \geq 0$ such that, on each parent E of dimension m:*

(i) *The updated functions have locally Lipschitz finite truncations and satisfy, almost everywhere where they are finite,*

$$2|\nabla \mu_j'| \leq a_r(\mu_j')^2 + \Pi_j + D_j. \tag{D.2}$$

All these data are exact compatible products. Each prescribed open set only enlarges, and its closure remains in the interior of its designated updated pole set. The original forced sets and the restrictions to $\{h_j > t_j\}$ or $\{h_j < t_j\}$ are preserved.

(ii) *With $G_j' = E \setminus (U_j'^{+} \cup U_j'^{-})$, the set*

$$H_E^* = \Big\{x : \#\{j \neq j_0 : x \in G_j'\} \geq m - 2\Big\} \tag{D.3}$$

is closed and misses Σ_E. In an exact collar over an end base B, its projection is contained in H_B^*. This holds for every further end base and every simultaneous product limit.

(iii) *For one common $\theta_* < .9$, all boundaries and joined sides satisfy*

$$\int 1.16 \sum_{j \neq j_0} D_j v^2 \leq \theta_*(\mathcal{N}(v) + e_0\|v\|_2^2) \leq .9(\mathcal{N}(v) + e_0\|v\|_2^2). \tag{D.4}$$

The additions are bounded with their derivatives on each post-radial component. There are only finitely many local additions on each such component and only finitely many extensions into it.

For the initial radial parent the same conclusions hold with locally finite additions over its compact transition shells; boundedness is asserted on the corresponding compact blocks and Equation (D.4) *holds on every compact test.*

The proof of Proposition D.2 is given at the end of this appendix. Lemma D.4 bounds the charge of a new shrinking shell by the larger of the previous charge and .78, up to a chosen error. Lemmas D.5 and D.6 preserve that bound and the function inequality on incident higher-dimensional components. Compactness of the singular points still in a set H_E^* then makes the number of additions on each component finite.

D.1 Compactness and the cost of an additional shell

Lemma D.3 (Compactness on joined regions)**.** *Let $m \geq 8$ and let Z have the matching form domain of Definition D.1. On every fixed compact region, a bounded sequence in $\mathcal{H}_Z$ has a strongly convergent L^2 subsequence. The following moving version also holds. Work in fixed smooth ambient geometry on each expanding region, as is the case after translation in an exact product end. Suppose positive weights and prescribed weighted mean curvatures converge smoothly there, and the minimizing sets and boundaries converge as in Theorem B.6 and Corollary B.7. Consider tests v_i with $\mathfrak{h}_{Z_i}(v_i) \leq 1$, all supported in a fixed compact region strictly inside a larger region of convergence. After passage to a subsequence there is a function v on the limiting raw piece Z_∞ such that:*

(i) *On every regular compact smaller region, the tests converge weakly in H^1 and strongly in L^2 in the smooth graphical identifications. At a limiting regular interface the two traces agree.*

(ii) *No L^2 mass is lost in neighborhoods of the limiting singular set. In particular, integrals of v_i^2 converge to those of v^2, including integrals against any fixed bounded continuous ambient potential and its pullback to the boundary and cylinder.*

(iii) *The limit belongs to $\mathcal{H}_{Z_\infty}$ and*

$$\mathfrak{h}_{Z_\infty}(v) \le \liminf_i \mathfrak{h}_{Z_i}(v_i). \tag{D.5}$$

These assertions include pure cylinders, translated cylinder intervals, and Euclidean product limits. Constants are local to the enlarged region; no uniform bound among all components is asserted. If the limiting boundary is smooth on the enlarged region, the same conclusions follow from ordinary Rellich compactness and the bounded trace maps in the smooth graphical charts, in every base dimension.

Proof. We give the estimates for tests first; they extend to the completions by density. On the enlarged region the weights are uniformly comparable to a positive constant, the ambient geometry is uniformly smooth, and the unweighted base mean curvature of the regular boundaries is bounded. The latter bound follows from the prescribed weighted mean curvature and the bounded first derivative of the logarithm of the weight. The perimeter varifolds have density at least one and no additional boundary term on this interior region, by Theorem B.6. They therefore meet the hypotheses of the Sobolev inequality used below.

We use the submanifold Sobolev inequality in the following precise form. For a d-dimensional Euclidean rectifiable varifold of density at least one with generalized mean curvature H, and a nonnegative compactly supported test f,

$$\left(\int f^{d/(d-1)}\right)^{(d-1)/d} \le C_d \int (|\nabla f| + |H| f). \tag{D.6}$$

This holds in arbitrary codimension; see Simon [36, Chapter 4, Section 5, Theorem 5.7]. This is the submanifold Sobolev inequality of Michael and Simon [31]. A smooth extension of the fixed ambient chart metric to a compact manifold and a fixed smooth isometric embedding [33] give the corresponding local inequality in our charts. The extra mean curvature of the embedding is bounded on this fixed compact geometry. For the moving-domain assertion the embedding is chosen once in the fixed ambient geometry; its constants are therefore independent of the boundary in the sequence. An initial test on the regular boundary has compact support away from the singular set, so it extends to an ambient test in finitely many smooth tubular charts. Thus the ambient-test formulation of the cited inequality applies. For $d > 2$, set

$$\gamma = \frac{2(d-1)}{d-2}, \qquad q_d = \frac{2d}{d-2}, \qquad 2(\gamma - 1) = q_d.$$

Apply (D.6) to $|v|^\gamma$, using smooth approximation at its zeros. The bounded mean curvature and Cauchy–Schwarz give

$$\|v\|_{q_d}^\gamma \le C \int (|v|^{\gamma-1}|\nabla v| + |v|^\gamma) \le C \|v\|_{q_d}^{\gamma-1} (\|\nabla v\|_2 + \|v\|_2).$$

Canceling the common power proves the H^1 estimate; a zero test needs no cancellation. The uniformly positive bounded weights give equivalent norms on the region. Applied to $S_{\mathrm{reg}} \times \mathbb{R}$, it

gives, with

$$q = \frac{2m}{m-2}, \qquad b = \frac{2(m-1)}{m-2}, \qquad 2(b-1) = q,$$

a local L^q bound on the cylinder. A test on the half-cylinder is reflected evenly at its base before this application; the reflection doubles its squared H^1 norm and is approximated by smooth tests in the regular product charts. The boundary alone has the stronger exponent $2(m-1)/(m-3) > 2$.

Here and below cutoffs are taken on the enlarged region and are pulled back to both interface sides. The cutoff errors are bounded by the local L^2 norm. Normalize $\hbar_Z(v) \le 1$. Choose a cylinder cutoff which is one at $s = 0$ and vanishes at the outer end of a fixed bounded interval. The fundamental theorem of calculus and Hölder give

$$\int_{S_{\mathrm{reg}}} |v(0)|^b \le C \int_{S\times[0,L]} \left(|v|^b + |v|^{b-1}|\partial_s v|\right) \le C. \tag{D.7}$$

The last step uses the cylinder L^q bound and $2(b-1) = q$.

The bulk need not have a uniform smooth boundary, so we use a BV calculation there. Extend $|v_{\mathrm{bulk}}|^b$ by zero in the ambient chart, including its compact cutoff. For an initial test its support misses the singular set, and the BV jump formula gives precisely the boundary contribution $\int_{S_{\mathrm{reg}}} |v(0)|^b$ in addition to the interior variation. This is also obtained directly by integration by parts against unit vector fields on the regular side. The ambient L^1 Sobolev inequality and Cauchy–Schwarz therefore yield, for $J = \int_\Omega |v|^q$,

$$J^{(m-1)/m} \le C\left(\int_\Omega |v|^b + \int_\Omega |v|^{b-1}|\nabla v| + \int_{S_{\mathrm{reg}}} |v(0)|^b\right) \le C(1 + J^{1/2}). \tag{D.8}$$

Here $bm/(m-1) = q$; the first integral on the right is at most $J^{1/2}\|v\|_2$, because $b = (q+2)/2$. Since $(m-1)/m > 1/2$, Equation (D.8) bounds J. Rescaling removes the normalization. Thus all pieces have a local L^p bound for some $p > 2$ controlled by their matching form norm.

We next exclude concentration in the moving version. On a slightly larger compact region let F be the compact portion of the limiting singular set. Its ambient volume and its limiting perimeter measure are zero. For a closed δ-neighborhood of F, perimeter measure convergence and the closed-set upper bound for weakly convergent measures imply

$$\limsup_i \mu_{S_i}(\overline{U_\delta(F)}) \le \mu_{S_\infty}(\overline{U_\delta(F)}),$$

with a fixed outer compact cutoff if necessary. The right side tends to zero as $\delta \downarrow 0$. The same conclusion for bulk measure follows from ambient volume bounds, and for a cylinder follows by multiplying the boundary measure by the bounded interval length. Hölder and the local L^p estimates give

$$\int_{A_i} |v_i|^2 \le C\mu_i(A_i)^{1-2/p}.$$

Thus the order is first $i \to \infty$ at a fixed singular neighborhood, then $\delta \downarrow 0$. Neither disappearing components nor a thin boundary layer can retain L^2 mass.

Off F, graphical convergence and the absence of stray boundary support supply smooth charts for both bulk Caccioppoli sets and their interfaces. Rellich compactness on the resulting bounded smooth chart domains [37, Lecture 5, Lemma 4] gives the asserted weak H^1 and strong L^2 limits. To check the trace bound, flatten a regular interface chart to $Q \times [0, a]$. The fundamental theorem of calculus and Cauchy–Schwarz, followed by integration in the normal variable, give

$$\int_Q |v(y,0)|^2\, dy \le \frac{2}{a}\int_{Q\times(0,a)} |v|^2 + 2a\int_{Q\times(0,a)} |\partial_t v|^2.$$

The smooth metric and weight change only the fixed chart constant. Density extends restriction to a bounded linear trace map on H^1; weak convergence of both traces therefore preserves their equality. Combining these regular limits with the no-concentration conclusion proves all the stated L^2 and bounded-potential convergence. Exhausting the regular pieces gives lower semicontinuity of their gradient integrals.

It remains to justify membership in the limit completion, rather than assume it. On this one limiting geometry, monotonicity and the local area bound give

$$\mu_S(B_{2a}(x)) \le Ca^{m-1}$$

for sufficiently small balls in the fixed region. Since $\dim_{\mathcal{H}} \Sigma \le m-8 < m-3$, a compact singular portion has ball covers, inside arbitrarily small neighborhoods, with $\sum_i a_i^{m-3}$ arbitrarily small. For cutoffs equal to one on these balls, supported on their doubles, and with gradients at most C/a_i, the maximum of the cutoffs has squared-gradient integral at most

$$C\sum_i a_i^{m-3}$$

on the boundary and on bounded cylinder intervals. The bulk cost is at most $C\sum_i a_i^{m-2}$ and is smaller. A finite subcover suffices. Use the same base cutoff on the bulk and cylinder, preserving interface matching. First truncate the limit function to a bounded function; then multiply it by one minus these cutoffs. The derivative of the cutoff has vanishing energy after truncation, and the other error tends to zero by absolute continuity of the already finite gradient integrals. Its L^2 error tends to zero as well. The resulting functions miss the singular set. Matching traces glue their two H^1 pieces across every regular interface chart; ordinary mollification in such charts, and a partition of unity, approximate them by the defining tests. This proves membership and Equation (D.5). The same proof applies to pure cylinders and product limits. For a product with a extra coordinates, use ambient dimension $m+a$ and singular-dimension bound $m+a-8$ in the preceding exponents; the strict codimension needed for the capacity estimate is unchanged. □

Lemma D.4 (Maximum of successive form costs)**.** *Fix a component and its boundary support and joined sides. Let $V \ge 0$ be a fixed potential, bounded on the compact base regions used below and pulled back to cylinders, satisfying*

$$\int_Z Vv^2 \le \theta \mathfrak{h}_Z(v) \tag{D.9}$$

on its boundary and both joined sides. Let $F \subset \Sigma$ be a compact piece and let D_ε be one of the nonnegative shell potentials of Lemma C.3 and Proposition C.5. In particular its base support lies in a fixed compact set and shrinks to F, and

$$\int_Z 1.16 D_\varepsilon v^2 \le .78\mathcal{N}_Z(v) + C_F\|v\|_2^2 \tag{D.10}$$

for all sufficiently small ε, on all three pieces. For every $\eta > 0$, a sufficiently small $\varepsilon > 0$ gives simultaneously

$$\int_Z (V + 1.16D_\varepsilon)v^2 \le (\max\{\theta, .78\} + \eta)\mathfrak{h}_Z(v). \tag{D.11}$$

In particular the assertion applies to a finite sum of compact additions on the initial radial parent.

Proof. Write $A_\varepsilon = 1.16D_\varepsilon$ for the potential charged in the form.

First $\int A_\varepsilon v^2 \to 0$ for every fixed $v \in \mathcal{H}_Z$. Indeed Equation (D.10) bounds these forms uniformly by a constant times $\mathfrak{h}_Z$, whereas for a defining test the integral is zero for small ε, since its support misses F. Density and Cauchy–Schwarz for the nonnegative form give the assertion.

We detail the localization needed on noncompact pieces. If $\sum_\alpha \chi_\alpha^2 = 1$ is a locally finite Lipschitz partition which has the same interface traces, then

$$\sum_\alpha \mathfrak{h}_Z(\chi_\alpha v) = \mathfrak{h}_Z(v) + \int_Z c_Z \sum_\alpha |\nabla \chi_\alpha|^2 v^2, \qquad c_Z = \begin{cases} .8 & \text{on bulk,} \\ 1.7 & \text{on boundary or cylinder.} \end{cases} \tag{D.12}$$

The mass term $e_0\|v\|_2^2$ allows the last integral to be at most $\tau \mathfrak{h}_Z(v)$ for any prescribed $\tau > 0$. For example use a sine/cosine pair of a Lipschitz cutoff angle, equal to $\pi/2$ on a fixed compact base set containing $\operatorname{supp} A_\varepsilon$ for all small ε, and zero outside a sufficiently large compact neighborhood. Its transition width makes its derivative arbitrarily small. The cosine part sees only V. Its bound is the assumed global form inequality for V. The sine part has compact base support, where the old potential is bounded. On a joined side also use a squared partition into cylinder intervals of a sufficiently large fixed length. The first member is identically one near $s = 0$ and extends as one on the bulk; all the others vanish near the interface and extend as zero there. Their squared gradients have an arbitrarily small sum. All cutoffs in the base are pulled back to the cylinder.

Choose the partition error so small that a violation of Equation (D.11) forces one localized test to have quotient larger than $\max\{\theta, .78\} + \eta/2$. This follows directly by summing the potential integrals and Equation (D.12); for instance it suffices to have $(\max\{\theta, .78\} + \eta/2)(1+\tau) < \max\{\theta, .78\} + \eta$. The selected test has compact base support and a bounded cylinder interval. Passing to a subsequence, its interval either stays in a fixed bounded range, or escapes to infinity. In the latter case it is translated to a fixed interval on the pure cylinder. Both the old bound and Equation (D.10) hold there by translating its tests into the original joined side. Thus in all cases the selected sequence lies in one fixed Hilbert space, in a fixed compact region.

Suppose now that $\varepsilon_i \downarrow 0$ gives such violating tests v_i, normalized by $\mathfrak{h}(v_i) = 1$. By Lemma D.3 and weak Hilbert compactness, $v_i \rightharpoonup v$ in this fixed space and $v_i \to v$ strongly in L^2. Write $w_i = v_i - v$. Then

$$\mathfrak{h}(w_i) = 1 - \mathfrak{h}(v) + o(1), \qquad \|w_i\|_2 \to 0.$$

The bounded old potential satisfies $\int V v_i^2 \to \int V v^2 \le \theta \mathfrak{h}(v)$. The new form vanishes on v, and its mixed term vanishes by Cauchy–Schwarz and the uniform bound on w_i. Consequently Equation (D.10) gives

$$\begin{aligned} \limsup_i \int (V + A_{\varepsilon_i}) v_i^2 &\le \theta \mathfrak{h}(v) + .78 \limsup_i \mathcal{N}(w_i) \\ &\le \theta \mathfrak{h}(v) + .78(1 - \mathfrak{h}(v)) \le \max\{\theta, .78\}. \end{aligned}$$

This contradicts the selected quotient. There are only three raw pieces, so the same sufficiently small scale works for all of them. □

We next extend a single chosen addition. The scale in Lemma D.4 is fixed before this next limit is taken.

D.2 Extension of potentials and functions

Lemma D.5 (Extension with late starting levels)**.** *Let B be a component, let $D \ge 0$ be a fixed smooth compactly supported potential addition on B, and let V denote the old cumulative potential on every component. Suppose these old potentials are exactly compatible and obey the common form bound $\theta\mathfrak{h}$. Suppose also that*

$$\int (V_B + 1.16D) v^2 \le \beta \mathfrak{h}_B(v), \qquad \beta \ge \theta,$$

on the boundary and joined sides over B. For the finitely many specified incident higher-dimensional component occurrences $B \times (T,\infty)^d \subset E$, consider smooth potentials D_E satisfying

$$0 \le D_E \le D \circ \pi_B, \qquad \operatorname{supp} D_E \subset K \times \prod_{i=1}^{d} [R_i,\infty), \qquad K \Subset B. \tag{D.13}$$

Here π_B is the projection of this specified product occurrence to its end base B. Assume the extensions use the same functions of the individually labelled normal coordinates under the commuting end base identifications, and hence are compatible. For every $\eta > 0$ all starts can be chosen sufficiently large that

$$\int (V_E + 1.16 D_E) v^2 \le (\beta + \eta) \mathfrak{h}_E(v) \tag{D.14}$$

on every affected raw piece. The conclusion concerns the same fixed boundary supports; it uses their product convergence from Corollary B.7.

Proof. Only finitely many occurrences are involved by Definition 6.1. Here and below a source is extended along the directed end-base-to-component relations, through their finite depth. For a common normal coordinate take the largest of the finitely many requested starts, so commuting chains give identical occurrences. It is enough to prove the assertion for any sequence of their starting levels with $\min_i R_i \to \infty$, passing to a fixed occurrence and piece if needed. Choose every localization width below using η and e_0 first, independently of the starting level positions.

Separate the new support from its complement with a sine/cosine pair as in Equation (D.12). More explicitly, an angle can be $\pi/2$ times a product of a compact base cutoff, equal to one on K, and normal cutoffs which equal one on $[R_i,\infty)$ and vanish before $R_i - L$. Enlarge the compact base region and L until the sum of squared derivatives is sufficiently small. Extend the angle by zero outside the product sector. Taking the starting levels beyond all collar thresholds makes this extension well defined. The cosine test misses the addition and has the old bound θ. Partition the sine part further into fixed large boxes in every normal coordinate, and into fixed large intervals in cylinder height when applicable. The total localization error is as small a fraction of $\mathfrak{h}$ as required.

If Equation (D.14) failed along starting levels tending out, the same quotient selection as in Lemma D.4 would give normalized tests on selected boxes with quotient at least $\beta + \eta/2$. Such a box must meet the new potential support, since otherwise the old bound $\theta \le \beta$ applies. Its center therefore has

$$t_i \ge R_i - O(L) \longrightarrow \infty \quad \text{for every } i = 1,\dots,d.$$

Its base region is fixed and compact. Recenter every normal coordinate. A bounded cylinder interval stays in a bounded-height joined region, or, after another subsequence, is translated to a pure cylinder. The ambient geometry on the recentered regions is exactly the product geometry, and the boundary supports converge to those over B times $\mathbb{R}^d$. The old potential is exactly $V_B \circ \pi_B$ there, because the starting levels also exceed its fixed product thresholds. The new potential is bounded above by the fixed bounded function $D \circ \pi_B$.

Apply Lemma D.3. For its limit v,

$$\limsup \int (V_E + 1.16 D_E) v_E^2 \le \int (V_B + 1.16 D) \circ \pi_B \, v^2, \qquad \mathfrak{h}_{\lim}(v) \le 1. \tag{D.15}$$

The end base form bound applies to the right side by slicing the $\mathbb{R}^d$ coordinates. For defining tests this is Fubini's theorem: the gradient terms in the extra directions are nonnegative, and all coefficients $.8, 1.7, e_0$ are the same in every base dimension. Approximation in the limit completion, proved in Lemma D.3, gives the result for v. In a pure-cylinder limit use the end base joined

bound after translation in cylinder height. Thus the right side of Equation (D.15) is at most β, a contradiction. Finite incidence permits simultaneous choices for all occurrences. □

Lemma D.6 (Compatible normal-coordinate functions and prescribed open sets)**.** *Every compact forcing update of Lemma C.3 admits extensions to its specified incident higher-dimensional component occurrences satisfying Equation* (D.13)*, preserving the function gradient inequality with coefficient a_r, and enlarging only the designated prescribed open sets. The extension is an exact product sufficiently far in every old end base collar and is made once per occurrence, independent of the order of commuting end base relations.*

Proof. Consider a plus addition on B, with forcing function u and shell potential D. Choose a compact base region $K \Subset \{h_j > t_j\}$ containing both $\overline{\{u > -\infty\}}$ and $\operatorname{supp} D$ in its interior. The first set includes the inner pole region; keeping only $\operatorname{supp} D$, which is annular, would not suffice for this extension. In an incident higher-dimensional component product over K put

$$u_E(x,t) = \min\{u(x), w_1(t_1), \dots, w_d(t_d)\}, \qquad D_E(x,t) = D(x) \prod_{i=1}^{d} \chi_i(t_i). \tag{D.16}$$

Each normal-coordinate function w_i is $-\infty$ up to a late starting level, then crosses to $+\infty$, and remains there, with

$$2|w_i'| \le .58 w_i^2 + p_j^*$$

where it is finite. For example integrate $2w_i' = .58w_i^2 + p_j^*$ where it is finite and extend by its poles. The crossing length is $2\pi/\sqrt{.58p_j^*} < \infty$; finite truncations remain locally Lipschitz at both ends. Only this label's positive floor is used. Each smooth cutoff χ_i takes values in $[0,1]$ and *finishes* increasing to one before w_i begins its crossing. Choose these cutoff starting levels to satisfy Lemma D.5 for the already fixed D and shell scale. Outside this product occurrence use the inactive value $-\infty$. The old function is replaced by its maximum with u_E.

For locally Lipschitz finite truncations, a finite minimum or maximum has the derivative of one active branch almost everywhere (on a tie, the branch derivatives agree almost everywhere on the tie set). If the active finite branch is u, every normal-coordinate function has already left its $-\infty$ plateau; hence every $\chi_i = 1$ and the entire shell potential D is available. If it is a normal-coordinate function, its derivative cost is paid by the preexisting baseline:

$$.58 w_i^2 + p_j^* \le a_r w_i^2 + \Pi_j + D_E.$$

The old branch retains its old inequality. Thus no sum of derivatives of the normal-coordinate functions, and no extra normal-coordinate function potential, enters the budget. For a minus update negate u_E and take a minimum with the old function. The sets $\{h_j > t_j\}$ and $\{h_j < t_j\}$ for opposite signs are disjoint and contain no opposite old poles by Lemma C.1.

Write T_i^+ for the end of the ith normal-coordinate function crossing. Enlarge the plus prescribed open set by the open set

$$\{x : \delta_B(x) < \varepsilon/2\} \times \prod_i (T_i^+ + b_i, \infty), \qquad b_i > 0.$$

The properly embedded closed tails ensure that its closure in the incident higher-dimensional component has $\delta_B \le \varepsilon/2$ and $t_i \ge T_i^+ + b_i$. It is therefore contained in the open set $\{\delta_B < \varepsilon\} \times \prod_i (T_i^+, \infty)$, where every branch of the minimum is $+\infty$. This proves closure confinement, including at the normal-coordinate function starting levels, with the corresponding statement for a

minus prescribed open set. The starting level positions are also chosen beyond the exact product thresholds, so its support retains the designated sign of $h_j - t_j$.

Use the same normal-coordinate functions and cutoffs for the same normal coordinates in every identified occurrence. On an end base obtained by selecting a star in one of these coordinates, that coordinate's normal-coordinate function is eventually $+\infty$ and its cutoff is one, and that factor disappears from Equation (D.16). If an unrelated coordinate is assigned a star, an intersection sufficiently far in its collar would require x to escape into a further end base of B, which the compact support forbids. Finite incidence allows all relevant intersection thresholds to be exceeded at once. These observations prove compatibility for all chains and show why a commuting product occurrence is not counted more than once. □

D.3 Completing the enlargement of the prescribed open sets

Proof of Proposition D.2. First consider the component system after the radial cut. Process its finite set of depths from the deepest end bases upward. The approximations h_j and thresholds of Lemma C.1 are fixed once, before any updates. Initially the cumulative addition is zero. Maintain a single common upper form bound at least .78; it is harmless to start this bookkeeping bound at .78 although the initial form vanishes. Enumerate the countable pairs consisting of a component and a possible successive local addition number by distinct integers $\nu \geq 1$. Reserve $\gamma_\nu = .1\,2^{-\nu}$ for that pair, allocating at most half to Lemma D.4 and half to the simultaneous extensions of Lemma D.5. The common bound after any collection of completed updates is at most

$$.78 + \sum_{\nu \text{ used}} \gamma_\nu \leq .88 < .9.$$

We may consequently take $\theta_* = .88$.

Suppose all strict further end bases of E have been treated and their updates extended. The remaining intersection of the singular set with H_E^* is

$$K_E = \Sigma_E \cap H_E^*$$

is compact. To prove this, if points in it escaped, the component escape property Lemma A.3 would give a subsequence over a deeper end base B, with compact convergent base coordinates and every escaping normal coordinate tending to infinity. The updated prescribed open sets are exact products there. There are finitely many relevant labels, so pass to a further subsequence with a fixed subset of $m-2$ undecided labels. Membership in each of their closed sets G_j' passes to the base limit. Thus the limit base point has at least $m-2$ undecided labels. This implies membership in H_B^*, because $\dim B - 2 \leq m-2$. By Corollary B.7 the translated boundary supports converge to the boundary support over B times the normal Euclidean factor. Support convergence puts the limiting base point on the boundary support over B. A regular limiting boundary point would have a neighborhood of smooth graphical convergence, which contains no singular points of the approximating boundaries. The limit must therefore be singular on the end base. This contradicts its completed enlargement of the prescribed open sets. Thus K_E has no escaping sequence; it is closed, and the compact-part property proves compactness. If $m < 8$ it is already empty.

Denote this initial compact set by $K_E^{(0)}$. For each remaining label and each sign of $h_j - t_j$, cover $K_E^{(0)}$ in that open region by the countable compact pieces with the plane-containment property of Lemma C.2. These pieces need not themselves be an open cover. Enumerate the signed pieces; their eventual inner forcing neighborhoods will provide the open cover needed for termination. For each, choose its shell sufficiently small by Lemma D.4, then hold that scale fixed and extend its

normal-coordinate functions, potential, and prescribed open set by Lemmas D.5 and D.6. Supports can be as small as needed to stay within the prescribed set $\{h_j > t_j\}$ or $\{h_j < t_j\}$. Opposite updates for one label therefore never conflict. Functions retain Equation (D.2) by their active-branch inequalities, and prescribed open sets enlarge monotonically.

To prove finite termination, after N additions set

$$O_N = \{x \in E : \#\{j \neq j_0 : x \in G_j^{(N)}\} < m - 2\}, \qquad K_E^{(N)} = K_E^{(0)} \setminus O_N.$$

Each O_N is open: with finitely many labels its complement is a finite union of intersections of closed $G_j^{(N)}$. Enlargement of the prescribed open sets implies $O_N \subset O_{N+1}$. Fix $x \in K_E^{(0)}$. For each of its strict inequalities $h_j > t_j$ or $h_j < t_j$, some enumerated compact piece contains x, and the addition on that piece puts x in an enlarged prescribed open set, irrespective of how small its positive shell scale was chosen. Only finitely many approximations h_j are involved. Once those additions have occurred, only threshold-equality labels can remain undecided at x. By Lemma C.1 there are at most $m - 8 < m - 2$ of those. Thus $x \in O_N$ for some finite N. The increasing open sets O_N cover $K_E^{(0)}$, so compactness supplies a finite subcover; taking its largest index gives $K_E^{(N)} = \varnothing$. Stop there. All subsequent updates only shrink the set H_E^*.

There may be countably many components at one depth. An originating component has finitely many directed incident higher-dimensional component occurrences, so each individual extension has already been chosen in finitely many places. Conversely, a fixed target component has finitely many source end bases, including through chains, so only finitely many components at that depth affect it. Each of those sources required finitely many additions. This uses both directions of finite incidence from Definition 6.1. Its updated functions are finite sums or finite minima and maxima; there is no infinite limiting function on that component. Its product thresholds are bounded by finitely many chosen thresholds, and all its potential derivatives remain bounded. The common form bound after a whole depth follows from the same summable allowance. This justifies proceeding to the next depth, and proves the asserted smoothness, compatibility, and form bounds throughout the system.

The new set H_E^* is closed because the number of labels is finite and each G_j' is closed. On a far collar, exact product prescribed open sets give the same count upstairs and downstairs. Lower end base dimension weakens the required count, proving the projection assertion in (ii). These are precisely the closed-set and product-limit properties needed for the regularization near H_E^* in Appendix E; a quantitative smooth tube is obtained there from regular graphical convergence.

For the initial radial cut, process its locally finite compact transition shells in order, choosing the supports of all additions inside buffered neighborhoods of their respective shells. By Lemmas 6.4 and B.4, the barrier regions separate these neighborhoods, and the boundary support lies in their interiors. The singular set of the boundary support in a shell is closed inside its compact buffered annulus, and the set H_E^* is closed. Their intersection is therefore compact. Apply the same increasing-open-cover argument there, keeping the closure of every forcing region inside this one shell buffer. There is no incident higher-dimensional component extension at this step. Before any addition only finitely many previous additions have been made, so their total is a bounded compactly supported potential and Lemma D.4 applies. The base localization is obtained from a proper distance cutoff; all estimates after localization use only fixed compact data. Use summable allowances as before. Local finiteness leaves smooth potentials and well-defined functions, and a compact test sees only finitely many shell additions. The form bound therefore persists after all shells are treated. This proves the radial assertion and completes the construction. □

E Regularization and restoration of the curvature invariant

The minimizing partition is smooth away from a singular set, whereas the next cutting step needs smooth complete components. Enlargement of the prescribed open sets has made the closed set H_E^* disjoint from the singular set; the first lemma below obtains the regular tubes needed for comparison. We construct globally smooth partitions that accurately approximate the partitions defined by the minimizing sets near these sets H_E^*. Throughout this section a stage r and its current label j_0 are fixed. The functions and prescribed open sets of the remaining labels have already been updated by Proposition D.2. Write

$$\widetilde{\Pi}_j = \Pi_j + D_j, \qquad G'_j = E \setminus (U_j^{+,\mathrm{updated}} \cup U_j^{-,\mathrm{updated}}),$$
$$H_E^* = \Big\{ x \in E : \#\{ j \in J_r \setminus \{j_0\} : x \in G'_j \} \geq m - 2 \Big\}.$$

Here $m = \dim E$ is the parent base dimension. The set H_E^* is closed and misses the singular set of the boundary support. A nonpositive counting threshold means the whole component. Scalar curvatures, mean curvatures, Laplacians, and measures include the old torus factors; functions and tests are invariant in those factors. Constants in this section may depend on a component and its finite incidence diagram. At the first, radial step the corresponding constants are assigned on enlarged compact radial blocks.

Use adjacent parameters $0 \leq t_r < t_{r+1} \leq 1$; for Theorem 6.5 they are $t_r = r/N_*$, which gives the stage constants of (6.2). Put

$$\Delta_r = t_{r+1} - t_r, \qquad a_r = .60 + .03 t_r, \qquad b_r = 1.12 - .05 t_r, \qquad \lambda_r = .25 - t_r/8,$$
$$\eta_r = \Delta_r/200, \qquad e_0 = (\lambda_r - \lambda_{r+1})/100 = \Delta_r/800, \qquad C = 3.99.$$

With the parameters of (7.3) they are those of Section 7, e_0 standing for e_r. Only positivity of the particular increment is needed. The proofs below are stated for finite label data. Their constant-preserving form, spelled out in Lemma E.6, will also be used for a finite exceptional list together with a summable constant tail.

Proposition E.1 (The full invariant step)**.** *Assume the stage r invariant and the conclusion of Proposition D.2. If every nonempty parent has base dimension at least two, there is a compatible split along regular zero sets of j_0 as in Definition 6.1, with the required forced signs and the specified continuous child-to-parent projection maps. There are smooth compatible old metrics and* 399 *new torus factors on every component such that the functions, prescribed open sets, potentials, and scalar curvature satisfy the stage $r+1$ invariant. This includes a possible one-dimensional output, on which the scalar lower bound is global. After the radial split the components have the complete compact-part/product-end geometry of Lemma A.2.*

To obtain the scalar conclusion, Lemmas E.4 and E.5 carry the form estimate before smoothing to the smooth split near its required set. Lemma E.7 gives the next function inequalities and the local positive form gap. The correction and supersolution statements Lemmas E.8 and E.10 extend this to an operator on each whole component; the proof of Proposition E.1 at the end of the appendix converts its supersolution into the 399 new circle factors.

E.1 Partitions near H_E^* and transfer of the form before smoothing

Lemma E.2 (Uniform control on the neighborhood of H_E^*)**.** *Fix an integer $J \geq 4$. For the fixed partitions defined by the minimizing sets and updated sets H_E^* on each parent component E, let ς_E*

be signed distance to the boundary support, truncated at ± 1*, positive in the plus Caccioppoli set. If the support is empty on a component, use the corresponding constant sign. After the radial split, the following choices can be made on each whole parent component.*

(i) There is a positive $\rho_E < 1/10$ *such that near*

$$\{x : \operatorname{dist}(x, H_E^*) < 4\rho_E,\ |\varsigma_E(x)| < 4\rho_E\}$$

the boundary support is regular, has an embedded normal tube, and signed distance has bounded derivatives through order $J+2$ *on a slightly smaller tube. The current closures of the prescribed open sets have positive uniform distance from the boundary support.*

(ii) On a collar with one normal factor $B \times (T, \infty)$*, signed distances converge uniformly to the pullback of* ς_B *as* $T \to \infty$*. They converge in* C^J *on the normal tubes near* H_E^**, with the incident higher-dimensional component tube chosen inside the end base's tube. These statements are uniform over the whole end base, including its noncompact ends.*

For the initial radial parent, assertion (i) *holds separately on each enlarged compact radial block, with its own radius and derivative bounds; assertion* (ii) *is not required. The radii depend on the partitions defined by the minimizing sets and the updated sets* H_E^**, and can be chosen before any smooth-approximation tolerances.*

Proof. On a compact set, closedness of H_E^* and its disjointness from the singular set give a regular neighborhood of every point where the set H_E^* meets the boundary. A finite subcover and the normal inverse function theorem give a positive tube radius and all the asserted finite-order bounds there. The same compactness argument applies to the separation of the current closures of the prescribed open sets, which lie inside the barrier domains of Lemma B.1 and Theorem B.6 and whose neighborhoods miss the support.

We explain why these statements remain uniform at infinity. If a sequence escapes in E, the finite end base description in Lemma A.3 gives, after passage to a subsequence, a product region over a further end base F in which the F coordinates converge in a compact set and all the remaining normal coordinates tend to infinity. Bounded metric balls lie in these product regions eventually. The minimizing sets converge there to the product of the minimizing set of F, by Corollary B.7. The updated prescribed open sets and their complements are exact products. Consequently a limit of points of H_E^* lies in H_F^*: a count of at least $m-2$ is at least $\dim F - 2$. The limiting boundary point is therefore regular. Smooth, single-sheet convergence near this point supplies an embedded normal tube of a fixed positive radius, and excludes additional support in a smaller neighborhood. In particular no sequence at distance tending to zero from both the set H_E^* and the support can have degenerating tube radius or unbounded prescribed finite derivatives. This proves the first assertion by contradiction, with buffers reduced if necessary. The barrier statement at infinity follows in the same way from its end base statement and set/support convergence.

For uniform convergence in an entire collar with one normal factor, suppose a discrepancy persists at (x_i, t_i) with $t_i \to \infty$. If x_i stays in a compact part of B, local product convergence gives a contradiction. Otherwise descend through the finite end base diagram of B. Commutation of its collars with the t collar puts both the partition of E and that of B over the same further end base F, with compact tangential coordinates and all escaping coordinates tending to infinity. Both limiting partitions are the same unique product partition. Local support convergence implies convergence of truncated distances: a nearest point at distance less than 1 is detected in a slightly larger fixed ball, and if there is no such point both truncated distances approach the saturated value. Away from the limiting support, Caccioppoli set convergence fixes the sign. This proves uniform C^0 convergence.

On a normal tube near H_E^* the limiting boundary is regular; smooth graph convergence and the normal inverse function theorem give C^J convergence of signed distance on smaller tubes.

Choose radii from the deepest end bases upward. Each incident higher-dimensional component radius is smaller than a fixed fraction of every relevant further-end base radius. For example, a point within ρ_E of H_E^* projects, far out, to within $2\rho_E$ of H_B^*, and $|\varsigma_E| < \rho_E$ implies $|\varsigma_B| < 2\rho_E$ there. These are statements about the fixed minimizing geometry and exact prescribed open sets, without approximation tolerances. This order also handles a simultaneous escape in several coordinates and proves the last assertion. At the radial step the same proof on compact blocks requires no assertion about an initial spatial end. □

Lemma E.3 (Compatible smooth partitions)**.** *The fixed partitions defined by the minimizing sets admit compatible smooth defining functions g_E with regular value 0, positive and negative on the respective closures of the current prescribed open sets, and with $L_E = g_E^{-1}(0) \subset \operatorname{int} G_{j_0}$. After the radial split they can be chosen uniformly as close as prescribed to ς_E in value and in C^J on smaller tubes near H_E^*. In a buffered neighborhood of H_E^*, a smooth diffeomorphism arbitrarily close to the identity in C^{J-1} identifies the smooth partition with the partition defined by the minimizing set. It uses the same map on the boundary and on the boundaries of its two sides. At the initial radial step, the approximation tolerances and comparison buffers are chosen separately on enlarged compact radial blocks. The defining functions, but not this comparison map, are exact products in sufficiently far old collars.*

Proof. First choose all the geometric radii in Lemma E.2. Prescribe positive value and C^J error targets on the neighborhoods of H_E^*, small enough also to preserve the prescribed open set signs. Allocate smaller targets to end bases to meet the requests of all their incident higher-dimensional components, including through chains, as in Lemma A.4. There are only finitely many such requests per end base and finite depth. This allocation does not change any tube radius.

Work now from deeper end bases upward. Smooth ς_E by local convolution and a smooth partition of unity, with arbitrarily small value error everywhere and derivative error on the regular buffers. These errors can be prescribed separately: use the smooth function itself on an inner regular buffer and blend to a value approximation outside a larger one. On a far collar blend to the previously chosen end base function. Lemma E.2 makes the discrepancy from the end base's signed distance to the boundary support arbitrarily small by moving the collar start. The discrepancy of the already chosen end base function from that distance is instead paid from its preallocated error target. These two errors are kept separate.

More explicitly, take cutoffs f_i from 0 to 1 in the one-factor normal coordinates, in valid far product ranges. Use weights

$$w_0 = \prod_i (1 - f_i), \qquad w_i = f_i \prod_{a<i} (1 - f_a).$$

They sum to 1. All derivatives through order J are bounded by constants fixed before the targets are allocated (translated cutoffs have the same bounds). Thus blending costs a fixed multiple of the participating errors. End base copies agree on all relevant overlaps once the starts exceed their product thresholds. When any $f_i = 1$, every participating copy equals that end base copy on the overlap; hence the result is exactly compatible. This proves the claimed approximations.

Near a possible zero in a normal tube near H_E^*, the defining function has nonzero normal derivative and its zero is a small normal graph over the boundary support. Off the tube, uniform value closeness preserves the Caccioppoli set where its sign is required. The copied end functions already have regular zeros. On the remaining compact region, a relative regular-value perturbation

makes all zeros regular without changing the copied ends. For completeness, if g is the preliminary function, choose a smooth $\chi \geq 0$, positive on its nonregular zeros and zero near the regions to be kept fixed. For an arbitrarily small regular value c of $-g/\chi$ on $\{\chi > 0\}$, the function $g + c\chi$ has regular zeros there: at such a zero,

$$d(g + c\chi) = -\chi\, d(-g/\chi) \neq 0.$$

The scalar function $-g/\chi$ is smooth on its open domain, so the same smooth Sard theorem used for the radial barriers supplies arbitrarily small choices of c. Where $\chi = 0$, its differential is that of g, since $d\chi = 0$ there. Bounded derivatives of χ and small c preserve all strict error and sign requirements. On the initial space use a positive smooth χ with locally prescribed small derivatives and the same regular-value argument; the radial separators are preserved.

If the perturbed boundary is $t = \alpha(x)$ in a normal tube of the boundary support, the map

$$(x, t) \longmapsto (x, t - \chi_1(x, t)\alpha(x))$$

with $\chi_1 = 1$ on the needed graph region and supported inside a larger regular buffer is a diffeomorphism for sufficiently small graph norm. It carries the graph to $t = 0$, preserves its two local sides, and tends to the identity in C^{J-1}. Choose the buffer cutoff on the regular boundary slightly beyond the projection of the smaller neighborhood of H_E^* and extend by the identity. Uniform regular charts give bounded cutoff derivatives. The same formula restricts to the boundary used as a cylinder base. □

Perform the split of Lemma A.1 using these defining functions. For the moment use the metric before smoothing: the old metric on the bulk and the frozen induced metric on each fresh infinite cylinder. Let p_A denote restriction/inclusion on a boundary or bulk and cylinder projection on a fresh cylinder. Pull back the remaining functions, prescribed open sets, and $\widetilde{\Pi}_j$ by p_A. These data are continuous at an interface and compatible in old ends. Define the new required set D_A from the remaining prescribed open sets and the new base dimension. The forms before smoothing of Proposition B.10 were defined on the minimizing boundary and its joined sides. We first estimate those forms, then use the comparison map to transfer the estimate to the metrics before smoothing of the smooth split just constructed.

Lemma E.4 (Counts and transfer of the form before smoothing)**.** *For sufficiently accurate defining functions in Lemma E.3, the scalar form before smoothing on each new component dominates the potential*

$$\lambda_{r+1} + (1 + 3\eta_r) b_{r+1} \sum_{j \neq j_0} \widetilde{\Pi}_j, \tag{E.1}$$

on a positive uniform neighborhood of D_A, with strictly positive mass and gradient reserves. The form has gradient coefficient $C = 3.99$ and includes the interface term $2 \int H_{\mathrm{out}} u(0)^2$ on a joined side. Tests are compactly supported, with matching interface traces.

Proof. For a side component $\dim A = m$, a point in D_A projects to at least $m - 1$ remaining undecided labels, hence into H_E^* and the parent's old required set. For a component of the zero set $\dim A = m - 1$, it projects to at least $m - 2$ remaining undecided labels. The boundary lies in G_{j_0}, which supplies the additional old undecided label. The projection is again in both required sets. These inclusions are unchanged on the fresh cylinder, where all relevant data are pulled back from its base. All these maps decrease lengths in the metrics before smoothing; therefore a small neighborhood of D_A projects into small neighborhoods of the two parent sets. The closed-set count, rather than a continuity claim for the counting function, is what is used here.

On compared pieces defined by the minimizers in a sufficiently small neighborhood of the old required set, continuity and the bounded derivatives of the fixed parent data give

$$R \geq \lambda_r + b_r \sum_j \Pi_j - e_0, \qquad e_0 = \frac{\lambda_r - \lambda_{r+1}}{100}.$$

By Proposition B.10 and Definition B.11, the form before smoothing on these pieces dominates $R - 1.02\Pi_{j_0}$, with spare gradient coefficients .94 on bulk and 1.99 on a boundary or cylinder. The bound in Proposition D.2 pays for the additional potential $1.16 \sum_{j \neq j_0} D_j$ from these spare gradient terms and a mass loss of at most $.9e_0$. Since $b_r > 1.02$, dropping the remaining nonnegative current-label term leaves

$$Q_{\min}(u) \geq \int \left(\lambda_r - 2e_0 + b_r \sum_{j \neq j_0} \Pi_j + 1.16 \sum_{j \neq j_0} D_j \right) u^2 + .2 \int |\nabla u|^2. \tag{E.2}$$

Indeed the unrounded gradient remainders are $.94 - .9 \cdot .8 = .22$ and $1.99 - .9 \cdot 1.7 = .46$, and the mass loss is at most $1.9e_0$. The exact numerical margins are

$$\begin{aligned} a_{r+1} - (1 + \eta_r) a_r &= \frac{3\Delta_r(180 - t_r)}{20000} \geq .02685\Delta_r > 0, \\ b_r - (1 + 3\eta_r) b_{r+1} &= \frac{\Delta_r[664 + 15(t_r + \Delta_r)]}{20000} \geq .0332\Delta_r > 0, \\ 1.16 - (1 + 3\eta_r) b_{r+1} &= \frac{800 + 1000 t_r + 664\Delta_r + 15\Delta_r(t_r + \Delta_r)}{20000} > 0. \end{aligned} \tag{E.3}$$

These identities use $0 \leq t_r < t_r + \Delta_r \leq 1$. Thus Equation (E.2) exceeds Equation (E.1) by at least $98e_0$ in mass and .2 in gradient, in addition to strictly positive potential margins.

Use the comparison diffeomorphism of Lemma E.3 on bulk and on the common boundary/cylinder base, leaving the cylinder coordinate and torus variables unchanged. A test supported in a smaller buffered neighborhood of D_A becomes a compact matching test on the regular pieces defined by the minimizers, extended by zero outside the comparison region. For a test in one component set it to zero on other components. In the bulk away from the boundary the map is the identity and the side signs already agree. All supports stay away from singular points and their cylinders.

The pulled-back metric, volume weight, potentials, intrinsic stabilized scalar curvature of the boundary, and outward mean curvature converge uniformly with the chosen finite derivatives. For example the induced scalar curvature can be computed from Gauss' equation, so the second fundamental form and the ambient curvature give the requisite bounds. Consequently, for arbitrarily small $\delta > 0$ the discrepancy of the compared forms is bounded by

$$\delta \left[\int (u^2 + |\nabla u|^2) + \int_{\text{interface}} u(0)^2 \right].$$

The last term is controlled using the first unit interval of the raw product cylinder:

$$\int_{\text{interface}} u(0)^2 \leq 2 \int_{S \times [0,1]} (u^2 + |\partial_t u|^2). \tag{E.4}$$

This follows by writing $u(0) = u(t) - \int_0^t \partial_t u$ and integrating in t. It needs no zero boundary value condition. Choosing all approximation errors sufficiently small leaves, for example, a reserve of $90e_0$ in mass and .15 in gradient. These choices can be prescribed on parent buffers before the individual new components are known: the projection inclusions place every relevant point there. At the radial step the enlarged compact blocks supply the bounds; preserved separators trap each new component in one such block before cylinder attachment. Finally impose all incident higher-dimensional component requests through the finite tolerance allocation already used in Lemma E.3. □

E.2 Interface smoothing and restriction of functions

The proof below uses scalar-curvature concentration at a mean-curvature jump, as in the corner smoothing of Miao [30, Sections 2–3]. Here the smoothing width varies tangentially, and the form estimate must preserve compatible product ends.

Lemma E.5 (Smoothing an interface with variable width)**.** *Each metric with a corner admits smooth compatible approximations with arbitrarily small C^0 error and scalar forms converging to its scalar form before smoothing as follows. On each joined component and for every $\delta > 0$, the smoothing can be chosen so that the absolute form error is at most*

$$\delta \int (u^2 + |\nabla u|^2)$$

for every compact matching test, using the metric before smoothing to measure the right side. The old warped-product structure and every sufficiently far old product end are preserved. The choices on a boundary can satisfy the requests of both incident sides at once.

Proof. In a normal collar of a smooth boundary write the old metric on the warped torus extension as $ds^2 + \beta_s$ for $s \le 0$, where s increases outward. Here β_s includes the old torus factors. The metric with a corner is $ds^2 + \beta_{\min(s,0)}$. Choose a smooth concave function Φ equal to t for $t \le -1$ and to 0 for $t \ge 1$, with $0 \le \Phi' \le 1$. Such a function is obtained by integrating a smooth decreasing step function symmetric about 0. It satisfies $\Phi \le 0$ and $\int \Phi'' = -1 = -\int |\Phi''|$.

For a smooth positive function $\epsilon(x)$ on the boundary put

$$\phi(x, s) = \epsilon(x)\Phi(s/\epsilon(x)), \qquad G_\epsilon = ds^2 + \beta_{\phi(x,s)}.$$

It equals the metric before smoothing outside $|s| < \epsilon(x)$ and differs from it in C^0 by $O(\sup \epsilon)$. In a fixed tangential chart the extra derivatives of ϕ satisfy

$$|\partial_x \phi| \le c|d\epsilon|,$$
$$|\partial_x^2 \phi| \le c\big(|d^2\epsilon| + |d\epsilon|^2/\epsilon\big).$$

Choose $\epsilon = \zeta^2$ with $\zeta > 0$ as small in C^2 as prescribed. Then $|d\epsilon|^2/\epsilon = 4|d\zeta|^2$ and $d^2\epsilon = 2\zeta d^2\zeta + 2d\zeta \otimes d\zeta$. Thus these tangential errors tend to zero. On a component, positive compatible choices have a positive infimum and bounded derivatives by the compact-part/product-end description. Their size, but not the resulting higher metric bounds, can be made arbitrarily small. Normal collar widths are also positive componentwise by the same description.

Let $B_\tau^{\mathrm{old}}, H_\tau^{\mathrm{old}}$ be the slice second fundamental form and mean curvature of $ds^2 + \beta_s$. Exactly,

$$B_s = \phi_s B_\phi^{\mathrm{old}}, \qquad H_s = \phi_s H_\phi^{\mathrm{old}}.$$

For a metric $ds^2 + \beta(x, s)$ the Gauss and normal variation identities give

$$R(G_\epsilon) = R(\beta(x, s)) - |B_s|^2 - H_s^2 - 2\partial_s H_s.$$

The tangential derivative bounds just proved show that all terms on the strip are uniformly bounded except

$$-2\phi_{ss} H_\phi^{\mathrm{old}}. \tag{E.5}$$

In particular the formula differentiates H_s only in s; potentially large mixed xs derivatives do not give another scalar spike. The remaining contribution from this derivative is $-2\phi_s^2 \partial_\tau H_\phi^{\mathrm{old}}$, which is bounded.

Let $J_\epsilon(x,s)$ be the density of β_ϕ relative to the zero slice. Uniformly on the strip, $J_\epsilon = 1 + O(\epsilon)$ and $H_\phi^{\mathrm{old}} = H_0^{\mathrm{old}} + O(\epsilon)$. For each fixed x,

$$\int -\phi_{ss}\, ds = 1, \qquad \int |\phi_{ss}|\, ds = 1.$$

Writing $E_x = \int_{-\epsilon(x)}^{\epsilon(x)} |\partial_s u|^2 ds$, the fundamental theorem of calculus gives

$$|u(s) - u(0)|^2 \le \epsilon(x) E_x, \quad |u(s)^2 - u(0)^2| \le 2\sqrt{\epsilon(x)}|u(0)|E_x^{1/2} + \epsilon(x)E_x.$$

After integrating against $|\phi_{ss}|ds$, the error in replacing $u(s)^2$ by $u(0)^2$ is at most $c\sqrt{\sup\epsilon}\,(u(0)^2 + E_x)$, for $\sup\epsilon \le 1$. The density and mean-curvature errors cost at most $c\sup\epsilon\,(u(0)^2 + E_x)$. Thus the form contribution of Equation (E.5) converges to $2\int H_0^{\mathrm{old}} u(0)^2$, with the outward sign required in Lemma B.9.

The bounded strip terms have vanishing form cost because

$$\int_{-\epsilon(x)}^{\epsilon(x)} u(s)^2 ds \le c\epsilon(x)u(0)^2 + c\epsilon(x)^2 E_x.$$

The metric and density errors in the gradient and fixed bounded potential terms tend to zero in H^1 form norm. Integrate these estimates in x and use Equation (E.4); the stated error bound follows. One first proves it for smooth matching tests and then for the matching H^1 completion by density.

Choose ζ on new components of the zero set, from deeper end bases upward, after propagating the two incident-side requests through their finite diagrams. The function is an exact pullback in old product ends, so the displayed smoothing is also an exact product there. It is unchanged far out in the fresh cylinder coordinate. Since the normal coordinate and old metric already split off every old normal factor, the operation commutes with all the inherited collar identifications. □

Lemma E.6 (Approximation that reproduces constants)**.** *On a component, continuous scalar functions admit smooth approximations with any prescribed positive value error, through approximation operators that reproduce constants. For compatible input on a component diagram, the approximations can be chosen compatible in all sufficiently far product ends of that diagram. On a component E, the data used in averaging and blending are its own input and the end base copies from its downward source diagram, as defined in Theorem 6.5(iii). If all these inputs equal the same constant c, the result equals c on E exactly. The same assertion holds componentwise for a finite list of functions and for a list having only finitely many nonconstant entries on each prescribed finite directed end-base/component diagram.*

Proof. Choose a locally finite coordinate cover, relatively compact smaller charts, and a smooth partition of unity $\{\psi_\alpha\}$ subordinate to the smaller charts. Extend each chart slightly and average the function against a smooth nonnegative kernel of integral one, at a sufficiently small chart-dependent scale. Call the resulting local function $T_\alpha f$. Uniform continuity on the chart closure makes $|T_\alpha f - f|$ smaller than the prescribed error on $\operatorname{supp}\psi_\alpha$. Thus

$$Tf = \sum_\alpha \psi_\alpha T_\alpha f$$

is a smooth approximation with that error. Every local average sends c to c, and $\sum\psi_\alpha = 1$, so $Tc = c$ exactly. Locally prescribed derivative accuracy is obtained in the same way wherever f is already smooth, using a slightly larger regular buffer.

Now use the finite tolerance allocation of Lemma A.4, and blend from deeper end bases upward with the weights w_i in the proof of Lemma E.3. Their sum is one. Consequently blending preserves a common constant as well as exact end base identities. The finite exceptional union over the component's downward source diagram is chosen before applying this construction. For every other entry the entire local input and all participating end base copies coincide with the same constant, so all its averaging and blending errors are zero. On the initial radial space the construction is local on buffered compact blocks and uses a locally finite chart cover. It has the same constant-reproduction property there. □

On a new component A, write R_{old} and Δ_{old} for the scalar curvature and invariant Laplacian of the smoothed split metric with its retained torus factors, before any new circle factors are added.

Lemma E.7 (Restricted functions and smooth potentials)**.** *The interface smoothing and potential approximations can be chosen so that the pulled-back functions and prescribed open sets satisfy the stage $r+1$ function conditions with smooth compatible potentials Π_j^{new} chosen so that*

$$(1+\eta_r)\widetilde{\Pi}_j \le \Pi_j^{\mathrm{new}} \le (1+2\eta_r)\widetilde{\Pi}_j.$$

If

$$V_A^* = \lambda_{r+1} + b_{r+1}\sum_{j\neq j_0}\Pi_j^{\mathrm{new}}, \qquad L_A^0 = -C\,\Delta_{\mathrm{old}} + R_{\mathrm{old}} - V_A^*, \tag{E.6}$$

then on some positive uniform neighborhood of D_A its form is bounded below by a positive multiple of $\|u\|_2^2$.

Proof. We justify restriction to a hypersurface, which does not follow merely by restricting an ambient almost-everywhere inequality. If f is locally Lipschitz on an open Riemannian set and $|df| \le F$ almost everywhere for continuous F, then at every point its local Lipschitz slope is at most F. In a small coordinate ball, convolve f on a slightly larger ball. The derivative bound of the convolutions is at most the supremum of F there times a metric distortion factor tending to 1 as the ball shrinks. Integrating this bound along straight segments, passing to the uniform limit, and shrinking the ball proves the assertion. It follows also that every rectifiable curve satisfies the corresponding integrated bound, by subdividing into such balls.

Apply this on each open finite-value locus of a function, with $F = (a_r\mu_j^2 + \widetilde{\Pi}_j)/2$. Restriction to the boundary, inclusion of bulk, and projection on a cylinder therefore preserve the bound everywhere in local slope, and hence almost everywhere for the restricted locally Lipschitz function. Finite truncations remain locally Lipschitz, including where a pole is present. Across an interface the same conclusion follows by splitting a path into its bulk and cylinder portions; traces agree. A sufficiently small C^0 metric change increases all these slopes by at most $1+\eta_r$.

For each label j, the raw potential has its own positive lower bound $p_j^* > 0$. Apply Lemma E.6 to $(1+3\eta_r/2)\widetilde{\Pi}_j$, with strict value error smaller than $\eta_r p_j^*/2$. This proves the displayed relative bounds. Different labels may require different accuracies. When the raw potential is a spatial constant c_j on the whole downward source diagram of the new component, choose its new value to be exactly $(1+3\eta_r/2)c_j$. By Equation (E.3),

$$2|\nabla\mu_j|_{\mathrm{new}} \le (1+\eta_r)(a_r\mu_j^2 + \widetilde{\Pi}_j) \le a_{r+1}\mu_j^2 + \Pi_j^{\mathrm{new}}.$$

Continuous projections preserve the inclusion of every closure of the prescribed open set inside its pole interior and preserve all prescribed q constraints.

Choose the interface form error small relative to the reserves of Lemma E.4. The potential in Equation (E.6) is at most that in Equation (E.1); the extra η_r supplies additional margin for the

potential smoothing. Metric equivalence allows the required-set neighborhood to be slightly reduced after smoothing. Thus one may retain, for example, an $e_0\|u\|_2^2$ lower bound for the form of L_A^0 on tests in that neighborhood. No compatibility of the neighborhoods themselves is asserted. □

We have obtained a smooth component metric and the next function inequalities. The scalar lower bound is still only a quadratic-form bound near the new required set. We next extend its positive gap to the whole component by a correction that vanishes near that set, solve a bounded elliptic equation, and approximate its solutions compatibly. The final group of 399 circle factors converts this form information into the pointwise scalar inequality.

E.3 Compatible corrections and positive supersolutions

At this point the smooth split components have compact truncations and exact product ends by Lemma A.1. At the radial step the preserved separating regions give the compact initial pieces. Hence Lemma A.2 already applies to these metrics and retained warping functions, before the following correction is constructed. In particular all bounds used below hold separately on each whole component.

Lemma E.8 (A compatible correction with a global gap)**.** *For the fixed operators L_A^0 of* (E.6)*, there are smooth nonnegative bounded compatible correction potentials K_A, with all derivatives bounded componentwise, such that $K_A = 0$ in a positive uniform neighborhood of D_A and*

$$\int (C|\nabla u|^2 + (R_{\mathrm{old}} - V_A^* + K_A)u^2) \geq g_A\|u\|_2^2 \qquad (g_A > 0) \tag{E.7}$$

for every compact smooth test. Let ω_A^{old} be the product of the retained warping functions for the smoothed metric on A, before the new circle factors at this split are appended. With K_A acting by multiplication, define

$$L_A = L_A^0 + K_A = -C\,\Delta_{\mathrm{old}} + R_{\mathrm{old}} - V_A^* + K_A.$$

All integrals use $\omega_A^{\mathrm{old}}\,\mathrm{d}\,\mathrm{vol}_A$. In an exact end over B, the base metric is a product and every retained warping function pulls back from B. Consequently

$$\Delta_{\mathrm{old},A} = \Delta_{\mathrm{old},B} + \Delta_{\mathrm{normal}}, \qquad L_A = L_B - C\,\Delta_{\mathrm{normal}}\,.$$

Proof. First disregard compatibility on a fixed component. Put $Q_A(u) = \int (C|\nabla u|^2 + (R_{\mathrm{old}} - V_A^* + K_A)u^2)$ in the measure weighted by the product of retained warping functions, and put $W_A = R_{\mathrm{old}} - V_A^*$. Write Q_A^0 for the same form with $K_A = 0$. Choose a smooth angle θ equal to 0 near D_A and $\pi/2$ outside the neighborhood of local positivity, with bounded derivatives and with the support of $\cos\theta$ inside that neighborhood. Uniform charts and the positive neighborhood width give such a function by smoothing a distance cutoff with buffers. Set

$$K_{\mathrm{ind}} = J\sin^2\theta + C|\nabla\theta|^2, \qquad J > \max\{0, -\inf W_A\}.$$

The identity for the squared partition $\cos\theta, \sin\theta$ shows that $Q_A^0 + K_{\mathrm{ind}}$ has gap $\gamma_A = \min\{e_0, J + \inf W_A\} > 0$. Indeed the gradient partition error is exactly $C|\nabla\theta|^2u^2$; local positivity controls the cosine part and $W_A + J > 0$ controls the sine part. The correction is zero on a smaller uniform neighborhood of D_A.

Proceed from deeper end bases upward. In sufficiently far collars with one normal factor the already constructed K_B copies agree on intersections: on a common quadrant each is the pullback of the same further-end base potential. They consequently define a *single* nonnegative function

K_{end} on the collar union. Copies are not added together on an intersection. The metric, measure, and V_A^* are likewise exact products there. In particular V^* has the same formula on an end base: C, λ_{r+1}, b_{r+1}, the label set, and the old torus list do not depend on base dimension, after padding unused flat factors on each finite compared diagram.

For the finitely many collars with one normal factor choose angles θ_i which rise from 0 to $\pi/2$ over a normal interval of length L, inside the valid common product ranges. Use the squared partition

$$\eta_0 = \prod_i \cos\theta_i, \qquad \eta_i = \sin\theta_i \prod_{a<i} \cos\theta_a.$$

It has $\sum_i \eta_i^2 = 1$ and $\sum_i |d\eta_i|^2 \le \sum_i |d\theta_i|^2 \le c_A L^{-2}$; the inequality follows successively from the two-term sine/cosine identity, so no number of intersections is being summed separately. The support of η_0 lies in a compact collar truncation.

Choose smooth multipliers $\alpha, \beta \in [0,1]$ with $\alpha = 1$ on $\operatorname{supp}\eta_0$ and $\beta = 1$ on the union of the end base-cutoff supports. Let α vanish only farther out than those transition intervals. Let β vanish on the inner side of a buffered collar union, so βK_{end} extends smoothly by zero. Both can be formed from products of one-coordinate cutoffs. Define

$$K_A = \alpha K_{\mathrm{ind}} + \beta K_{\mathrm{end}}.$$

On the central support this dominates K_{ind}; on each collar support it dominates the single end base copy. Eventually $\alpha = 0, \beta = 1$, giving exact compatibility. Nonnegativity permits the harmless overlap of these two terms in the finite transition region.

This correction vanishes in a uniform neighborhood of D_A. The individual correction already does. For the end term, a point of D_A in a product collar projects into D_B, because an end base has a smaller required counting threshold. Start the support of β with a buffer from the collar entry. A point sufficiently close to D_A in this support then has a nearby required point in the same collar, and their end base projections are just as close. It lies in the vanishing neighborhood of K_B. Take the minimum of the finitely many end base widths and entry buffers; at corners the copies agree, so the same conclusion applies.

Apply the individual gap to $\eta_0 u$. Apply each end base gap by slicing to $\eta_i u$; its support is away from the collar entry, and the normal gradient term is nonnegative. If $g_* = \min(\gamma_A, \{g_B\}) > 0$, the finite localization identity gives

$$Q_A(u) \ge (g_* - C c_A L^{-2}) \|u\|_2^2.$$

Choose L so that the error is at most $g_*/2$. All quantities used for this choice are fixed before solving any elliptic equation. Boundedness and the derivative assertions follow from compact parts and finitely many product copies. Empty required sets or absent end bases cause no change: one may take $\theta = \pi/2$ or omit the corresponding cutoff. This completes the compatible induction and proves the operator identity. □

Lemma E.9 (Bounded solutions of $Ls = 1$)**.** *Let A carry the complete component geometry and finite retained warping function weight ω of Lemma A.2, and let $\Delta_\omega = \omega^{-1}\operatorname{div}(\omega \operatorname{grad}\cdot)$. Suppose W is smooth and bounded with bounded derivatives, and $L = -C\,\Delta_\omega + W$ has form gap $g > 0$. Then there is a unique bounded smooth solution s of $Ls = 1$. It satisfies $0 < \inf s \le \sup s < \infty$ and has bounded derivatives of every fixed order. Bounded smooth supersolutions of $Lw \ge 0$ are nonnegative. The same conclusions hold on Euclidean products of A.*

Proof. Use the positive injectivity radius, uniform bounds on curvature and all its covariant derivatives, positive bounded weight, and center-uniform polynomial volume estimate of Lemma A.2.

The form gap and boundedness of W make the closed form strictly coercive on H^1: combine $Q \geq g\|u\|_2^2$ with $Q \geq C\|\nabla u\|_2^2 - \|W_-\|_\infty \|u\|_2^2$. It therefore has a self-adjoint inverse on L^2, of norm at most g^{-1}. If $f \geq 0$, testing $Lu = f$ with the negative part gives $Q(u_-) \leq 0$, so the inverse is positive.

Here is the decay estimate needed for bounded sources. We use an Agmon-type exponential-weight argument [1], proved here for the weighted component operator. If f is supported in a metric unit ball I and $u = L^{-1}f$, put $\chi = \exp(\delta \min\{d(\cdot, I), D\})$, where $C\delta^2 < g$. The form identity for a bounded Lipschitz multiplier gives

$$Q(\chi u) = \langle Lu, \chi^2 u\rangle + C\int |\nabla\chi|^2 u^2.$$

It holds by compact approximation in H^1; on the support of f, $\chi = 1$. Hence

$$(g - C\delta^2)\|\chi u\|_2^2 \leq \langle f, u\rangle \leq g^{-1}\|f\|_2^2. \tag{E.8}$$

Let $D \to \infty$. The polynomial volume bound, uniform in the center of I, implies

$$\int_A e^{-2\delta d(x,I)}\,\mathrm{d}\mu(x) \leq C_{A,\delta} < \infty.$$

Indeed sum the exponential bound on successive unit annuli against their polynomial volume upper bound.

Choose smooth compactly supported $0 \leq f_R \leq 1$ equal to 1 on an increasing compact exhaustion, and put $u_R = L^{-1}f_R \geq 0$. These are initially H^1 weak solutions. For each fixed R, the coefficients and f_R are smooth, and weak interior regularity makes u_R smooth [37, Lecture 6, Theorem 1 and Corollary 1].

For any $h \in L^2(I)$, self-adjointness, Equation (E.8), and Cauchy–Schwarz with opposite exponential weights give

$$|\langle u_R, h\rangle| = |\langle f_R, L^{-1}h\rangle| \leq \frac{C_{A,\delta}^{1/2}}{\sqrt{g(g - C\delta^2)}}\|h\|_2.$$

Thus u_R has a uniform local L^2 bound on every unit ball, with constant independent of its center and of R. Choose the uniform chart radius small enough that each larger chart ball lies in a metric unit ball. Writing B_1 and $B_{1/2}$ for the fixed rescaling of two such nested chart balls, interior local boundedness gives

$$\|u_R\|_{L^\infty(B_{1/2})} \leq c_A\big(\|u_R\|_{L^2(B_1)} + \|f_R\|_{L^\infty(B_1)}\big).$$

Here we use Simon [37, Lecture 18, Theorem 2, with $p = 2$]; the fixed radius and coordinate changes are absorbed in c_A. The principal coefficients are uniformly elliptic and bounded, the lower-order coefficients are bounded, and the weight is bounded above and below. Since $0 \leq f_R \leq 1$, this gives a bound independent of the center and R.

Fix two nested relatively compact coordinate domains. For all sufficiently large R, the source is exactly 1 on the larger domain. The interior Schauder estimate [37, Lecture 12, Theorem 2], followed by differentiation, therefore bounds every derivative of u_R on the smaller domain, uniformly for these R. A diagonal subsequence converges smoothly on compact subsets to a bounded nonnegative solution s of $Ls = 1$. Applying the same estimates to this limiting equation in the uniform component charts gives bounded derivatives of every fixed order on A.

The maximum argument must allow W to have either sign. Let w be a bounded smooth supersolution, put $w_- = \max\{-w, 0\}$, and fix $0 < \delta < \sqrt{g/C}$. Write $\psi = e^{-\delta d(x,x_0)}$ and take

compact Lipschitz cutoffs ζ_R equal to 1 on $B(x_0, R)$, supported in $B(x_0, 2R)$, with $|d\zeta_R| \le 2/R$. Testing the supersolution with $(\psi\zeta_R)^2 w_-$, and expanding the gradient, gives directly

$$g\|\psi\zeta_R w_-\|_2^2 \le Q(\psi\zeta_R w_-) \le C\int |d(\psi\zeta_R)|^2 w_-^2.$$

All tests are compact. Boundedness of w and polynomial volume growth make ψw_- square integrable. The terms containing $d\zeta_R$ tend to zero, including the cross term by Cauchy–Schwarz. Dominated convergence therefore yields

$$g\|\psi w_-\|_2^2 \le C\delta^2\|\psi w_-\|_2^2,$$

so $w_- = 0$. On an open subdomain the same compact-test argument applies when the negative part has zero boundary trace, using the Dirichlet form gap. In particular it applies to nonnegative boundary values. Apply comparison to both signs of the difference of two bounded solutions to obtain uniqueness. If $M = 1 + \|W\|_\infty$, then $L(s - M^{-1}) = 1 - W/M \ge 0$, and comparison gives $s \ge M^{-1}$.

Euclidean products retain a positive injectivity radius and uniform bounds on curvature and all its covariant derivatives and polynomial volume, and their form gap follows by slicing. The preceding proof therefore applies verbatim; uniqueness identifies their solution with the pullback of s. □

Lemma E.10 (Exact solutions and compatible approximations)**.** *Fix all the corrections K_A and hence the operators L_A of Lemma E.8. There are positive smooth compatible functions v_A such that*

$$L_A v_A \ge \tfrac{1}{2} \tag{E.9}$$

on every component. Each v_A has positive infimum and bounded derivatives of every fixed order on its component.

Proof. For each fixed L_A, let s_A be the unique bounded solution $L_A s_A = 1$ from Lemma E.9. To approximate s_A by compatible functions v_A, we first estimate $s_A - s_B$ uniformly over a whole end base, including its deeper ends. On an exact collar with one normal factor $B \times [T, \infty)$, the difference $w = s_A - s_B$ is bounded and satisfies $L_A w = 0$. Set $m_B = \inf s_B > 0$ and choose $a > 0$ with $Ca^2\|s_B\|_\infty \le 1/2$. The function

$$b(x,t) = e^{-a(t-T)} s_B(x)$$

satisfies $L_A b = e^{-a(t-T)}(1 - Ca^2 s_B) \ge \frac{1}{2} e^{-a(t-T)}$. Comparison on the half collar, using its sliced Dirichlet gap and the Dirichlet comparison argument in the proof of Lemma E.9, yields

$$|s_A - s_B|(x,t) \le \frac{\|s_A - s_B\|_\infty}{m_B}\, e^{-a(t-T)} s_B(x). \tag{E.10}$$

The boundary inequalities hold at $t = T$ for both signs. Uniform interior elliptic estimates for the homogeneous difference give the same exponential decay in C^2 on $t \ge T + 1$. All constants are uniform in $x \in B$, including its deeper ends. Thus the estimate holds on each collar with one normal factor, also where collars overlap. Exact agreement of the approximations on those overlaps will come from the compatible end base copies used in the following construction.

All operator coefficients and every exact s_A are now fixed. Let M_A be a finite bound for $L_A : C^2 \to C^0$ in uniform charts, and choose a target

$$\tau_A < \min\{\tfrac{1}{4}\inf s_A,\ (4M_A)^{-1}\}.$$

The finite family of collar blending cutoffs has a C^2 operator bound $H_A \geq 1$, independent of translation of their starts. Include in H_A the fixed finite factor for the number of copies. For each component B, let ε_B be the minimum of

$$\frac{\tau_A}{\prod_{i=0}^{\ell-1}(4H_{A_i})} \qquad (A = A_0 > A_1 > \cdots > A_\ell = B),$$

over all directed component-to-end-base chains ending at B, including the length-zero chain. There are finitely many such chains, so this minimum is positive. Appending the edge $A > B$ to a minimizing chain for A gives $\varepsilon_B \leq \varepsilon_A/(4H_A)$. All these numbers are fixed after the corrections and operator norms, and before constructing any v_A.

Construct v_A from deeper end bases upward. Keep s_A in the central part and use the ordinary weights in the proof of Lemma E.3 to blend to the already constructed copies of v_B. Move the starts until $\|s_A - s_B\|_{C^2} < \varepsilon_A/(4H_A)$ on all participating collars, using Equation (E.10). The persistent difference $v_B - s_B$ is paid by its previously allocated tolerance, not by moving the start. Start also beyond the finitely many compatibility thresholds for the v_B's, so their copies agree on overlaps. The derivative bounds for the weights then give

$$\|v_A - s_A\|_{C^2} \leq \varepsilon_A \leq \tau_A.$$

The result is an exact end base copy sufficiently far out, is smooth, and has bounded higher derivatives by compact parts and finitely many smooth products. Finally $v_A \geq \frac{3}{4} \inf s_A > 0$ and $L_A v_A \geq 1 - M_A \tau_A \geq 3/4$, which implies Equation (E.9). □

E.4 Restoring the invariant

The passage from a positive solution to scalar-curvature control by a warped product with a circle fiber appears in Fischer-Colbrie and Schoen [16, Theorem 4(ii)]. Here bounded positive supersolutions must also be exactly compatible with every component end; Lemma E.10 supplies these functions.

Proof of Proposition E.1. Use Lemmas E.3, E.5 and E.7 for the smooth base metrics and the function data, and Lemmas E.8 and E.10 for v_A. Append $k = 399$ circle variables, each with warping function $v_A^{2/(k+1)}$. The warped-product identity is

$$R_{\mathrm{new}} = R_{\mathrm{old}} - \frac{4k}{k+1} \frac{\Delta_{\mathrm{old}} v_A}{v_A}.$$

To include the interaction with the old factors, write $U = \sum_\alpha \log f_\alpha$ for the sum of their logarithms and $z = \log f$ for the common new logarithm of the warping function. The scalar formula in Lemma 5.3, applied to the combined list, gives

$$\begin{aligned} R_{\mathrm{new}} - R_{\mathrm{old}} &= -2k\,\Delta_h z - 2k\langle dU, dz\rangle - k(k+1)|dz|_h^2 \\ &= -2k\,\Delta_{\mathrm{old}} z - k(k+1)|dz|_h^2. \end{aligned}$$

Here $\Delta_{\mathrm{old}} z = \Delta_h z + \langle dU, dz\rangle$ is the Laplacian of the old warped metric on invariant functions, equivalently the base Laplacian with weight $e^U = \prod_\alpha f_\alpha$. In terms of f, the same expression is $-2kf^{-1}\,\Delta_{\mathrm{old}} f - k(k-1)f^{-2}|df|_h^2$. Substituting $z = 2\log v_A/(k+1)$ and using $\Delta_{\mathrm{old}} \log v_A = v_A^{-1}\,\Delta_{\mathrm{old}} v_A - v_A^{-2}|dv_A|_h^2$ cancels the two $|dv_A|^2$ terms. Here k counts only the newly appended circle factors. Any unit factors used earlier to compare old lists on a finite diagram have zero logarithmic

derivative and leave both Δ_{old} and this coefficient unchanged. Since $4k/(k+1) = C$, Equations (E.6) and (E.9) give

$$R_{\mathrm{new}} = V_A^* - K_A + \frac{L_A v_A}{v_A} \geq V_A^* \quad \text{on } D_A.$$

Here $K_A = 0$ on a neighborhood of D_A. Adding these factors does not change base distances or function gradients. The new factors are positive, bounded above and away from zero, and exactly compatible. The same number is appended in every index at the stage, so all end identities compare the same list of factors. This proves the analytic invariant, including when the new required threshold is zero. The componentwise metric operations preserve every unused old circle length identically equal to 1: restriction, product extension, and $\beta_s \mapsto \beta_{\phi(x,s)}$ all fix that constant entry. The 399 appended lengths are the only common new entries. Likewise restriction and cylinder pullback preserve a constant pole function and its full prescribed open set, and Lemma E.6 preserves the specified multiplied constant potentials.

The zero set lies in $\operatorname{int} G_{j_0}$ and all current forced signs hold by Lemma E.3; earlier prescribed open sets are carried by the continuous projection and retain their pole neighborhoods. Thus this is exactly the regular compatible splitting operation required by the component construction. At the radial split, the preserved alternating compact barrier regions separate all boundary components into bounded transition shells and trap every closed sign-side component between two separating regions (with a compact initial block). These components are compact before prolongation. The normal collars and smooth metrics used above are products in all old ends. Lemmas A.1 and A.2 therefore supply the claimed geometry after prolongation. □

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